CATALAN MERTENS, VERSION 9: CATALAN FUSION, ARITHMETIC RESTRICTION SPECTRA, AND THE FACTOR-FOUR RETURN PROBLEM

Contents

1. Introduction 3
2. The selection wall 4
3. Exact Catalan decomposition of the compressed Mertens walk 5
3.1. Compressed time 5
3.2. Return skeleton and factorization 5
4. The empirical factor-four return phenomenon 6
5. Open Catalan sectors: meanders and pointwise control 7
6. Catalan free energy and a sufficient selection principle 8
7. Brownian excursion and meander as continuum Catalan mechanics 9
7.1. A variational action 9
8. Transfer matrices and the quantum Catalan picture 10
8.1. Height confinement as a spectral problem 10
8.2. Fredkin spin chains 10
9. Statistical mechanics of Dyck paths: pressure, area, and q-Catalan weights 11
10. Non-equilibrium transport and the Burgers route 11
11. The deterministic arithmetic selector 12
11.1. Björner's filtered complex 12
11.2. Subset-sum energy representation 12
12. Three possible selector mechanisms 13
12.1. Hyperplane-chamber entropy 13
12.2. Shadow inequalities and parity restoration 13
12.3. Persistent topology 14
13. Structural viability of the physical routes 14
14. Potential obstructions 14
14.1. The selector may remain unconstrained at the needed scale 14
14.2. Static topology may not control filtered dynamics 14
14.3. Parity barrier 14
14.4. Canonical collapse to a classical Dirichlet object 15
15. A hierarchy of possible sufficient principles 15
16. The prime-delay selector and its exact interference algebra 15
16.1. Simultaneous prime-shell identities 15
16.2. Threshold kernel and transfer recursion 16
16.3. A maximal-parity Walsh coefficient 17
16.4. Uniform prefix energy: a corrected sufficient principle 18
16.5. The exceptional Catalan class is exactly the RH obstruction 18
17. All-short-interval cancellation as a mesoscopic Catalan selector 20
17.1. The exact prime-coloured anti-lift 22
17.2. The Catalan–Ramaré stopping problem 22
17.3. The Catalan occupation-energy hierarchy 22
17.4. The exact occupation spectrum 24

17.5. Catalan tent conversion: peaks and area 26
17.6. The correctly calibrated frontier 27
17.7. Prime dilations: exact gain and exact failure 27
17.8. Narayana peaks: a complementary local operator 29
17.9. No-go audit: what the exact identities do not prove 33
18. An aggregate route that does not predict M pointwise 35
18.1. A hierarchy of moments equivalent to RH 35
18.2. Exact excursion decomposition of the moment budget 36
18.3. The arithmetic correlation target 37
18.4. Finite evidence and its exact limitation 37
18.5. A staged proof program 37
19. Version 9: the Catalan fusion program 38
19.1. Decision: fuse first, specialize to M"obius second 38
19.2. Universal state space and the master spectrum 39
19.3. Prefix-profile polynomial and insertion algebra 40
19.4. Catalan and lattice-path matroid fusion 40
19.5. Sectoral entropy and the zero-count boundary 42
19.6. Arithmetic specialization and its exact limitations 44
19.7. Collective restrictions: circuits, capacities, and prime bands 45
19.8. Operator, determinant, and causal-cutoff modules 46
19.9. Fused work packages 46
19.10. Milestones and fail gates 47
19.11. Relation to the factor-four phenomenon 47
19.12. Logical boundary with RH 47
20. Research program retained from Version 8 47
20.1. Exact finite combinatorics 48
20.2. Finite selector chambers 48
20.3. Dynamic shadows 48
20.4. Transfer operators 48
20.5. Hydrodynamic coarse graining 48
21. Relation to existing literature 48
22. Conclusions and open questions 49
Appendix A. Reproducible computations and archived outputs 50
Appendix B. Python source: moment hierarchy audit 51
Appendix C. Python source: exact selector identities 53
Appendix D. Python source: excursion heights and Catalan tails 55
Appendix E. Python source: finite τ-dephasing illustration 58
Appendix F. Python source: publication figures 59
Appendix G. Python source: Version 5 no-go audit retained in Version 6 61
Appendix H. Python source: Version 6 mesoscopic Catalan audit 62
Appendix I. Python source: Version 6 causal Ramaré prefix audit 64
Appendix J. Python source: Version 7 Catalan-energy exponent ladder 66
Appendix K. Python source: Version 8 spectrum, Narayana, frame, and return audit 68
Appendix L. Python source: Version 9 Catalan fusion and synergy certificate 74
References 78

Remark 0.1. This document is an exploratory mathematical notebook. Its purpose is to isolate an exact Catalan structure inside the Mertens walk, to develop physical and probabilistic tools that act directly on that Catalan state space, and to identify a precise obstruction between ambient Catalan rarity and the deterministic arithmetic trajectory. Rigorous statements are labelled as such; proposed mechanisms are labelled as heuristics, conjectures, or questions.

Remark 0.2 (Version 9). Version 9 preserves the introduction, the Catalan theory, the physical routes, the factor-four computation, the analytic audit, the bibliography, and every reproducibility appendix of Version 8. Its new contribution is a research architecture built around a single refined object: a partially restricted Catalan class together with its complete prefix profile, return spectrum, height spectrum, and sectoral entropy flow. The labels [LIT], [DED], [ADAPT], and [REC] distinguish literature, deduction, adaptation, and research proposal. No proposed bridge to M"obius is called a theorem unless it is proved independently of the desired Mertens bound.

1. Introduction

Let

$$M(x) = \sum_{n \le x} \mu(n) \tag{1.1}$$

be the Mertens function. The classical square-root target is

$$M(x) = O_\varepsilon\Big(x^{1/2+\varepsilon}\Big) \qquad (\varepsilon > 0), \tag{1.2}$$

which is equivalent to the Riemann hypothesis; see, for example, the standard analytic literature and the discussion in Björner's topological formulation [12]. The present notebook does not pursue an explicit-formula, zero-density, or Dirichlet-series proof strategy. The target (1.2) is retained as the benchmark against which a different mechanism is measured.

The starting observation is elementary but exact. Remove the zero values of the Möbius function. If

$$1 = q_1 < q_2 < q_3 < \cdots \tag{1.3}$$

are the squarefree positive integers and

$$X_m = \sum_{j \le m} \mu(q_j), \tag{1.4}$$

then every increment of X_m is exactly $+1$ or -1, and

$$X_m = M(q_m). \tag{1.5}$$

Consequently, each interval between consecutive zeros of X_m is a primitive lattice excursion. If such an excursion has length $2n$, then, after fixing its sign, the number of possible excursion shapes is exactly the Catalan number

$$C_{n-1} = \frac{1}{n}\binom{2n-2}{n-1}. \tag{1.6}$$

Thus the real Mertens excursion is one state inside a finite Catalan state space.

The largest finite anomaly found in the supplied data is itself a Catalan observable. At $m = 276{,}638{,}778$ compressed steps the path has 53,078 returns, and its zero local time is 3.9996 times the iid mean scale $\sqrt{2m/\pi}$. Section 4 shows that this excess persists across many finite scales, while carefully separating the exact data from any unproved asymptotic factor four. Because each return creates a new Catalan factor, explaining this local-time phenomenon is a central problem of the notebook rather than a descriptive afterthought.

The usefulness of this observation depends entirely on a second question. Catalan combinatorics describes the ambient set of admissible excursion shapes, but the arithmetic trajectory is a deterministic selection from that set. A rare Catalan shape remains possible unless the arithmetic selector itself satisfies an additional restriction. The central problem of this notebook is therefore to identify a deterministic mechanism that controls the location of the Mertens trajectory inside Catalan state space.

The physical methods considered here are useful because they quantify geometry inside that state space. Brownian excursion theory identifies the diffusive $\sqrt{L}$ scale; transfer matrices convert height restrictions into spectral gaps; path-integral arguments assign an action of order H^2/L to an excursion of height H and duration L; area-weighted Dyck paths provide genuine partition functions and free energies; Fredkin spin chains realize uniform superpositions of Dyck paths as quantum ground states; and TASEP provides a route from excursion statistics to non-equilibrium hydrodynamics and the Burgers equation [13, 14, 16, 17, 18, 20, 21].

The arithmetic side also has a nontrivial exact structure. Björner associated to each cutoff x a simplicial complex of squarefree integers and proved that the Mertens function is its reduced Euler characteristic [12]. In that representation, the compressed walk is the one-dimensional trace of a filtered, shifted simplicial complex. This supplies a deterministic selector which is absent from a generic random walk.

The project therefore separates into three layers:

(1.7) Mertens filtration $\longrightarrow$ Catalan excursion state $\longrightarrow$ physical/entropic cost of its geometry.

The difficult arrow is the first-to-second selection law: why should the filtration select a non-exceptional region of the Catalan space? The rest of the notebook develops several concrete formulations of this question.

2. The selection wall

This section states the main obstruction at the outset.

Definition 2.1. Let $\mathcal{D}_n^{\mathrm{prim}}$ denote the set of positive primitive excursions of length $2n$: paths $D = (D_j)_{0\le j\le 2n}$ satisfying

$$D_0 = D_{2n} = 0, \qquad D_j > 0 \quad (1 \le j \le 2n-1), \qquad D_{j+1} - D_j \in \{-1, +1\}. \tag{2.1}$$

Proposition 2.2. *For every $n \ge 1$,*

$$\left|\mathcal{D}_n^{\mathrm{prim}}\right| = C_{n-1}. \tag{2.2}$$

Proof. Delete the initial up-step and the final down-step and shift the remaining path down by one unit. This is a bijection with ordinary Dyck paths of length $2n-2$, counted by C_{n-1}. □

If a Mertens excursion has length $2n$, Proposition 2.2 places its shape in a set of size C_{n-1}. Under the uniform Catalan measure, high excursions occupy a small fraction of this set; the maximum of a uniform Dyck path is naturally of order $\sqrt{n}$, and exponential moments and moderate/large deviations are well studied [16, 17].

Warning 2.3. Ambient rarity is not an arithmetic restriction. A deterministic selector may choose a member of an exponentially small subset. Therefore a statement of the form

$$\frac{\#\{D \in \mathcal{D}_n^{\mathrm{prim}} : \max D \ge H\}}{C_{n-1}} \ll \exp\Big(-cH^2/n\Big) \tag{2.3}$$

becomes relevant to M only after one proves a property of the arithmetic selection rule.

This warning is the main logical checkpoint of the project. The physical theory can compute or estimate the cost of a region of Catalan space. The arithmetical task is to show that the real trajectory cannot systematically occupy regions whose cost is too large.

3. Exact Catalan decomposition of the compressed Mertens walk

3.1. **Compressed time.** Let

$$Q(x) = \#\{n \le x : \mu(n) \neq 0\} \tag{3.1}$$

be the squarefree counting function. The standard asymptotic is

$$Q(x) = \frac{6}{\pi^2}x + O(\sqrt{x}), \tag{3.2}$$

so compressed time and ordinary integer time are linearly comparable.

Proposition 3.1. *For every $m \ge 1$,*

$$X_m = M(q_m), \tag{3.3}$$

and for every real $x \ge 1$,

$$M(x) = X_{Q(x)}. \tag{3.4}$$

Proof. The terms omitted from (1.4) are exactly those for which $\mu(n) = 0$. □

3.2. **Return skeleton and factorization.** Let

$$0 = z_0 < z_1 < z_2 < \cdots \tag{3.5}$$

be the successive return times of the compressed walk to zero, whenever they exist. Put

$$L_r = z_r - z_{r-1} = 2n_r. \tag{3.6}$$

Parity forces each complete excursion length to be even.

Proposition 3.2. *Fix a finite return skeleton $0 = z_0 < z_1 < \cdots < z_R$ and fix the sign of each excursion. The number of nearest-neighbour paths realizing exactly these complete excursions is*

$$\mathcal{N}(z_1, \dots, z_R) = \prod_{r=1}^{R} C_{n_r - 1}. \tag{3.7}$$

Proof. The excursion interiors are independent combinatorial choices once their durations and signs are specified. Proposition 2.2 applies to each factor. □

Equation (3.7) is an exact entropy associated with the zero-return skeleton:

$$S_{\mathrm{Cat}}(z_R) = \log \mathcal{N}(z_1, \dots, z_R) = \sum_{r=1}^{R} \log C_{n_r - 1}. \tag{3.8}$$

Using Stirling's formula,

$$\log C_n = n \log 4 - \frac{3}{2} \log n + O(1). \tag{3.9}$$

Thus each completed excursion carries an entropy deficit of order $\frac{3}{2} \log n$ relative to unrestricted ± 1 words of comparable length.

4. The empirical factor-four return phenomenon

The return count is the discrete local time at zero of the compressed path,

$$R(m) = \#\{1 \le j \le m : X_j = 0\}. \tag{4.1}$$

It is not auxiliary data: $R(m)$ is exactly the number of completed Catalan factors in (3.7), and the full gap sequence entering S_{Cat} refines it.

At a return time z_R, the resulting entropy connection is already exact. Since $C_{n-1} \le 2^{2n-2}$,

$$-\log_2 \frac{\mathcal{N}(z_1, \dots, z_R)}{2^{z_R}} = \sum_{r \le R} \left(L_r - \log_2 C_{L_r/2-1} \right) \ge 2R. \tag{4.2}$$

Thus an excess of returns forces a corresponding minimum loss of compatible sign bits, although R alone does not determine the sharper entropy because the individual gaps still enter logarithmically.

For a simple symmetric random walk,

$$\mathbb{E}R_{\mathrm{rw}}(m) = \sum_{1 \le k \le m/2} \frac{\binom{2k}{k}}{2^{2k}} \sim \sqrt{\frac{2m}{\pi}}. \tag{4.3}$$

Moreover the normalized number of returns has a half-normal limiting law, not a concentrated Gaussian fluctuation around its mean [9]. Therefore a ratio to the expectation is a useful scale comparison but must not be read as a conventional number of standard deviations.

For the audited prefix with 276,638,778 nonzero steps,

$$R = 53{,}078, \qquad \frac{R}{\sqrt{m}} = 3.1912293779, \qquad \boxed{\frac{R}{\sqrt{2m/\pi}} = 3.9996128947.} \tag{4.4}$$

The corresponding tail of the limiting half-normal benchmark is about 1.42×10^{-3}. This is a finite model comparison, not a probability assigned to the deterministic Möbius sequence and not evidence for an asymptotic constant 4.

The excess is visible across several compressed scales rather than being created only at the last row:

m	$R(m)$	$R(m)/\sqrt{2m/\pi}$
10^3	93	3.686
10^4	339	4.249
10^5	1206	4.780
10^6	4353	5.456
10^7	9590	3.801
$5 \cdot 10^7$	21866	3.876
10^8	34232	4.290
$2 \cdot 10^8$	47215	4.184
276638778	53078	3.9996

At the same time, 26,511 of the 53,078 completed excursions have length 2, a proportion 0.49947, and the sign changes between successive excursions have proportion 0.50342. Both are close to their simplest iid benchmarks. Thus the visible anomaly is not an obvious excess in the shortest-excursion fraction or a bias in the next excursion sign; it lies in the accumulated renewal/local-time realization and its heavy-tailed gap sequence.

Indeed the ambient Catalan first-return law is exactly

$$\mathbb{P}_{\mathrm{rw}}(L = 2n) = \frac{2C_{n-1}}{2^{2n}} = \frac{C_{n-1}}{2^{2n-1}}, \qquad \mathbb{P}_{\mathrm{rw}}(L > u) \sim \sqrt{\frac{2}{\pi u}}. \tag{4.5}$$

The observed gap survival ratios relative to the last asymptotic expression are 1.0035 at $u = 10^2$, 0.9752 at 10^3, 0.9681 at 10^4, 0.9334 at 10^5, 0.8973 at 10^6, and 0.5974 at 10^7. Hence the factor

four is not explained by a visibly different short-gap law; the depletion appears in the extreme tail, precisely where a renewal with index $1/2$ is least self-averaging. Algebraically,

$$\frac{m}{R(m)^2} = 0.0981938 \approx \frac{\pi}{32}, \tag{4.6}$$

which is the square-scale restatement of (4.4), not an independent coincidence.

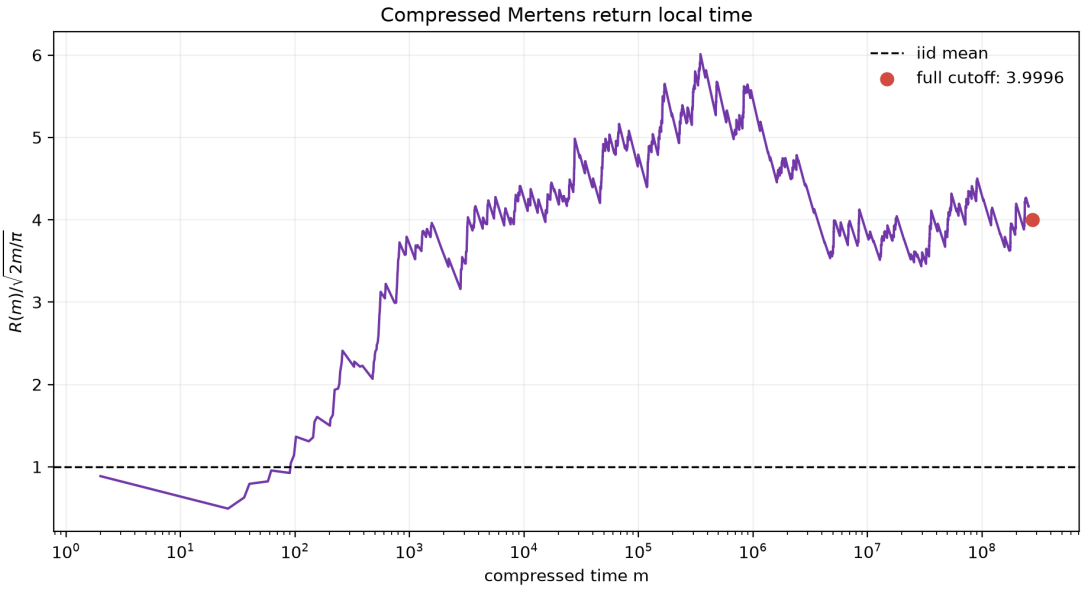


FIGURE 1. The finite return-local-time ratio along completed compressed Mertens excursions. The dashed line is the iid expectation scale. The red point uses the full cutoff, including the open meander after the last return. The curve is a data audit, not an asymptotic fit.

There is an exact Fourier formulation of the proof problem:

$$\boxed{R(m) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \sum_{j\le m} \exp\left(it \sum_{r\le j} \mu(q_r)\right) dt.} \tag{4.7}$$

Proving an arithmetic law for (4.7) would use the whole Catalan return skeleton rather than only a terminal Mertens sum. It is also noncircular: a lower bound of order $\sqrt{m}$ for the number of returns does not by itself bound the height inside the intervening excursions and hence does not imply RH. The honest next task is first to determine whether the elevated ratio persists under substantially larger cutoffs and then to seek a local-limit or renewal theorem for the deterministic compressed Möbius path. Equivalently, one may attack the arithmetic suppression of the extreme Catalan gap tail. At present neither an asymptotic factor 4 nor such a theorem is claimed.

5. Open Catalan sectors: meanders and pointwise control

Complete excursions alone do not control a point lying inside a very long unfinished excursion. The natural pointwise object is therefore a Catalan prefix, or Bernoulli meander. This is also the correct continuum bridge to Brownian meander [14, 15].

Definition 5.1. For $t \ge 1$ and $h > 0$ with $t \equiv h \pmod 2$, let $\mathcal{M}_{t,h}$ be the set of paths $S_0 = 0$, $S_t = h$, with increments ± 1 and

$$S_j > 0 \qquad (1 \le j \le t). \tag{5.1}$$

Let $\mathcal{M}_t = \bigcup_h \mathcal{M}_{t,h}$.

Proposition 5.2. *The ballot theorem gives*

$$|\mathcal{M}_{t,h}| = \frac{h}{t}\binom{t}{(t+h)/2}, \tag{5.2}$$

and

$$|\mathcal{M}_t| = \binom{t-1}{\lfloor (t-1)/2 \rfloor}. \tag{5.3}$$

Remark 5.3. The meander is the open-boundary sector of the same Catalan geometry. Closing a positive meander by a first return produces a primitive Dyck excursion, while cutting a primitive Dyck excursion before its terminal return produces a positive meander. For pointwise estimates on M, the meander formulation avoids any dependence on a future return time.

For a compressed time m let

$$a(m) = \max\{z_r : z_r \le m\}, \qquad t(m) = m - a(m), \qquad h(m) = |X_m|. \tag{5.4}$$

Whenever $h(m) > 0$, the segment from $a(m)$ to m is an element of $\mathcal{M}_{t(m),h(m)}$ after reflection if necessary.

Proposition 5.4. *There exist absolute constants $c, C > 0$ such that, under the uniform measure on $\mathcal{M}_t$,*

$$\mathbb{P}_t\{S_t \ge h\} \le C\left(1 + \frac{h}{\sqrt{t}}\right)\exp\left(-c\frac{h^2}{t}\right) \tag{5.5}$$

for admissible t, h.

Proof. Equation (5.2) reduces the endpoint law to binomial coefficients. Standard Chernoff bounds for the symmetric binomial distribution give Gaussian decay in h^2/t, while (5.3) is of order $2^t/\sqrt{t}$. Summing the endpoint masses from h upward gives (5.5) after adjustment of absolute constants. □

This proposition supplies a rigorous ambient energy scale at the current time, not only at completed returns.

6. Catalan free energy and a sufficient selection principle

Definition 6.1. For the actual compressed Mertens path at a time m with $h(m) > 0$, define its open Catalan selection energy by

$$\mathcal{I}_{\mathrm{Cat}}(m) = -\log \mathbb{P}_{t(m)}\{S_{t(m)} \ge h(m)\}, \tag{6.1}$$

where $\mathbb{P}_t$ is the uniform measure on $\mathcal{M}_t$.

This is a deterministic number attached to the actual path: probability is used only to measure the volume of the ambient Catalan/meander sector containing an endpoint at least as extreme as the observed one.

Proposition 6.2. *There are constants $c_1, c_2, c_3 > 0$ such that*

$$\mathcal{I}_{\mathrm{Cat}}(m) \ge c_1\frac{h(m)^2}{t(m)} - c_2\log\left(2 + \frac{h(m)}{\sqrt{t(m)}}\right) - c_3. \tag{6.2}$$

Proof. Take minus logarithms in Proposition 5.4. □

Theorem 6.3. *Assume that for some constant $A > 0$ and all sufficiently large compressed times m,*

$$\mathcal{I}_{\mathrm{Cat}}(m) \le A \log m. \tag{6.3}$$

Then

$$|X_m| = O\Big(\sqrt{m \log m}\Big), \tag{6.4}$$

and consequently

$$M(x) = O\Big(\sqrt{x \log x}\Big). \tag{6.5}$$

In particular, (1.2) *follows.*

Proof. By Proposition 6.2 and (6.3), the quantity $h(m)^2/t(m)$ is $O(\log m)$, up to a lower-order logarithmic correction. Since $t(m) \le m$, this gives (6.4). Equation (3.4) and the linear comparability (3.2) then give (6.5). □

Warning 6.4. The hypothesis (6.3) is precisely the unresolved selector statement. Theorem 6.3 is useful because it isolates the target in a finite combinatorial language: the arithmetic path must be prevented from entering meander sectors whose relative Catalan volume is smaller than a power of the current scale.

7. Brownian excursion and meander as continuum Catalan mechanics

Kaigh proved an invariance principle for a recurrent random walk conditioned on a late first return to zero: after diffusive rescaling the path converges to Brownian excursion [13]. Durrett, Iglehart, and Miller established weak convergence results for Brownian meander and excursion in related conditional settings [14]. These results provide a continuum mechanics for the Catalan state space rather than an independent model imposed on the arithmetic data.

For an excursion of discrete duration L, the diffusive scaling is

$$y_L(s) = L^{-1/2} D_{\lfloor Ls \rfloor}, \qquad 0 \le s \le 1. \tag{7.1}$$

The limiting excursion has order-one height, so the natural discrete height scale is

$$H \asymp \sqrt{L}. \tag{7.2}$$

The exponential moments of the normalized Dyck maximum are uniformly controlled and converge to the corresponding Brownian-excursion quantity [16]. Kousha further developed large and moderate deviation principles for the Dyck maximum [17].

7.1. **A variational action.**

Heuristic 7.1. This is a physical large-deviation heuristic. For a continuum path $y : [0, L] \to \mathbb{R}_+$ with $y(0) = y(L) = 0$, assign the Euclidean action

$$\mathcal{A}[y] = \frac{\kappa}{2} \int_0^L \dot{y}(t)^2 \, dt. \tag{7.3}$$

Among paths that reach height H, the piecewise-linear triangular profile minimizes (7.3). Its action is

$$\mathcal{A}_{\min}(H, L) = 2\kappa \frac{H^2}{L}. \tag{7.4}$$

Thus the physical rate variable is H^2/L, matching the Gaussian scale already present in the exact ballot estimate (5.5).

The importance of (7.4) is conceptual: the physical energy scale is not being guessed from Mertens data. It is the continuum expression of a rate already visible in Catalan/ballot combinatorics.

8. Transfer matrices and the quantum Catalan picture

8.1. **Height confinement as a spectral problem.** Let A_h be the adjacency matrix of the path graph on heights $0, 1, \ldots, h$. The number of Dyck paths of length $2n$ confined to this height interval is

$$Z_n(h) = (A_h^{2n})_{00}. \tag{8.1}$$

The eigenvalues are

$$\lambda_k(h) = 2\cos\left(\frac{k\pi}{h+2}\right), \qquad 1 \le k \le h+1. \tag{8.2}$$

Consequently,

$$Z_n(h) = \frac{2}{h+2}\sum_{k=1}^{h+1} \sin^2\left(\frac{k\pi}{h+2}\right)\lambda_k(h)^{2n}. \tag{8.3}$$

This spectral representation is standard in the analysis of Dyck-path maxima and underlies related deviation results [17].

For large h,

$$\lambda_1(h) = 2 - \frac{\pi^2}{(h+2)^2} + O(h^{-4}), \tag{8.4}$$

so confinement below an atypically low ceiling has a free-energy scale of order

$$\Delta F_{\mathrm{conf}} \asymp \frac{n}{h^2}. \tag{8.5}$$

This is complementary to the high-excursion action H^2/n in (7.4). The two scales are the spectral and variational faces of the same diffusive geometry.

8.2. **Fredkin spin chains.** The Fredkin spin chain provides a genuine quantum realization of the Catalan ensemble. In the spin-1/2 model, the ground state can be written as an equal-weight superposition of Dyck paths; related models and their correlation structure are developed in [21, 22]. Abstractly, if $\mathcal{D}_n$ is the Dyck basis,

$$|\Psi_n\rangle = \frac{1}{\sqrt{C_n}}\sum_{D\in\mathcal{D}_n} |D\rangle. \tag{8.6}$$

For any subset $\mathcal{A} \subseteq \mathcal{D}_n$ with projector $\Pi_{\mathcal{A}}$,

$$\langle\Psi_n|\Pi_{\mathcal{A}}|\Psi_n\rangle = \frac{|\mathcal{A}|}{C_n}. \tag{8.7}$$

Thus a Catalan free energy

$$\mathcal{F}_{\mathrm{Cat}}(\mathcal{A}) = -\log\frac{|\mathcal{A}|}{C_n} \tag{8.8}$$

has a literal interpretation as the negative logarithm of a quantum-sector weight in an equal-amplitude Catalan state.

Question 8.1. Can the arithmetic selector be represented by a local or quasi-local deformation of a Catalan transfer matrix or Fredkin-type Hamiltonian whose ground-state or low-energy sector contains the Mertens excursion? A positive answer would turn the selector problem into a spectral-energy estimate rather than a statement of random sampling.

9. Statistical mechanics of Dyck paths: pressure, area, and q-Catalan weights

Dyck paths are established physical configuration spaces for directed polymers and vesicles. Owczarek and Prellberg study area-weighted Dyck paths as a two-dimensional vesicle model and define pressure through finite-size free-energy differences [18]. Let $A(D)$ be the area under a Dyck path and define

$$Z_n(q) = \sum_{D \in \mathcal{D}_n} q^{A(D)}. \tag{9.1}$$

At $q = 1$,

$$Z_n(1) = C_n, \tag{9.2}$$

while $q \neq 1$ biases the ensemble by area. The asymptotics of area-weighted Dyck paths exhibit Airy-type scaling and have been analyzed in detail [19].

For a Mertens excursion D_r, define its absolute area

$$A_r = \sum_{j=z_{r-1}}^{z_r} |X_j|. \tag{9.3}$$

The pair (H_r, A_r) separates a sharp spike from a sustained displacement. A physical route based on q-Catalan weights would seek an arithmetic inequality on the partition weight of the sector containing (H_r, A_r).

Heuristic 9.1. In a Catalan polymer interpretation, excluding a region of path space produces an entropic pressure equal to a free-energy difference. An arithmetic excursion that remains at large height for a long interval can therefore be assigned a pressure or free-energy cost determined entirely by the number of Catalan paths compatible with that geometric constraint. The open problem is to connect this cost to the deterministic arithmetic selector.

10. Non-equilibrium transport and the Burgers route

The asymmetric exclusion process provides a distinct physical use of Brownian excursions. Derrida, Enaud, and Lebowitz showed that in the maximal-current phase of open TASEP, the non-Gaussian part of density fluctuations can be described in terms of Brownian excursion statistics [20]. At hydrodynamic scale, TASEP is governed by a conservation law of Burgers type,

$$\partial_t \rho + \partial_x(\rho(1-\rho)) = 0. \tag{10.1}$$

This creates a rigorous physical chain linking constrained lattice paths, excursion fluctuations, current transport, and non-linear hydrodynamics.

The present project does not identify the Mertens walk with TASEP. The possible use is structural: construct a dynamics on Catalan states whose conserved current or entropy production encodes the arithmetic selector. If the selector can be expressed as a constrained transport process, tools from hydrodynamic limits, entropy solutions, shocks, and large-deviation functionals become available.

Question 10.1. Is there a local state variable on the filtered squarefree complex for which the parity imbalance X_m is a conserved or approximately conserved current, and for which the admissible Catalan updates induce a non-equilibrium transport law?

A positive answer could replace a direct estimate on $\mu(n)$ by an entropy-production inequality on the Catalan dynamics. This is speculative, but the TASEP/Brownian-excursion/Burgers chain shows that such a passage from discrete constrained paths to hydrodynamic free energies is a real mechanism in statistical physics, not merely a metaphor.

11. The deterministic arithmetic selector

11.1. **Björner's filtered complex.** For each squarefree integer k, let $P(k)$ be the set of its prime divisors and define

$$\Delta_x = \{P(k) : k \le x,\ k \text{ squarefree}\}. \tag{11.1}$$

This is an abstract simplicial complex because divisors of a squarefree integer correspond to subsets of its prime set. Björner observed and exploited the exact identity

$$M(x) = -\widetilde{\chi}(\Delta_x), \tag{11.2}$$

where $\widetilde{\chi}$ is the reduced Euler characteristic [12]. He further showed that these complexes are shifted and studied their Betti numbers, wedge-of-spheres homotopy type, and combinatorial shadow inequalities. Björner explicitly noted that these results did not by themselves yield new control of the growth of $M(x)$ [12].

The filtration

$$\Delta_1 \subseteq \Delta_2 \subseteq \Delta_3 \subseteq \cdots \tag{11.3}$$

adds a simplex precisely when the cutoff crosses a squarefree integer. The change in Euler characteristic is therefore the sign of the corresponding Möbius value. The compressed Mertens walk is exactly the Euler-characteristic trace of this filtration.

11.2. **Subset-sum energy representation.** A squarefree integer corresponds to a finite prime subset S, and its logarithm is additive:

$$E(S) = \log \prod_{p\in S} p = \sum_{p\in S} \log p. \tag{11.4}$$

The sign is the parity

$$\mu\left(\prod_{p\in S} p\right) = (-1)^{|S|}. \tag{11.5}$$

Thus the selector can be stated exactly as follows: order all finite prime subsets by the additive energy (11.4), then record the parity (11.5). Mertens is the cumulative parity imbalance.

For finitely many primes $p_1, \dots, p_d$, let

$$w = (w_1, \dots, w_d), \qquad w_j = \log p_j, \tag{11.6}$$

and

$$E_w(S) = \sum_{j\in S} w_j. \tag{11.7}$$

The relative order of two subsets changes on a hyperplane

$$H_{S,T} = \{w : \langle \mathbf{1}_S - \mathbf{1}_T, w\rangle = 0\}. \tag{11.8}$$

Unique factorization implies that the prime-log vector does not lie on any hyperplane $H_{S,T}$ with $S \neq T$.

Proposition 11.1. *If a and b are distinct positive integers with $a, b \le X$, then*

$$|\log a - \log b| \ge \log\left(1 + \frac{1}{X}\right). \tag{11.9}$$

Consequently, the ordering of any fixed finite family of squarefree products $\le X$ is separated from its subset-sum collision hyperplanes by a positive deterministic margin.

Proof. Assume $a > b$. Then $a \geq b+1$ and $b \leq X$, so

$$\log(a/b) \geq \log(1+1/b) \geq \log(1+1/X). \tag{11.10}$$

□

This hyperplane-arrangement formulation gives a concrete geometric meaning to "selection": a finite parity word, and hence a finite Catalan excursion, is stable on a chamber of weight space. The arithmetic vector $(\log p_j)$ lies in one such chamber.

Definition 11.2. For a finite family $\mathcal{S}$ of prime subsets, define the selector chamber of the prime-log vector to be the connected component containing $w = (\log p_j)$ of

$$\mathbb{R}^d_{>0} \setminus \bigcup_{S \neq T \in \mathcal{S}} H_{S,T}. \tag{11.11}$$

The chamber solid angle or normalized volume is a possible deterministic analogue of statistical weight. A useful theorem would compare the chamber volume of a Catalan-extreme sign word with that of ordinary sign words and then prove a lower bound for the chamber containing the actual prime-log vector.

12. Three possible selector mechanisms

12.1. **Hyperplane-chamber entropy.** The first route uses Definition 11.2. For a finite excursion involving a finite collection of subset energies, let Ω_r be the region of positive weight space that preserves the relevant ordering and hence the same parity word. Define a geometric selector energy

$$\mathcal{G}_r = -\log \nu(\Omega_r), \tag{12.1}$$

where ν is a normalized reference measure on an appropriate projective section of positive weight space.

Question 12.1. Does a high Catalan excursion necessarily correspond to a small union of ordering chambers, with a bound of the form

$$\nu\{w : \text{induced excursion has height} \geq H\} \leq \exp(-\Psi(H,L)), \tag{12.2}$$

where $\Psi(H,L)$ is at least comparable to H^2/L in the relevant regime?

If so, the remaining arithmetic problem becomes a Diophantine non-concentration statement for the prime-log vector relative to those chambers. Proposition 11.1 gives a first, very weak, deterministic separation from the collision walls.

12.2. **Shadow inequalities and parity restoration.** Björner obtains number-theoretic inequalities from combinatorial shadow functions associated with the shifted complex [12]. A family of many high-dimensional faces forces a sufficiently large lower-dimensional shadow. Since adjacent dimensions contribute opposite signs to Euler characteristic, this is a plausible source of parity compensation.

Warning 12.2. A shadow inequality is primarily static: it says that certain lower-dimensional faces exist when a higher-dimensional face exists. The Mertens problem is temporal along the filtration. Many shadow faces may have entered much earlier. The useful statement must therefore be a filtered shadow inequality controlling how much opposite parity is forced to enter within a bounded future energy interval.

The corresponding research target is a dynamic inequality of the schematic form

$$\text{current parity imbalance } H \quad \Longrightarrow \quad \text{future opposite-parity influx} \geq \Phi(H,L), \tag{12.3}$$

with a scale strong enough to feed into the Catalan energy H^2/L.

12.3. **Persistent topology.** Equation (11.2) can be refined by the Euler–Poincaré formula. Writing $\beta_k(\Delta_x)$ for reduced Betti numbers,

$$M(x) = \sum_{k\geq 0} (-1)^{k-1} \beta_k(\Delta_x). \tag{12.4}$$

Björner studied the asymptotic behaviour of these Betti numbers and proved that Δ_x has the homotopy type of a wedge of spheres [12]. A large Mertens excursion is therefore a sustained imbalance between even- and odd-dimensional homological contributions.

The filtration suggests studying not only $\beta_k(\Delta_x)$ at a fixed x but the birth and death structure across x. Persistent-homology constraints could in principle control the duration for which one parity dominates. This route preserves information discarded by the scalar Euler characteristic.

Question 12.3. Can one prove an interlacing, pairing, or energy inequality between persistent homology classes of adjacent parity whose projection onto Euler characteristic yields a bound on Catalan excursion height?

13. Structural viability of the physical routes

The physical structures above do not by themselves prove a new estimate for M. They are nevertheless mathematically relevant for four specific reasons.

First, the Catalan state space is exact, not fitted. Proposition 2.2 follows from the deterministic zero-return geometry of the compressed Mertens walk.

Second, the physical rate H^2/L is already visible in exact combinatorics. Proposition 5.4 supplies a finite discrete Gaussian tail, while Brownian action and large-deviation theory explain the same scale in the continuum.

Third, the arithmetic selector has additional exact structure. Equations (11.2) and (11.4) show that the path is generated by a shifted simplicial filtration and an additive prime-log ordering. Therefore the selector problem is not an arbitrary deterministic choice from Catalan space.

Fourth, Theorem 6.3 converts a concrete selector-energy inequality into square-root control. This isolates a finite combinatorial target that can be attacked by geometry, topology, spectral theory, or statistical mechanics.

14. Potential obstructions

14.1. **The selector may remain unconstrained at the needed scale.** The strongest obstruction is that the known simplicial and subset-sum structure may fail to prevent the prime-log ordering from entering Catalan sectors with very large selection energy. Establishing such a prevention is the substantive new theorem required by the program.

14.2. **Static topology may not control filtered dynamics.** Euler characteristic and Betti-number identities can exhibit large cancellation only after many filtration steps. RH-scale control requires cancellation at every sufficiently large scale. Any topological approach must therefore retain filtration time and cannot rely solely on the terminal homotopy type.

14.3. **Parity barrier.** The selector records the parity $(-1)^{|S|}$ of prime subsets. Many coarse counting and sieve mechanisms are weak precisely when asked to distinguish even from odd numbers of prime factors. In the present language, a successful selector theorem must see parity at the level of the ordered subset-sum geometry rather than only the density of faces.

14.4. **Canonical collapse to a classical Dirichlet object.** A canonical partition function built directly from the signed energies (11.4) gives

$$\sum_S (-1)^{|S|} e^{-sE(S)} = \prod_p (1 - p^{-s}) = \frac{1}{\zeta(s)} \tag{14.1}$$

for $\Re s > 1$. Equation (14.1) is exact, but it collapses the excursion geometry into the familiar Euler product. The new information sought here lies in pathwise objects—return times, Catalan shapes, heights, areas, chamber geometry, and filtered topology—before such a transform removes the temporal structure.

Remark 14.1. Equation (14.1) is used as a diagnostic boundary: a proposed physical construction that immediately reduces to this canonical partition function has returned to the classical analytic representation and has not exploited the Catalan excursion structure.

15. A hierarchy of possible sufficient principles

The following principles are ordered from closest to the desired estimate to more structural formulations.

Conjecture 15.1 (Open Catalan selection principle)**.** *For the compressed Mertens walk,*

$$\mathcal{I}_{\mathrm{Cat}}(m) = O(\log m). \tag{15.1}$$

By Theorem 6.3, Conjecture 15.1 implies (6.5).

Conjecture 15.2 (Chamber non-concentration principle)**.** *For a natural finite-dimensional normalization of the prime-log selector, the chamber containing the arithmetic weight vector has geometric energy at most logarithmic in the cutoff, and Catalan-extreme endpoint sectors have chamber energy bounded below by a positive multiple of H^2/L.*

Conjecture 15.3 (Filtered parity-restoration principle)**.** *There is a deterministic filtered shadow or persistence functional $\mathcal{R}$ such that a parity imbalance H sustained for L compressed steps forces*

$$\mathcal{R} \geq c\frac{H^2}{L} - O(\log L), \tag{15.2}$$

and the arithmetic filtration satisfies $\mathcal{R} = O(\log m)$ up to compressed time m.

These conjectures are not independent claims of truth. They are candidate formulations of the same missing selector mechanism at different levels: direct Catalan volume, prime-log chamber geometry, and filtered topology.

16. The prime-delay selector and its exact interference algebra

This section makes the selector more explicit than the chamber description. The identities are exact; the passage from them to local Catalan moment bounds is not known and will be isolated separately.

16.1. **Simultaneous prime-shell identities.** For a prime p, define

$$A_p(x) = \sum_{\substack{n \leq x \\ p \nmid n}} \mu(n). \tag{16.1}$$

Terms divisible by p can be written uniquely as pm with $p \nmid m$; for these terms $\mu(pm) = -\mu(m)$. Therefore

$$M(x) = A_p(x) - A_p(x/p) = \sum_{\substack{x/p < n \leq x \\ p \nmid n}} \mu(n) \qquad (p \text{ prime}). \tag{16.2}$$

Thus a value $M(x)$ must be reproduced simultaneously as a filtered parity imbalance in every prime shell $(x/p, x]$. For $p = 2$ this is the shell underlying Björner's odd-squarefree Betti-number formula [12]. Identity (16.2) is elementary, but it is not a generic property of a Catalan path.

16.2. **Threshold kernel and transfer recursion.** Fix $y \geq 2$, let $P_y = \{p : p \leq y\}$, and put

$$K_y(t) = \sum_{\substack{S \subseteq P_y \\ \sum_{p \in S} \log p \leq t}} (-1)^{|S|}. \tag{16.3}$$

Unique factorization gives, for $0 \leq t \leq \log y$,

$$K_y(t) = M(e^t). \tag{16.4}$$

Here and below $M(u)$ for real u means $\sum_{n \leq u} \mu(n)$. The restriction $t \leq \log y$ is essential: outside it the finite prime set P_y no longer contains every prime that may occur below the threshold.

Enumerate the primes as $p_1 < p_2 < \cdots$ and define K_j by using only $p_1, \ldots, p_j$. Splitting subsets according to whether they contain p_j gives the exact recursion

$$K_j(t) = K_{j-1}(t) - K_{j-1}(t - \log p_j). \tag{16.5}$$

With $(T_a f)(t) = f(t - a)$ and $\mathcal{T}_a = I - T_a$,

$$K_j = \mathcal{T}_{\log p_j} K_{j-1}. \tag{16.6}$$

This is a deterministic delayed anti-copy operation: every new prime inserts a translated copy with the opposite sign.

Let dK_j denote the atomic jump measure. Distributionally,

$$dK_j = \mathop{*}_{\ell \leq j} (\delta_0 - \delta_{\log p_\ell}), \qquad \widehat{dK_j}(\tau) = \prod_{\ell \leq j} (1 - e^{-i\tau \log p_\ell}), \tag{16.7}$$

and hence

$$|\widehat{dK_j}(\tau)| = 2^j \prod_{\ell \leq j} \left| \sin \frac{\tau \log p_\ell}{2} \right|. \tag{16.8}$$

Equations (16.5)–(16.8) are exact finite identities [DED].

For an integer $q \geq 1$, define the normalized frequency moment

$$\mathcal{D}_{j,q}(T) = \frac{1}{2T} \int_{-T}^{T} \left(2^{-j} |\widehat{dK_j}(\tau)| \right)^{2q} d\tau \tag{16.9}$$

and let $P_j = \prod_{\ell \leq j} p_\ell$.

Theorem 16.1 (Exact Cesàro dephasing law)**.** *For every fixed $j, q \geq 1$,*

$$\lim_{T \to \infty} \mathcal{D}_{j,q}(T) = \left(\frac{\binom{2q}{q}}{4^q} \right)^j. \tag{16.10}$$

Moreover,

$$\left| \mathcal{D}_{j,q}(T) - \left(\frac{\binom{2q}{q}}{4^q} \right)^j \right| \leq \min\left\{ 1, \frac{1}{T \log(1 + P_j^{-q})} \right\}. \tag{16.11}$$

Proof. The elementary Fourier expansion

$$\sin^{2q}(\theta/2) = 4^{-q} \sum_{m=-q}^{q} (-1)^m \binom{2q}{q+m} e^{im\theta} \tag{16.12}$$

has constant coefficient $\binom{2q}{q}/4^q$ and coefficient ℓ^1 norm equal to one. Multiplying these expansions for $\theta = \tau \log p_\ell$ produces frequencies

$$\lambda_m = \sum_{\ell \le j} m_\ell \log p_\ell = \log(A/B), \qquad |m_\ell| \le q. \tag{16.13}$$

Unique factorization makes every nonzero vector m a nonzero frequency. Its symmetric Cesàro average is $\sin(T\lambda_m)/(T\lambda_m)$, proving (16.10). After cancellation, $A, B \le P_j^q$; if $A > B$ then $\log(A/B) \ge \log(1 + P_j^{-q})$. Summing with the unit ℓ^1 norm gives (16.11). □

Warning 16.2 (What the τ theorem does and does not control). The same rational independence and Kronecker's theorem give

$$\sup_{\tau \in \mathbb{R}} 2^{-j} |\widehat{dK_j}(\tau)| = 1. \tag{16.14}$$

Thus Theorem 16.1 is an averaged statement, not pointwise decay. Its elementary finite-window error becomes useful only for very large T because P_j^q is enormous. More importantly, the product (16.7) contains all subsets of $p_1, \dots, p_j$, including products beyond the arithmetic cutoff y. Restriction to the causal region $t \le \log y$ destroys that product factorization. These two facts prevent the infinite-window average from being used directly as a proof of (16.20).

Figure 2 illustrates the theorem for the first few primes. The dashed levels are the exact $q = 1$ limits 2^{-j}; the curves are finite computations and have no asymptotic evidentiary role.

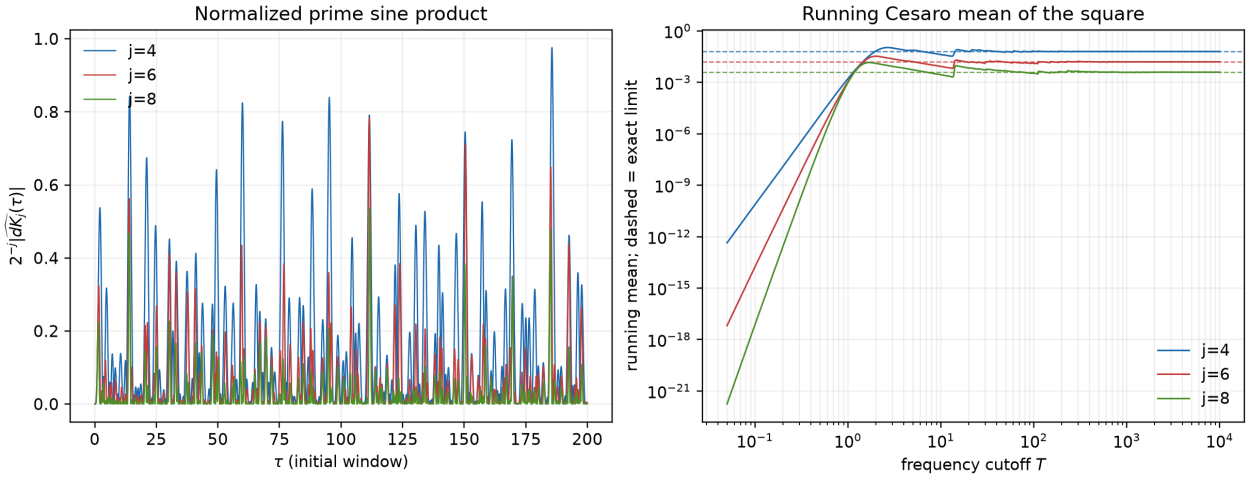


FIGURE 2. Normalized prime sine products and running Cesàro square means.

16.3. **A maximal-parity Walsh coefficient.** Put $j = \pi(y)$ and define the Boolean halfspace

$$f_{y,t}(\varepsilon) = \mathbf{1}\left\{ \sum_{\ell=1}^{j} \varepsilon_\ell \log p_\ell \le t \right\}, \qquad \varepsilon \in \{0,1\}^j. \tag{16.15}$$

Under the normalized Walsh convention,

$$\widehat{f}_{y,t}([j]) = 2^{-j} \sum_{\varepsilon \in \{0,1\}^j} (-1)^{\varepsilon_1 + \cdots + \varepsilon_j} f_{y,t}(\varepsilon), \tag{16.16}$$

we have

$$K_y(t) = 2^j \widehat{f}_{y,t}([j]), \qquad M(y) = 2^{\pi(y)} \widehat{f}_{y,\log y}([\pi(y)]). \tag{16.17}$$

Fourier analysis of Boolean halfspaces is well developed [26], but the coefficient in (16.17) has maximal degree and the threshold lies in a highly sparse tail of the cube. Standard low-degree and balanced-halfspace estimates do not directly give the required bound.

16.4. **Uniform prefix energy: a corrected sufficient principle.** Let $0 = z_0 < z_1 < \cdots$ be the return times of the compressed walk. On a complete excursion set

$$D_r(u) = |X_{z_r+u}|, \qquad 0 \le u \le L_r := z_{r+1} - z_r, \tag{16.18}$$

and for every prefix $1 \le v \le L_r$ define

$$\mathcal{A}_q(r; v) = \sum_{u=1}^{v} D_r(u)^{2q}. \tag{16.19}$$

The same definition is used for every prefix of the currently open excursion, even when its next return time is not yet known.

Conjecture 16.3 (Uniform prime-Catalan prefix bound)**.** *For every fixed integer $q \ge 1$ and every $\varepsilon > 0$,*

$$\mathcal{A}_q(r; v) \ll_{q,\varepsilon} v^{q+1}(z_r + v)^{\varepsilon} \tag{16.20}$$

uniformly over all completed excursions, all their prefixes, and all prefixes of the final open meander.

Theorem 16.4 (Uniform prefix moments imply RH)**.** *Conjecture 16.3 implies the Riemann hypothesis.*

Proof. Fix a compressed time N, let $z_* \le N$ be its last return time, and put $v = N - z_*$. If $v = 0$ there is nothing to prove. Otherwise the prefix containing N is included in Conjecture 16.3, so

$$|X_N|^{2q} \le \mathcal{A}_q(r; v) \ll_{q,\varepsilon} v^{q+1} N^{\varepsilon} \le N^{q+1+\varepsilon}. \tag{16.21}$$

Therefore

$$|X_N| \ll_{q,\varepsilon} N^{1/2+1/(2q)+\varepsilon/(2q)}. \tag{16.22}$$

Given $\delta > 0$, choose a fixed q sufficiently large and then choose $\varepsilon > 0$ sufficiently small. This gives $X_N = O_\delta(N^{1/2+\delta})$. Since the squarefree counting function $Q(x) \sim 6x/\pi^2$ and $M(x) = X_{Q(x)}$, the classical Mertens criterion for RH follows. □

Remark 16.5 (Logical strength). Theorem 16.4 is a rigorous implication, not an equivalence. The uniform prefix conjecture can be strictly stronger than RH. Its purpose is to expose a structured sufficient target that controls complete excursions, internal prefixes, and the open meander without predicting their individual signs.

16.5. **The exceptional Catalan class is exactly the RH obstruction.** Let $\mathfrak{C}$ denote the space of infinite nearest-neighbour paths, decomposed at their returns to zero into signed primitive Catalan excursions and, at every finite cutoff, a possibly open terminal meander. Define the RH-exceptional Catalan class by

$$\mathfrak{E}_{\mathrm{RH}} = \left\{ Y \in \mathfrak{C} : \exists \varepsilon_0 > 0 \text{ such that } \sup_{m \ge 1} \frac{|Y_m|}{m^{1/2+\varepsilon_0}} = \infty \right\}. \tag{16.23}$$

Equivalently, membership is tested over every internal excursion prefix and every open terminal meander. This last clause cannot be omitted.

Theorem 16.6 (Exceptional-class characterization)**.** *Let $X_m = \sum_{j \le m} \mu(q_j)$ be the squarefree-compressed Mertens walk. Then*

$$\boxed{X \notin \mathfrak{E}_{\mathrm{RH}} \quad \Longleftrightarrow \quad \mathrm{RH}.} \tag{16.24}$$

Thus, for the arithmetic path, not belonging to the exceptional Catalan class is exactly the Riemann hypothesis.

Proof. By squarefree counting,

$$Q(x) = \frac{6}{\pi^2}x + O(\sqrt{x}), \qquad q_m \asymp m, \tag{16.25}$$

and $M(x) = X_{Q(x)}$. Therefore

$$M(x) = O_\varepsilon(x^{1/2+\varepsilon}) \quad (\varepsilon > 0) \tag{16.26}$$

if and only if

$$X_m = O_\varepsilon(m^{1/2+\varepsilon}) \quad (\varepsilon > 0). \tag{16.27}$$

The latter statement is exactly $X \notin \mathfrak{E}_{\mathrm{RH}}$ by Equation (16.23); the former is the classical Mertens criterion for RH. □

Proposition 16.7 (Exact global class versus stronger local class)**.** *In excursion coordinates, the condition*

$$D_r(v) \ll_\varepsilon (z_r + v)^{1/2+\varepsilon} \qquad \textit{for every } r, v \textit{ and every } \varepsilon > 0 \tag{16.28}$$

including the open meander, is exactly equivalent to $X \notin \mathfrak{E}_{\mathrm{RH}}$ and hence to RH. It is neither stronger nor weaker. By contrast, a bound measured at the internal scale,

$$D_r(v) \ll_\varepsilon v^{1/2+\varepsilon}, \tag{16.29}$$

or the uniform local moment bound (16.20)*, is a strictly stronger property on the space of nearest-neighbour paths and is not known to follow from RH for the Mertens path.*

Proof. At the prefix (r, v) the current compressed time is $m = z_r + v$ and $D_r(v) = |X_m|$, which proves the first assertion point by point. To separate the two path properties, place at arbitrarily large times N a primitive mountain excursion of length $L \asymp \sqrt{N}$ and height $L/2$, separating successive mountains by unit excursions that return immediately to zero. The resulting nearest-neighbour path obeys a global bound $|Y_m| = O(\sqrt{m})$, but at the mountain peak it has height comparable to L, violating (16.29) and, for fixed q, the Brownian local moment scale. Thus local Brownian control contains information not present in the global RH-scale condition. □

Warning 16.8 (Criterion is not mechanism). Theorem 16.6 is an exact characterization and, by itself, proves no new estimate. Nevertheless, its equivalence to RH does not make every attempted proof circular. Circularity occurs only if exclusion from $\mathfrak{E}_{\mathrm{RH}}$ is assumed, renamed, or recovered by immediate substitution. A proof from an independent arithmetic theorem about the prime selector would be a non-circular proof of an equivalent criterion, just as for any other valid RH criterion.

There are therefore two distinct assertions:

$$\underbrace{X \notin \mathfrak{E}_{\mathrm{RH}} \iff \mathrm{RH}}_{\text{exact criterion [DED]}} \qquad \text{and} \qquad \underbrace{\text{prime selector} \Longrightarrow X \notin \mathfrak{E}_{\mathrm{RH}}}_{\text{open mechanism [ADAPT]}}. \tag{16.30}$$

The first statement does not close the Catalan programme. It identifies its target. The programme remains open precisely because the second statement is neither proved nor disproved.

Controlling only completed excursions by their future return time is not enough for the reverse implication: a return may lie far beyond the point being estimated, and an indefinitely open meander would escape the condition. The correct local formulation is the uniform prefix hierarchy (16.20), which acts at the present scale $z_r + v$.

17. All-short-interval cancellation as a mesoscopic Catalan selector

The first unconditional arithmetic theorem in this notebook that acts uniformly at every position is due to Matomäki and Teräväinen [4]. It must be translated with care: it controls the chord of the Mertens path across a long interval, not the largest height attained between the two endpoints.

Theorem 17.1 (Matomäki–Teräväinen [LIT]). *Fix $\theta > 0.55$ and $\eta > 0$. Uniformly for all sufficiently large x and all $H \geq x^{\theta}$,*

$$\sum_{x<n\leq x+H} \mu(n) \ll_{\theta,\eta} \frac{H}{(\log x)^{1/3-\eta}}. \tag{17.1}$$

In particular the left-hand side is $o(H)$ for every initial point x.

The next statement is the exact Catalan content of this theorem.

Proposition 17.2 (Uniform mesoscopic chord flatness [DED]). *Fix $0.55 < \theta < \vartheta < 1$ and $0 < c < 1/3$. Then, uniformly for m sufficiently large and $\ell \geq m^{\vartheta}$ with $m + \ell \leq 2m$,*

$$|X_{m+\ell} - X_m| \ll_{\theta,\vartheta,c} \frac{\ell}{(\log m)^c}. \tag{17.2}$$

Consequently every sufficiently long chord of every Mertens excursion or open meander in a dyadic band has slope tending uniformly to zero.

Proof. Put $x = q_m$ and $H = q_{m+\ell} - q_m$. Squarefree counting gives

$$\ell = Q(x + H) - Q(x) = \frac{6}{\pi^2} H + O(\sqrt{x}). \tag{17.3}$$

Since $q_m \asymp m$, $\ell \geq m^{\vartheta}$, and $\vartheta > \theta > 1/2$, we have $H \asymp \ell$ and $H \geq x^{\theta}$ for all sufficiently large m. Apply Theorem 17.1 with $\eta = 1/3 - c$ and use

$$X_{m+\ell} - X_m = M(q_{m+\ell}) - M(q_m). \tag{17.4}$$

□

Corollary 17.3 (Excursion envelope at the proved resolution [DED]). *Let a complete excursion occupy compressed times $[a, a + L]$ with $a + L \leq 2a$, and put $h = (2a)^{\vartheta}$. For $0 \leq v \leq L$,*

$$D(v) \ll h + \frac{\min\{v, L - v\}}{(\log a)^c}. \tag{17.5}$$

For an open meander the corresponding one-sided estimate is

$$D(v) \ll h + \frac{v}{(\log a)^c}. \tag{17.6}$$

Proof. For $v < h$ use the nearest-neighbour bound $D(v) \leq v$; for $v \geq h$ use Proposition 17.2. For a complete excursion apply the same argument backwards from the terminal zero and take the better of the two bounds. □

Thus Theorem 17.1 excludes a wide linear mountain, but it does not control a spike whose width is below a^{ϑ}, nor a long excursion that accumulates a small bias over many mesoscopic blocks. The following construction shows that this is a genuine logical obstruction, not merely a weakness of the proof above.

Proposition 17.4 (Mesoscopic Catalan no-go theorem [DED]). *For every $0 < \delta < 1$ and every scale h, there are arbitrarily long primitive Dyck excursions Y of length L such that*

$$|Y_{t+s} - Y_t| \leq 2\delta s + O(\delta^{-1}) \qquad (s \geq h), \tag{17.7}$$

while

$$\max_t Y_t \asymp \delta L. \tag{17.8}$$

Taking $\delta = (\log N)^{-c}$, $h = N^{\vartheta}$, and $L \asymp N$, and concatenating these excursions at rapidly increasing scales, produces an RH-exceptional Catalan path satisfying the same qualitative mesoscopic chord restriction as Equation (17.2)*. Therefore primitive-Catalan geometry plus Theorem 17.1 does not imply RH.*

Proof. Choose an integer $r \asymp \delta^{-1}$ and form

$$W = (U^{r+1}D^r)^K. \tag{17.9}$$

Every complete block has length $2r+1$ and net rise one. Concatenate W with the reverse path that retraces W to zero. The resulting path is a primitive Dyck excursion. Its height is $K + O(r)$ and its length is $2K(2r+1)$, proving Equation (17.8). Any interval contains complete blocks plus at most two partial blocks, so its increment is bounded by $s/(2r+1) + O(r)$. The same estimate holds across the turning point, which proves Equation (17.7). Taking K large compared with h completes the construction. □

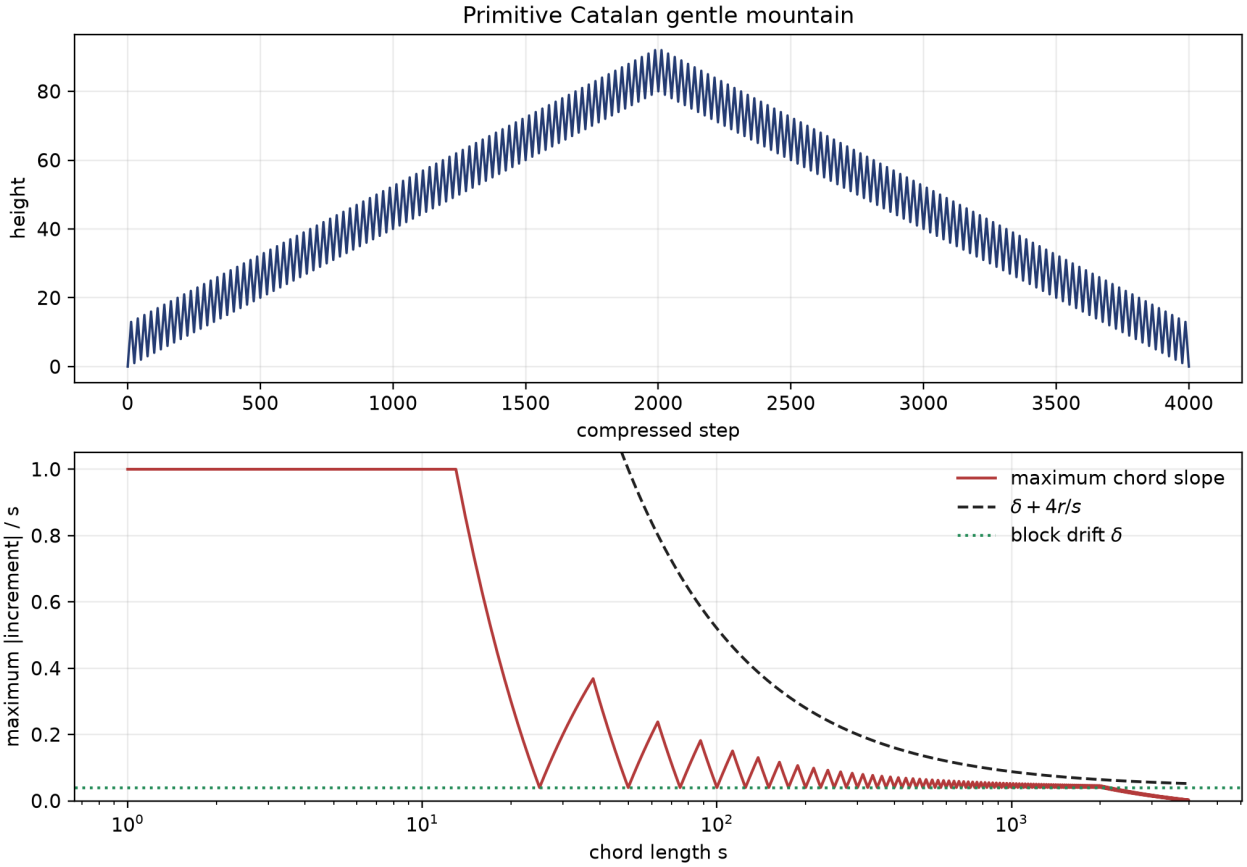


FIGURE 3. Finite audit of the gentle-mountain construction. Long chords have small mean slope although the primitive excursion accumulates a height proportional to δL. The figure is a logical counterexample for the ambient Catalan class, not Mertens data.

17.1. **The exact prime-coloured anti-lift.** The useful part of the proof of Theorem 17.1 is not merely the final chord estimate. Let $\mathcal{P} = (P, Q] \cap \mathbb{P}$ and

$$\nu_{\mathcal{P}}(n) = \#\{p \in \mathcal{P} : p \mid n\}. \tag{17.10}$$

For every squarefree n with $\nu_{\mathcal{P}}(n) > 0$,

$$\boxed{\mu(n) = -\sum_{\substack{p \in \mathcal{P} \\ p \mid n}} \frac{\mu(n/p)}{\nu_{\mathcal{P}}(n)}.} \tag{17.11}$$

This is the squarefree form of the Ramaré extraction: deleting any selected prime reverses the Möbius sign, and the denominator normalizes the number of available colours. Unlike the delayed inclusion–exclusion recursion (16.5), Equation (17.11) extracts one prime colour with a partition-of-unity normalization.

For an integer interval beginning at a, define the coloured prefix currents

$$C_p(u) = -\sum_{\substack{a < n \le a+u \\ p \mid n \\ \mu(n)^2 = 1 \\ \nu_{\mathcal{P}}(n) > 0}} \frac{\mu(n/p)}{\nu_{\mathcal{P}}(n)}, \tag{17.12}$$

$$R_{\mathcal{P}}(u) = \sum_{\substack{a < n \le a+u \\ \nu_{\mathcal{P}}(n) = 0}} \mu(n). \tag{17.13}$$

Then the following vector identity is exact for every prefix simultaneously:

$$\boxed{M(a+u) - M(a) = R_{\mathcal{P}}(u) + \sum_{p \in \mathcal{P}} C_p(u).} \tag{17.14}$$

The current $C_p(u)$ is a weighted Möbius sum over the dilated interval $(a/p, (a+u)/p]$. Hence Equation (17.14) is a multiscale anti-lift of the entire Catalan prefix, not just its endpoint.

17.2. **The Catalan–Ramaré stopping problem.** For a path prefix $F(1), \dots, F(v)$ define its occupation norm by

$$\|F\|_{\mathrm{Cat},2q;v}^{2q} = \sum_{u=1}^{v} |F(u)|^{2q}. \tag{17.15}$$

The next candidate mechanism retains the prime colours until after this norm is formed.

Conjecture 17.5 (Prime-colour square function at Catalan stopping times [ADAPT])**.** *For each fixed q, suitable prime bands can be chosen so that, uniformly on every actual Mertens excursion prefix and open meander,*

$$\sum_{u \le v} \left| \sum_{p \in \mathcal{P}} C_p(u) \right|^{2q} \le N^{o(1)} \sum_{u \le v} \left(\sum_{p \in \mathcal{P}} |C_p(u)|^2 \right)^q, \tag{17.16}$$

with a companion estimate for $R_{\mathcal{P}}$ that does not use its cardinality by the triangle inequality.

17.3. **The Catalan occupation-energy hierarchy.** Write

$$S_x(h) = M(x+h) - M(x), \qquad \mathcal{E}_q(x, H) = \sum_{h=1}^{H} |S_x(h)|^{2q}. \tag{17.17}$$

This is the occupation energy of the ordinary-time Mertens chord. In compressed squarefree time the corresponding quantity is

$$\mathcal{E}_q^{\#}(m,L)=\sum_{u=1}^{L}|X_{m+u}-X_m|^{2q}. \tag{17.18}$$

The two energies must not be identified silently. Ordinary time repeats a height whenever $\mu(n)=0$, whereas compressed time records only genuine ±1 jumps. They are comparable only after the squarefree-time change and its multiplicities have been specified.

For a diffusive path of duration H, height has scale $H^{1/2}$, so its $2q$-occupation energy has scale

$$H\,(H^{1/2})^{2q}=H^{q+1}. \tag{17.19}$$

This is the Catalan/Brownian prediction, not an arithmetic estimate. In the continuum excursion ensemble, the Laplace functional

$$\mathbb{E}_{\mathrm{exc}}\exp\!\left(-\lambda\int_0^1 e(t)^{2q}\,dt\right) \tag{17.20}$$

is represented by a half-line Feynman–Kac operator with potential $V(y)=\lambda y^{2q}$: harmonic for $q=1$ and anharmonic for $q>1$ [11]. This fixes the natural exponent in (17.19); it does not select the arithmetic path.

Definition 17.6 (Dyadic Catalan energy)**.** For fixed $q\ge1$, let $\mathrm{CE}_q^{\mathrm{dyad}}$ denote

$$\boxed{\mathcal{E}_q(x,x)\ll_{q,\varepsilon}x^{q+1+\varepsilon}\quad(x\ge2).} \tag{17.21}$$

Theorem 17.7 (Exact exponent ladder and RH-completeness [DED])**.** *For each fixed $q\ge1$,*

$$\mathrm{CE}_q^{\mathrm{dyad}}\quad\Longrightarrow\quad M(X)\ll_{q,\varepsilon}X^{\frac{q+1}{2q+1}+\varepsilon}. \tag{17.22}$$

Moreover,

$$\boxed{[\mathrm{CE}_q^{\mathrm{dyad}}\ \textit{for every fixed } q]\quad\Longleftrightarrow\quad\mathrm{RH}.} \tag{17.23}$$

Proof. Put $A=|S_x(x)|$. Since $|S_x(h+1)-S_x(h)|\le1$, at least $\lfloor A/2\rfloor$ of the final prefix values have modulus at least $A/2$. Consequently

$$A^{2q+1}\ll_q\mathcal{E}_q(x,x)\ll_{q,\varepsilon}x^{q+1+\varepsilon}. \tag{17.24}$$

This gives $|M(2x)-M(x)|\ll x^{(q+1)/(2q+1)+\varepsilon}$. A dyadic telescoping sum proves (17.22). If (17.21) holds for every fixed q, then for any $\delta>0$ one chooses q so that $(q+1)/(2q+1)<1/2+\delta/2$; the Mertens criterion follows.

Conversely, RH is equivalent to $M(y)\ll_\eta y^{1/2+\eta}$ for every $\eta>0$. Uniformly for $x\le y\le2x$ this yields $|S_x(y-x)|\ll_\eta x^{1/2+\eta}$, and therefore

$$\mathcal{E}_q(x,x)\ll_{q,\eta}x^{q+1+2q\eta}. \tag{17.25}$$

Taking $\eta=\varepsilon/(2q)$ proves (17.21) for that fixed q. □

Definition 17.8 (Dyadic exceptional occupation class)**.** For a nearest-neighbour path $Y\in\mathfrak{C}$, put

$$\mathfrak{E}_{\mathrm{CE}}^{\#}=\left\{Y:\ \exists q\ge1,\ \exists\varepsilon_0>0,\quad\sup_{m\ge2}\frac{\sum_{u=1}^{m}|Y_{m+u}-Y_m|^{2q}}{m^{q+1+\varepsilon_0}}=\infty\right\}. \tag{17.26}$$

Corollary 17.9 (Perfect calibration of non-exceptionality [DED])**.** *For the compressed Mertens path,*

$$\boxed{X\notin\mathfrak{E}_{\mathrm{CE}}^{\#}\quad\Longleftrightarrow\quad X\notin\mathfrak{E}_{\mathrm{RH}}\quad\Longleftrightarrow\quad\mathrm{RH}.} \tag{17.27}$$

Thus this dyadic occupation definition of non-exceptional Catalan is exactly RH, not a more restrictive condition.

Proof. The argument of Theorem 17.7 applies verbatim to the compressed path because its increments are ± 1. Non-membership in $\mathfrak{E}^{\#}_{\mathrm{CE}}$ is exactly the fixed-q dyadic hierarchy for every q and every positive ε. It is therefore equivalent to $X_m \ll_\varepsilon m^{1/2+\varepsilon}$, which is non-membership in $\mathfrak{E}_{\mathrm{RH}}$. Finally $q_m \asymp m$ and $M(x) = X_{Q(x)}$ give the last equivalence. □

q	1	2	3	4	5
$(q+1)/(2q+1)$	2/3	3/5	4/7	5/9	6/11

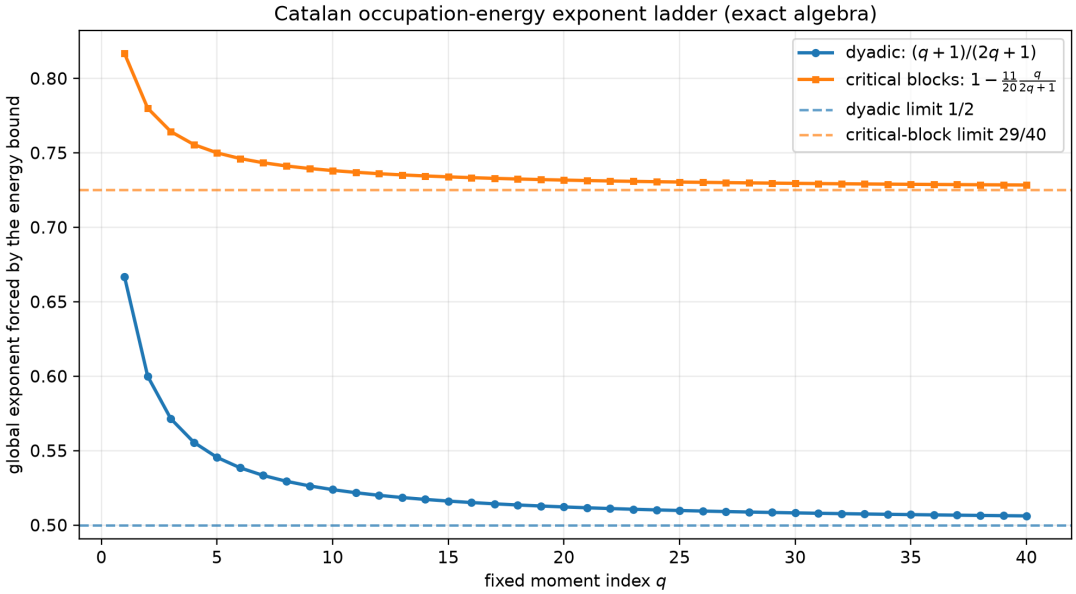


FIGURE 4. The exact dyadic exponent ladder $(q+1)/(2q+1) \downarrow 1/2$ and, for comparison only, the exponent obtained by summing critical blocks of length $x^{11/20}$. The curves visualize algebraic consequences, not numerical evidence for the conjectured bounds.

Warning 17.10 (The local hierarchy is stronger than RH). The superficially similar all-short-scale assertion

$$\mathcal{E}_q(x,H) \ll_{q,\varepsilon} H^{q+1}x^{\varepsilon} \qquad (1 \le H \le x) \tag{17.28}$$

is not the criterion in Theorem 17.7. It is a stronger local regularity statement, and no implication from RH to (17.28) is known. Thus the phrase non-exceptional Catalan is exactly RH only when exceptional means failure of the full fixed-q dyadic hierarchy (17.21); a uniform local Brownian law is more restrictive.

17.4. **The exact occupation spectrum.** The quadratic member of the hierarchy admits an exact diagonalization. Put $a_n = \mu(x+n)$ for $1 \le n \le H$ and let J_H be the summation matrix

$$(J_H a)_h = \sum_{n \le h} a_n. \tag{17.29}$$

Then

$$\mathcal{E}_1(x,H) = \|J_H a\|_2^2 = \langle a, K_H a\rangle, \qquad K_H = J_H^* J_H, \tag{17.30}$$

where

$$(K_H)_{rs} = H + 1 - \max(r, s). \tag{17.31}$$

Proposition 17.11 (Exact spectrum of the occupation kernel [DED])**.** *For $1 \le k \le H$, set*

$$\vartheta_k = \frac{(2k-1)\pi}{2H+1}, \qquad v_k(n) = \frac{2}{\sqrt{2H+1}} \sin\big((H+1-n)\vartheta_k\big). \tag{17.32}$$

The vectors v_k form an orthonormal eigenbasis of K_H, with

$$\boxed{\lambda_k = \frac{1}{4\sin^2\Big(\frac{(2k-1)\pi}{4H+2}\Big)}.} \tag{17.33}$$

Consequently

$$\boxed{\mathcal{E}_1(x, H) = \sum_{k=1}^{H} \lambda_k |\langle a, v_k\rangle|^2.} \tag{17.34}$$

For $k \ll H$ one has $\lambda_k \asymp H^2/(2k-1)^2$.

Proof. The inverse of K_H is the second-difference matrix associated with the first-difference operator, with one Dirichlet and one discrete Neumann boundary condition. Substitution of the displayed sine vectors gives $K_H v_k = \lambda_k v_k$; the finite sine orthogonality identity gives the stated normalization. Equation (17.34) is Parseval in this basis. □

After reversing the coordinate order, the kernel (17.31) becomes $\min(r, s)$, exactly the covariance matrix of a discrete unit-variance random walk. Under diffusive rescaling it converges to the Brownian covariance kernel $\min(t, s)$, whose Karhunen–Loève eigenfunctions are the corresponding half-integer sine modes [8]. Thus the Brownian operator is literal at the level of the quadratic kernel, although the arithmetic increment vector remains deterministic.

Let $e(t) = e^{2\pi i t}$ and

$$A_{x,H}(\alpha) = \sum_{n=1}^{H} \mu(x+n) e(\alpha n). \tag{17.35}$$

Each sine coefficient in (17.34) is a normalized difference of two values of (17.35). Therefore

$$|\langle a, v_k\rangle| \le \frac{2}{\sqrt{2H+1}} \sup_{\alpha\in\mathbb{R}} |A_{x,H}(\alpha)|. \tag{17.36}$$

Since $\sum_k \lambda_k = \operatorname{tr} K_H = H(H+1)/2$, we obtain the exact deterministic comparison

$$\boxed{\mathcal{E}_1(x, H) \le \frac{2H(H+1)}{2H+1} \sup_{\alpha\in\mathbb{R}} |A_{x,H}(\alpha)|^2 < (H+1) \sup_{\alpha} |A_{x,H}(\alpha)|^2.} \tag{17.37}$$

The ordinary-time quantity remains intrinsically Catalan. Indeed, with $m_0 = Q(x)$ and

$$w_j(x, H) = \#\{1 \le h \le H : Q(x+h) = j\}, \tag{17.38}$$

one has the exact holding-time identity

$$\boxed{\mathcal{E}_1(x, H) = \sum_{j} w_j(x, H) |X_j - X_{m_0}|^2.} \tag{17.39}$$

The nonzero steps give the same Dyck/meander skeleton as before; the zeros of μ supply the arithmetically determined holding weights, including truncation at both ends of the window.

Theorem 17.12 (Unconditional anti-ballistic occupation bounds [LIT+DED])**.** *Let $\eta > 0$.*

(a) If $\theta > 3/5$ and $H \geq x^{\theta}$, then

$$\mathcal{E}_1(x,H) \ll_{\theta,\eta} \frac{H^3}{(\log x)^{2/3-2\eta}}. \tag{17.40}$$

(b) For every $A > 0$, if $x^{5/8+\eta} \leq H \leq x^{1-\eta}$, then

$$\mathcal{E}_1(x,H) \ll_{A,\eta} \frac{H^3}{(\log x)^{A}}. \tag{17.41}$$

Proof. Matomäki and Teräväinen prove, uniformly in α, that

$$A_{x,H}(\alpha) \ll_{\theta,\eta} H(\log x)^{-1/3+\eta} \tag{17.42}$$

in the range of part (a) [4, Theorem 1.5]. Equation (17.37) gives (17.40). Matomäki, Shao, Tao, and Teräväinen prove uniform orthogonality of μ to every fixed-complexity nilsequence, with arbitrary logarithmic saving, in the range of part (b) [5]. Linear phases are one-step nilsequences, so the same deterministic comparison gives (17.41), after renaming the logarithmic exponent. □

Part (a) proves $\mathcal{E}_1 = o(H^3)$ in its range. It is important to separate the published twisted-sum estimate [LIT] from the occupation-energy corollary [DED]; no novelty claim is made for the latter without a dedicated bibliographic search. The untwisted theorem at the better threshold $\theta > 0.55$ controls only the terminal chord. Consequently the gentle-mountain construction above remains a no-go for scalar chord information alone, but not for Theorem 17.12.

17.5. **Catalan tent conversion: peaks and area.** The occupation estimate has a direct pathwise consequence which the terminal chord estimate does not have. Put

$$\Delta(x,H) = \max_{0\leq h\leq H} |M(x+h) - M(x)|. \tag{17.43}$$

Lemma 17.13 (One-sided tent inequality [DED])**.** *For every integer-valued path $S(0) = 0$ with $|S(h) - S(h-1)| \leq 1$,*

$$\boxed{\frac{\Delta^3}{3} \leq \sum_{h=1}^{H} |S(h)|^2, \qquad \Delta = \max_{0\leq h\leq H} |S(h)|.} \tag{17.44}$$

Consequently $\Delta(x,H) \leq (3\mathcal{E}_1(x,H))^{1/3}$.

Proof. Choose h_0 with $|S(h_0)| = \Delta$. Since $S(0) = 0$ and the path is one-Lipschitz, $h_0 \geq \Delta$ and $|S(h_0 - t)| \geq \Delta - t$ for $0 \leq t \leq \Delta$. Hence

$$\sum_{h=1}^{H} |S(h)|^2 \geq \sum_{t=0}^{\Delta-1} (\Delta - t)^2 = \frac{\Delta(\Delta+1)(2\Delta+1)}{6} \geq \frac{\Delta^3}{3}. \tag{17.45}$$

No right-hand margin at the maximizing time is required. □

Corollary 17.14 (Unconditional interior-peak bounds [LIT+DED])**.** *For $H \geq x^{\theta}$ and $\theta > 3/5$,*

$$\boxed{\Delta(x,H) \ll_{\theta,\eta} \frac{H}{(\log x)^{2/9-2\eta/3}}.} \tag{17.46}$$

For every $A > 0$, uniformly when $x^{5/8+\eta} \leq H \leq x^{1-\eta}$,

$$\boxed{\Delta(x,H) \ll_{A,\eta} \frac{H}{(\log x)^{A}}.} \tag{17.47}$$

The exponent A in the second display has been renamed after taking a cube root.

The same energy also controls the q-Catalan area already defined in Section 9. For the window path,

$$\sum_{h\le H} |M(x+h)-M(x)| \le H^{1/2}\mathcal{E}_1(x,H)^{1/2}. \tag{17.48}$$

Thus Theorem 17.12 gives a logarithmic saving over the ballistic area H^2, and by (17.39) this is exactly a holding-time-weighted area statement on the compressed Catalan skeleton. This is a rigorous bridge from the proved $J_H^* J_H$ estimate back to Catalan peak and area geometry; it supplies logarithmic anti-ballistic information, not a power saving and not an RH implication. No further "Stokes" consequence is asserted here without an explicit identity and boundary audit.

17.6. **The correctly calibrated frontier.** For every increment sequence with $|a_n| \le 1$,

$$\mathcal{E}_1(x,H) \le \sum_{h\le H} h^2 \ll H^3, \tag{17.49}$$

and a monotone mountain saturates the exponent. The established and open levels are therefore

$$\boxed{\begin{array}{llll} \mathcal{E}_1 & \ll & H^3 & \text{trivial,} \\ \mathcal{E}_1 & \ll & H^3(\log x)^{-2/3+o(1)} & \theta > 3/5 \quad \text{[LIT+DED]}, \\ \mathcal{E}_1 & \ll_A & H^3(\log x)^{-A} & \theta > 5/8 \quad \text{[LIT+DED]}, \\ \mathcal{E}_1 & \overset{?}{\ll} & H^{3-\delta} & \text{first power-saving frontier [open]}, \\ \mathcal{E}_1 & \overset{?}{\ll} & H^{2+\varepsilon} & \text{Brownian/CRSF target [open]}. \end{array}} \tag{17.50}$$

Thus CE_1 remains a valuable calibration, but it is not adopted as the next lemma. The next major target is any fixed power saving over the ballistic occupation scale.

Proposition 17.15 (Propagation of a mesoscopic power saving [DED])**.** *Suppose, uniformly in the initial point on dyadic ranges, that for $H \asymp X^\theta$,*

$$\mathcal{E}_1(x,H) \ll H^{3-\delta}X^\varepsilon \tag{17.51}$$

with fixed $\theta, \delta > 0$. Then

$$M(X) \ll X^{1-\theta\delta/3+\varepsilon}. \tag{17.52}$$

If (17.51) *is available directly with $H \asymp x$, the exponent improves to $1-\delta/3$.*

Proof. The Lipschitz plateau argument gives $|S_x(H)|^3 \ll \mathcal{E}_1(x,H)$, hence $|S_x(H)| \ll H^{1-\delta/3}X^\varepsilon$. Partition $[X,2X]$ into $O(X/H)$ such windows and use the triangle inequality:

$$|M(2X)-M(X)| \ll XH^{-\delta/3}X^\varepsilon = X^{1-\theta\delta/3+\varepsilon}. \tag{17.53}$$

Dyadic telescoping proves the claim. No cross-window cancellation is used. □

17.7. **Prime dilations: exact gain and exact failure.** Let $b=(b_1,\dots,b_L)$ and place b_j at the shifted positions $r+p(j-1)$ in $\{1,\dots,H\}$, with zeros elsewhere. Write $B_j=\sum_{i\le j} b_i$ and

$$\ell = H - \big(r+p(L-1)\big)+1, \qquad 1 \le \ell \le p. \tag{17.54}$$

Directly counting the holding time of each partial sum gives

$$\boxed{\mathcal{E}_1(D_{p,r}b;H) = p\,\mathcal{E}_1(b;L) - (p-\ell)|B_L|^2 \le p\,\mathcal{E}_1(b;L).} \tag{17.55}$$

The residue r, which is determined by $x \bmod p$ in an arithmetic window, cannot be omitted from the exact statement. From the trivial child estimate $\mathcal{E}_1(b;L) \ll L^3$ and $L \asymp H/p$, one gets $\mathcal{E}_1(D_{p,r}b;H) \ll H^3/p^2$. If the child already has Brownian energy $O(L^2)$, the lift has energy $O(H^2/p)$.

Proposition 17.16 (No abstract dilation frame [DED])**.** *There is no sequence-independent bound*

$$\left\| J_H \sum_{\gamma=1}^{R} a_\gamma \right\|_2^2 \le H^{o(1)} \sum_{\gamma=1}^{R} \|J_H a_\gamma\|_2^2 \tag{17.56}$$

for arbitrary families of dilation channels.

Proof. Take $a_d(n) = \mathbf{1}_{d|n}$ with distinct $d \asymp D$ and $D = o(H)$. Then

$$(J_H a_d)(h) = \lfloor h/d \rfloor, \qquad \|J_H a_d\|_2^2 \sim \frac{H^3}{3d^2}. \tag{17.57}$$

For R such channels, $J_H(\sum_d a_d)(h) = h \sum_d d^{-1} + O(R)$, so the squared norm is asymptotic to $R^2 H^3/D^2$, whereas the sum of the individual squared norms has order RH^3/D^2. Their ratio has order R. □

This no-go applies directly to an abstract version of the frame proposed in Version 7. Any valid frame must use the signed arithmetic coefficients, not dilation sparsity alone.

Lemma 17.17 (Low-mode reduction [DED])**.** *For every a with $|a_n| \le 1$ and every $1 \le K < H$,*

$$\sum_{k>K} \lambda_k |\langle a, v_k \rangle|^2 \ll \frac{H^3}{K^2}. \tag{17.58}$$

Consequently, a target $O(H^{3-\delta})$ only requires new information on $k \le H^{\delta/2}$.

Proof. The eigenvalues decrease and $\lambda_{K+1} \ll H^2/K^2$. Bessel's identity and $\|a\|_2^2 \le H$ give

$$\sum_{k>K} \lambda_k |\langle a, v_k \rangle|^2 \le \lambda_{K+1} \|a\|_2^2 \ll H^3/K^2. \tag{17.59}$$

□

Corollary 17.18 (Explicit low-mode target [DED])**.** *Writing $c_k = \langle a, v_k \rangle$, for every $K < H$,*

$$\mathcal{E}_1(x, H) \ll H^2 \sum_{k \le K} \frac{|c_k|^2}{(2k-1)^2} + \frac{H^3}{K^2}. \tag{17.60}$$

Hence, with $K = H^{\delta/2}$, the power-saving target follows from

$$\sum_{k \le H^{\delta/2}} \frac{|c_k|^2}{(2k-1)^2} \ll H^{1-\delta}. \tag{17.61}$$

This is an exact organization of the target, not an arithmetic mechanism.

Conjecture 17.19 (Signed low-mode Ramaré–Heath–Brown recombination [ADAPT])**.** *There exist a causal, weighted prime-factor decomposition $a = E + \sum_{\gamma \in \Gamma} a_\gamma$, fixed $\delta > 0$, and $K = H^{\delta/2}$ such that the rough current, the channel diagonal, and the signed cross-channel sum obey*

$$\|J_H P_K E\|_2^2 + \sum_{\gamma} \|J_H P_K a_\gamma\|_2^2 \ll H^{3-\delta+o(1)}, \tag{17.62}$$

$$2\Re \sum_{\gamma<\gamma'} \langle J_H P_K a_\gamma, J_H P_K a_{\gamma'} \rangle \ll H^{o(1)} \sum_{\gamma} \|J_H P_K a_\gamma\|_2^2 + H^{3-\delta+o(1)}. \tag{17.63}$$

Here P_K denotes projection onto the first K eigenmodes of K_H.

This is not an abstract frame: Proposition 17.16 rules that out. It is a signed Type-II statement for the actual Ramaré–Heath–Brown coefficients in the low-frequency Green kernel

$$\sum_{r,s} b_\gamma(r)\overline{b_{\gamma'}(s)}K_H(r_\gamma + d_\gamma r, r_{\gamma'} + d_{\gamma'}s). \tag{17.64}$$

Conjecture 17.19, together with Lemma 17.17, implies the first open line $\mathcal{E}_1 \ll H^{3-\delta+o(1)}$ of (17.50).

There is also a one-layer scale tension. The standard Ramaré sieve error has cardinality $O(H \log P/\log Q)$. If $P = x^\rho$ and $Q \le x^\sigma$, cardinality alone leaves a constant-order proportion ρ/σ. If $\log P/\log Q \to 0$, then $P = Q^{o(1)}$ and the diagonal gain P^{-1} or P^{-2} is only subpower on polynomial ranges. A fixed power saving therefore appears to require a multilevel, nearly exhaustive causal factorization rather than a single prime band plus a discarded rough set. This deduction is a design constraint [DED], not a no-go theorem for every possible Ramaré decomposition.

Proposition 17.20 (A sufficient factor-frame budget [DED])**.** *Suppose the genuine factor channels have the shifted form $a_\gamma = D_{d_\gamma, r_\gamma} b_\gamma$ with $|b_\gamma| \le w_\gamma$, the rough remainder has J_H-energy $O(H^{3-\delta})$, and an arithmetic frame with loss H^η holds for their recombination. If*

$$H^\eta \sum_\gamma w_\gamma^2 \left(\frac{1}{d_\gamma^2} + \frac{1}{H}\right) \ll H^{-\delta}, \tag{17.65}$$

then $\mathcal{E}_1(x, H) \ll H^{3-\delta}$.

Proof. The shifted dilation estimate and the trivial bound on each child give

$$\|J_H a_\gamma\|_2^2 \ll w_\gamma^2 \left(\frac{H^3}{d_\gamma^2} + H^2\right). \tag{17.66}$$

Sum over γ, apply the assumed arithmetic frame, use (17.65), and add the rough remainder. □

This proposition records the quantitative burden without asserting that the Ramaré–Heath–Brown decomposition satisfies it. In particular, the subpower prime extracted in the published proof cannot by its d^{-2} geometry alone produce a fixed power saving; polynomial-scale factors created later in the factorization must remain visible in both the diagonal budget and the signed low-mode recombination.

17.8. **Narayana peaks: a complementary local operator.** For an oriented Dyck path of length $L = 2n$, let σ_j be its ± 1 increments and let $P(\sigma)$ be its number of peaks. The classical Narayana number

$$N(n, k) = \frac{1}{n}\binom{n}{k}\binom{n}{k-1} \tag{17.67}$$

counts all such paths with k peaks. For primitive excursions, removing the first up-step and last down-step leaves an arbitrary Dyck path of semilength $n-1$; hence, for $n \ge 2$, the primitive peak distribution is $N(n-1, k)$, not $N(n, k)$. Direct expansion of the peak indicators gives the exact local quadratic identity

$$\boxed{P(\sigma) = \frac{L + 1 - \sum_{j=1}^{L-1} \sigma_j \sigma_{j+1}}{4}.} \tag{17.68}$$

Thus peak count does not produce a second copy of the Green operator $J_H^* J_H$: it is a nearest-neighbour variation statistic, concentrated at the opposite, local end of the spectral description. The joint data

$$(\mathcal{E}_1,\ P,\ \text{run-length profile}) \tag{17.69}$$

may nevertheless distinguish a coherent mountain from a path with the same length and many local reversals.

A literature-transfer warning is essential. Hildebrand's theorem concerns prescribed patterns of $\lambda(n), \lambda(n+1), \lambda(n+2)$ at consecutive integers [10]; it does not concern $\mu(q_j), \mu(q_{j+1})$ at consecutive squarefree integers. The latter indices have variable gaps, so no restriction on Narayana peaks per Mertens excursion follows from that theorem.

There is, however, a closer and rigorous neighbouring construction. In the Möbius sign-pattern argument of Matomäki, Radziwiłł, and Tao, the first graph has squarefree vertices and joins n to $n+1$ when both integers are squarefree; its disconnection, caused in particular by multiples of 4, forces an enlarged multiplicative graph with edges of the form n to $n+p$ under divisibility and squarefreeness constraints [6, 7]. This must not be confused with the ordered compressed edge q_j to q_{j+1}: the latter can cross an arbitrary gap and is the edge relevant to (17.68).

The resulting classification is therefore three-layered. The zero pattern of any fixed collection of integer shifts is governed by squarefree sieve densities [LIT]. Conditional on Chowla, the nonzero Möbius signs are asymptotically equidistributed after conditioning on that zero pattern [LIT, conditional]. The published graph argument also supplies a genuine consistency obstruction: an assumed missing two-letter Möbius pattern would force near constancy across large portions of the enlarged squarefree graph, contradicting the Matomäki–Radziwiłłshort-interval theorem [LIT]. None of these statements presently yields the distribution of $\mu(q_j)\mu(q_{j+1})$ along a specified Mertens excursion. Consequently a Narayana law for arithmetic excursions remains a well-posed open transfer problem [ADAPT], now anchored to sieve/Chowla sign-pattern machinery but not proved by it.

For comparison with Version 7, partition $\mathcal{P}$ into logarithmic prime blocks $\mathcal{P}_\nu$ and put

$$B_\nu(u) = \sum_{p \in \mathcal{P}_\nu} C_p(u), \qquad E_a(u) = R_{\mathcal{P}}(u). \tag{17.70}$$

Conjecture 17.21 (Long-term arithmetic Brownian CRSF [ADAPT]). *At a prescribed scale H, the actual signed Ramaré prime bands and their logarithmic subdivision can be chosen so that, uniformly in the initial point a,*

$$\boxed{\left\| \sum_\nu B_\nu \right\|^2_{\ell^2[1,H]} \le (\log x)^A \sum_\nu \|B_\nu\|^2_{\ell^2[1,H]}} \tag{17.71}$$

and

$$\boxed{\|E_a\|^2_{\ell^2[1,H]} + \sum_\nu \|B_\nu\|^2_{\ell^2[1,H]} \ll_\varepsilon H^{2+\varepsilon}.} \tag{17.72}$$

This is retained only as the Brownian endpoint of the programme. Proposition 17.16 shows that no corresponding statement is possible for arbitrary dilation channels, and (17.72) is not the first open milestone.

There are two distinct targets. The first asks only for $H \asymp x^\theta$, $\theta > 11/20$, and leads after block summation to the mesoscopic exponent below. The second asks directly for $H \asymp x$; it is precisely the $q = 1$ dyadic-energy target and gives exponent $2/3$. They must not be merged into a statement for all $H \ge x^\theta$, because that range already contains $H = x$. Version 8 also removes the absolute sum of every off-diagonal correlation used in the earlier draft. That condition was unnecessarily strong: the required object is an almost-orthogonal frame bound for the sum itself.

Proposition 17.22 (Exact consequence and scale audit for CRSF [DED]). *Conjecture 17.21 implies*

$$\sum_{u \le H} |M(a+u) - M(a)|^2 \ll_{\varepsilon,\theta} H^{2+\varepsilon} \tag{17.73}$$

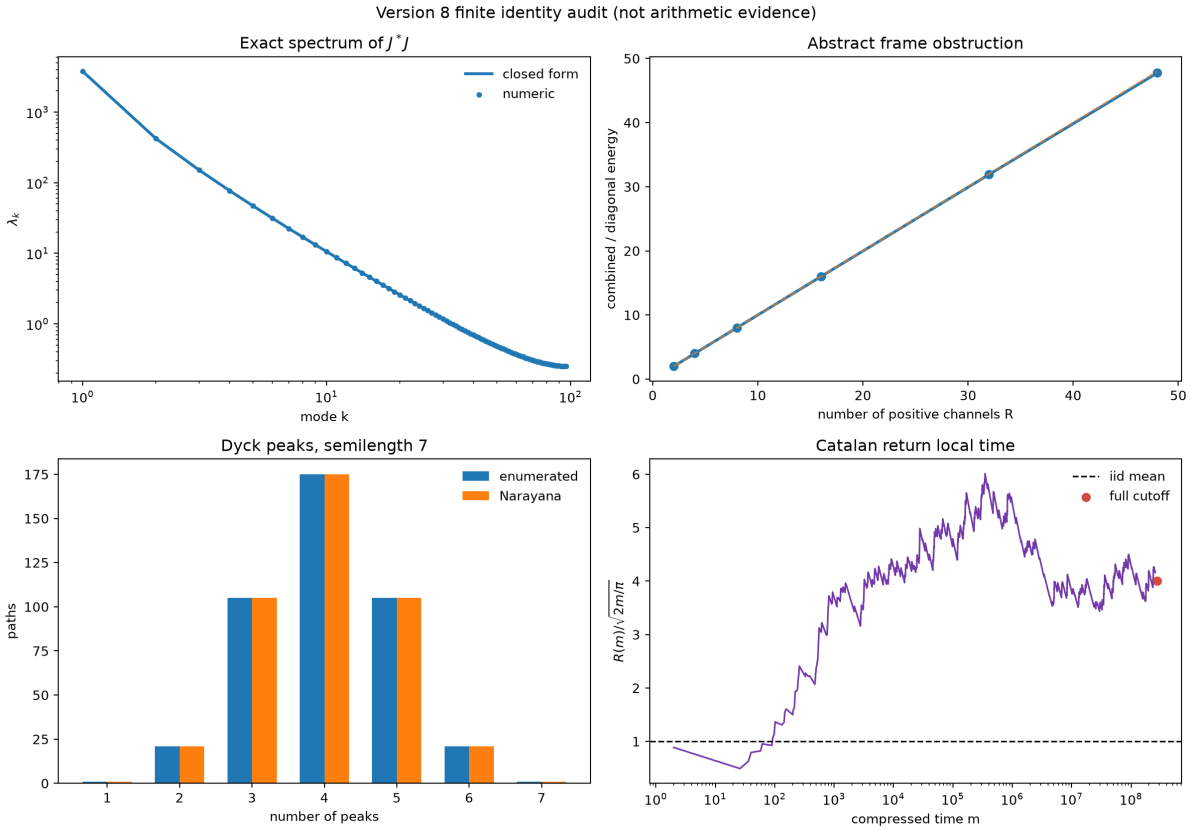


FIGURE 5. Finite Version 8 audits: exact $J_H^*J_H$ spectrum, failure of an unsigned abstract frame, Narayana peak enumeration, and compressed return local time. The first three panels verify finite identities or counterexamples; the fourth is arithmetic data. None tests the open power-saving estimate.

and hence

$$|M(a+H)-M(a)| \ll_{\varepsilon,\theta} H^{2/3+\varepsilon}. \tag{17.74}$$

At the limiting scale $\theta \downarrow 11/20$, subdivision into critical blocks gives

$$M(X) \ll_{\varepsilon} X^{49/60+\varepsilon}. \tag{17.75}$$

If instead the same CRSF estimate were available with $H \asymp x$, then

$$M(X) \ll_{\varepsilon} X^{2/3+\varepsilon}. \tag{17.76}$$

Proof. Equation (17.14) and expansion of the square show, by (17.71), (17.72), and $\|U+V\|_2^2 \le 2\|U\|_2^2+2\|V\|_2^2$, that the left-hand side of Equation (17.73) is $O(H^{2+\varepsilon})$ after absorbing the logarithm. Put $A = |M(a+H)-M(a)|$. Since Mertens increments have modulus at most one, for at least $\lfloor A/2 \rfloor$ values of u immediately preceding H the prefix has modulus at least $A/2$. Thus

$$A^3 \ll \sum_{u \le H} |M(a+u)-M(a)|^2 \ll H^{2+\varepsilon}, \tag{17.77}$$

which proves Equation (17.74) after renaming ε.

Partition a dyadic interval of length X into blocks of length $H \asymp X^{\theta}$. The triangle inequality is used only at this final inter-block step and gives

$$\frac{X}{H} H^{2/3+\varepsilon} = X^{1-\theta/3+\varepsilon}. \tag{17.78}$$

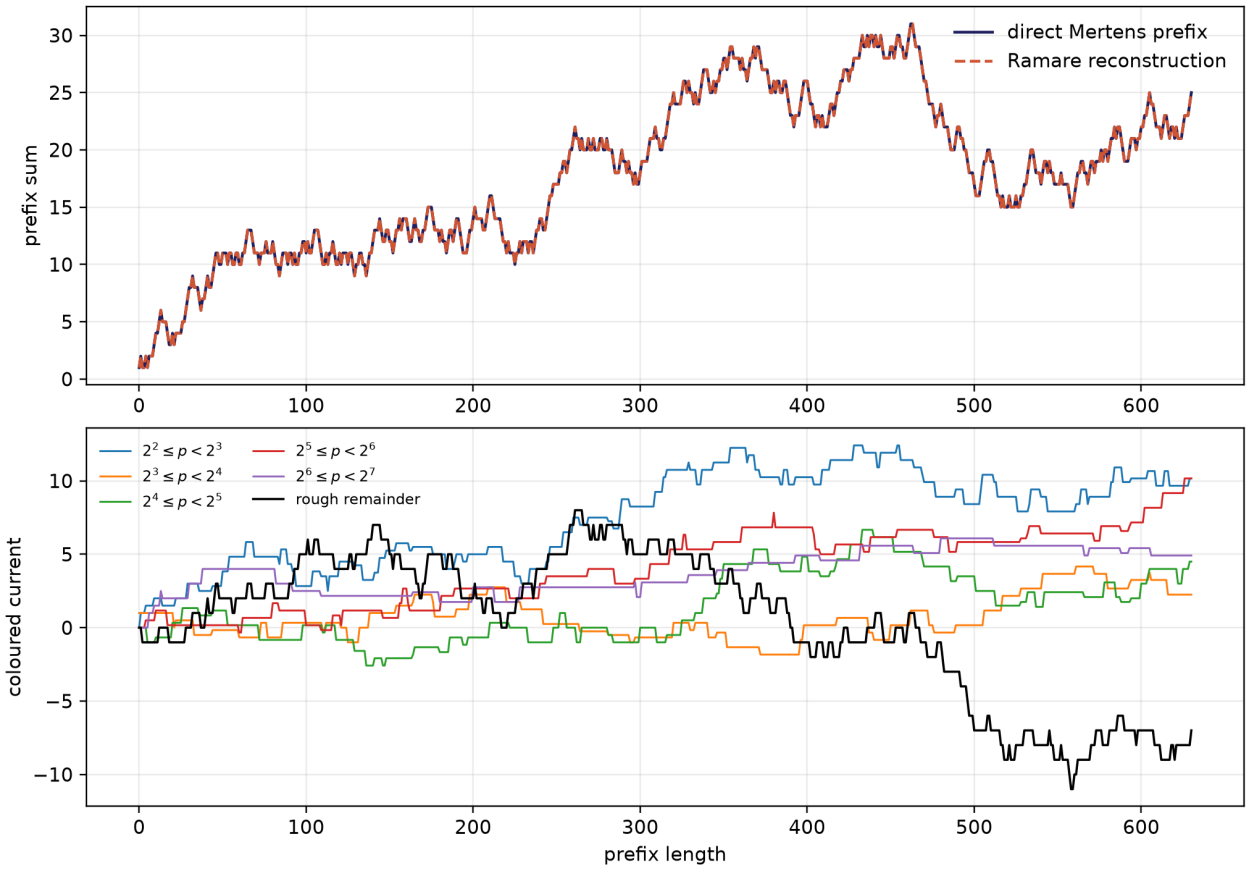


FIGURE 6. Finite exact reconstruction of all prefixes in one computational prime band. The lower panel retains the prime-block currents and the rough remainder before scalar projection. This is an implementation audit of Equation (17.14), not asymptotic evidence for Conjecture 17.21.

Letting $\theta \downarrow 11/20$ gives $1 - \theta/3 = 49/60$. Summation over dyadic intervals preserves the exponent. If $H \asymp X$ is available directly, no critical-scale subdivision is needed and the exponent is $2/3$. □

Conjecture 17.23 (Higher Catalan frame hierarchy [ADAPT])**.** *For every fixed $q \ge 1$, the same causal prime blocks satisfy*

$$\left\| \sum_{\nu} B_\nu \right\|_{\ell^{2q}[1,H]} \ll_q (\log x)^A \left\| \left(\sum_{\nu} |B_\nu|^2 \right)^{1/2} \right\|_{\ell^{2q}[1,H]}, \tag{17.79}$$

$$\|E_a\|_{\ell^{2q}[1,H]}^{2q} + \sum_{u \le H} \left(\sum_{\nu} |B_\nu(u)|^2 \right)^q \ll_{q,\varepsilon} H^{q+1+\varepsilon}. \tag{17.80}$$

At the dyadic scale $H \asymp x$, Conjecture 17.23 implies $\mathrm{CE}_q^{\mathrm{dyad}}$. If it holds for every fixed q, Theorem 17.7 gives RH. This was the specific open selector bridge in Version 7: Catalan theory supplies the exponent H^{q+1}, while the arithmetic task is to prove that the causally coloured Ramaré currents obey the frame and diagonal-energy bounds. Neither inequality is asserted in the cited short-interval literature.

Remark 17.24 (The all-moment mesoscopic ceiling). Even a mesoscopic hierarchy

$$\sum_{u\le H} |M(a+u)-M(a)|^{2q} \ll_{q,\varepsilon} H^{q+1+\varepsilon} \tag{17.81}$$

would give, by the same Lipschitz plateau argument and critical-block subdivision,

$$M(X) \ll_{q,\varepsilon} X^{\,1-\theta q/(2q+1)+\varepsilon}. \tag{17.82}$$

As $q\to\infty$ and $\theta\downarrow 11/20$, this approaches $X^{29/40+\varepsilon}$, not the RH exponent. Thus mesoscopic all-prefix control must be supplemented either by cross-block cancellation or by an all-scale occupation estimate reaching $H\asymp x$.

This is a vector-valued adaptation of the small-prime separation in Matomäki–Teräväinen. It is not asserted in their paper, and a targeted search did not locate this stopped Catalan form in the literature; that absence is only weak evidence of novelty. The condition is genuinely Catalan because the norm is taken over all nested prefixes up to an arithmetic first-return or open-meander stopping time. Taking the absolute value before the prefix norm, or summing the colours by the triangle inequality, destroys the proposed gain.

Equation (17.11) also exposes two precise obstructions. First, the unrestricted prime-deletion operator has μ as an eigenfunction of eigenvalue -1, so no contraction can follow from deletion algebra alone. Second, the rough current $R_{\mathcal{P}}$ is too large to discard by cardinality at the square-root scale. Any successful argument must therefore obtain cancellation from the localized, stopped cross-colour terms and must retain the rough layer in the multiscale system.

Ramaré colouring does not provide a Fredkin matching rule. A squarefree integer can have several eligible extracted primes, whereas its matched up/down partner in the Catalan arch is indexed by an unrelated later squarefree integer. Multiplicativity does not force the matched integers to share a prime colour. Coloured-Dyck, Fredkin, and quenched-polymer models are therefore retained as established vocabulary and diagnostics, not as proved arithmetic bridges.

The proposed non-circular chain is now

$$\begin{aligned}&\text{Ramaré partition of unity [LIT/DED]}\\ &\quad\Longrightarrow \text{stopped prime-colour square function [open ADAPT]}\\ &\quad\Longrightarrow \text{uniform Catalan occupation moments}\\ &\quad\Longrightarrow X\notin \mathfrak{C}_{\mathrm{RH}} \Longleftrightarrow \mathrm{RH}.\end{aligned} \tag{17.83}$$

The unresolved statement is no longer the already known scalar estimate on one terminal short sum. It is the preservation of prime-colour cancellation after passing to the nonlinear occupation norm over every Catalan stopping prefix.

17.9. **No-go audit: what the exact identities do not prove.** The exact Fourier and prime-shell identities above remain useful diagnostics, but Version 8 removes them from the proposed proof mechanism. The following statements isolate the reason.

Proposition 17.25 (Catalan universality obstruction)**.** *For every $n\ge 1$ there is a primitive Dyck excursion of length $2n$ and height n. More generally, every finite concatenation of primitive Dyck excursions followed by a strict meander is realized by a unique finite word of increments in $\{-1,+1\}$.*

Proof. The mountain word $+^n-^n$ is primitive and has height n. For the second statement, read the successive differences of the prescribed paths and concatenate them. The return skeleton and the final meander recover the decomposition uniquely. □

Corollary 17.26 (Catalan-only no-go theorem)**.** *No estimate that uses only the nearest-neighbour property, the return skeleton, primitive Catalan enumeration, or uniform Catalan large deviations can imply a sublinear bound for the selected path. An additional arithmetic statement excluding the mountain words is logically necessary.*

The corollary does not deny the exact Catalan structure. It says that ambient counting and typicality cannot select the Mertens word from that structure.

Proposition 17.27 (Redundant prime lift)**.** *Let $a = q_{z_r}$ be a return anchor, $x = q_{z_r+v}$, and*

$$A_p(y) = \sum_{\substack{n \le y \\ p \nmid n}} \mu(n). \tag{17.84}$$

For every prime p,

$$\big(A_p(x) - A_p(a)\big) - \big(A_p(x/p) - A_p(a/p)\big) = X_{z_r+v}. \tag{17.85}$$

Consequently, averaging any power of the left-hand side over primes reproduces the corresponding Catalan prefix moment exactly and gives no estimate.

Proof. Subtract the exact shell identity $M(y) = A_p(y) - A_p(y/p)$ at $y = x$ and $y = a$, then use $M(a) = X_{z_r} = 0$. The final assertion follows because the difference is independent of p. □

This proposition invalidates the prime-dilation "alignment" proposal as an independent bridge: after definitions are expanded, its desired inequality is precisely Conjecture 16.3 again.

Proposition 17.28 (Stability floor for smoothed chambers)**.** *Fix $x \ge 2$ and the squarefree integers not exceeding $2x$. Perturb their prime weights by*

$$w_p(\theta, \xi) = \log p + \theta \xi_p, \qquad |\xi_p| \le 1. \tag{17.86}$$

Put

$$\delta_x = \frac{\log(1 + 1/(2x))}{4 \log_2(2x)}. \tag{17.87}$$

For every choice of ξ, the complete order of those squarefree integers is unchanged whenever $|\theta| \le \delta_x$. Hence any bad-excursion event true for the arithmetic order occupies at least the interval $[-\delta_x, \delta_x]$ in every perturbation direction.

Proof. Distinct integers $m, n \le 2x$ satisfy

$$|\log m - \log n| \ge \log(1 + 1/(2x)). \tag{17.88}$$

The perturbation of their energy difference is at most

$$|\theta|\big(\omega(m) + \omega(n)\big) \le 2|\theta| \log_2(2x). \tag{17.89}$$

At the radius (17.87) this is at most half the deterministic gap, so no pair reverses order. □

If θ is uniform on a fixed interval $[-\eta, \eta]$, an actual bad excursion therefore forces bad probability at least

$$\frac{\delta_x}{\eta} \asymp \frac{1}{x \log x}. \tag{17.90}$$

A smoothed Catalan tail smaller than this floor cannot be obtained merely by replacing the arithmetic order with a typical perturbed order: it must already control the rigid neighbourhood containing the arithmetic point. Version 5 therefore does not promote smoothed analysis to a proof route.

Definition 17.29 (Admissible Catalan–arithmetic bridge)**.** A future bridge is called admissible only if it satisfies all four tests:

(i) it is false for at least one primitive Dyck mountain word;
(ii) it uses a proved arithmetic relation not recoverable from the values of the single prefix path alone;
(iii) after all definitions are substituted, it is not the desired uniform prefix bound itself;
(iv) it is uniform over internal prefixes and the final open meander.

No admissible bridge is proved in this notebook. This is the precise boundary left after the audit. The exact target remains Conjecture 16.3; calling it a selector, alignment, dephasing, or free-energy estimate does not constitute progress unless an admissible arithmetic mechanism proves it.

18. An aggregate route that does not predict M pointwise

The pointwise selection principle in Conjecture 15.1 is deliberately strong. It asks for a restriction at every time and can therefore reproduce, in disguised form, the original difficulty of controlling $M(x)$. This section replaces that hypothesis by averaged positive quantities. A large value cannot hide from sufficiently high moments because the Mertens walk has increments of absolute value at most one.

Remark 18.1 (Provenance). The use of moments to recover a point value from local regularity is a classical analytic mechanism. The specific time-moment criterion and its decomposition over Catalan excursions below are deductions in the present notebook [DED]. Verjovsky's local Fourier-moment criterion is a distinct recent realization of a related moment-to-point principle [25] [LIT]; it averages a Möbius Fourier polynomial over a short frequency arc, whereas the criterion below averages the Mertens walk over time. The weak Mertens conjecture is another classical averaged sufficient condition, but it is stronger than RH and also forces simplicity of the zeros [24] [LIT].

18.1. **A hierarchy of moments equivalent to RH.** For an integer $k \geq 1$, define

$$\mathcal{B}_k(X) = \sum_{n \leq X} |M(n)|^{2k}. \tag{18.1}$$

Theorem 18.2 (Aggregate moment criterion)**.** *The Riemann hypothesis is equivalent to the following family of estimates: for every fixed integer $k \geq 1$ and every $\varepsilon > 0$,*

$$\mathcal{B}_k(X) \ll_{k,\varepsilon} X^{k+1+\varepsilon}. \tag{18.2}$$

Proof. Assume RH. For every $\eta > 0$ the classical Mertens criterion gives $|M(n)| \ll_\eta n^{1/2+\eta}$. Given k and ε, take $\eta = \varepsilon/(2k)$. Summation gives (18.2), after an inessential adjustment of ε.

Conversely, suppose (18.2) holds for every fixed k. Fix N and write $H = |M(N)|$. Since

$$|M(n+1) - M(n)| \leq 1, \tag{18.3}$$

we have $|M(N+j)| \geq H/2$ for $0 \leq j \leq \lfloor H/2 \rfloor$. Moreover $H \leq N$, so these indices are at most $2N$. Hence, for $H \geq 4$,

$$\mathcal{B}_k(2N) \geq c_k H^{2k+1} \tag{18.4}$$

with an absolute admissible choice such as $c_k = 2^{-2k-2}$. It follows that

$$H \ll_{k,\varepsilon} N^{(k+1+\varepsilon)/(2k+1)}. \tag{18.5}$$

Given $\delta > 0$, choose k so large that $1/(4k+2) < \delta/2$, and then choose ε so that $\varepsilon/(2k+1) < \delta/2$. Equation (18.5) yields $M(N) = O_\delta(N^{1/2+\delta})$, which is equivalent to RH. □

The first RH-equivalent statement in the new route is therefore Theorem 18.2. No Catalan, physical, or topological statement preceding it proves RH. This is the dependency-graph checkpoint for the remainder of the program.

Corollary 18.3 (Finite-order milestones)**.** *For a single fixed k, estimate* (18.2) *implies*

$$M(X) \ll_{k,\varepsilon} X^{\frac{k+1}{2k+1}+\varepsilon}. \tag{18.6}$$

In particular, $k = 1, 2, 3$ give the successive exponents $2/3$, $3/5$, and $4/7$.

Warning 18.4. Even the case $k = 1$ would be a major unconditional advance: it would give a fixed zero-free half-plane $\Re s > 2/3$. The hierarchy avoids pointwise sign prediction, but it does not make the arithmetic cancellation automatic. Its advantage is that it supplies auditable intermediate targets rather than an all-or-nothing selector axiom.

18.2. **Exact excursion decomposition of the moment budget.** In compressed time put

$$\mathcal{B}_k^{\#}(Y) = \sum_{m \le Y} |X_m|^{2k}. \tag{18.7}$$

The proof of Theorem 18.2, together with $q_m \asymp m$, gives the analogous equivalence

$$\text{RH} \iff \mathcal{B}_k^{\#}(Y) \ll_{k,\varepsilon} Y^{k+1+\varepsilon} \quad \text{for every fixed } k \text{ and every } \varepsilon > 0. \tag{18.8}$$

For an excursion D of length L, define its integrated $2k$-energy

$$\mathcal{E}_k(D) = \sum_{u=1}^{L-1} |D(u)|^{2k}. \tag{18.9}$$

At a cutoff Y, let $D_1, \dots, D_R$ be the completed excursions and let U_Y be the open terminal meander of length t_Y. Then the identity

$$\boxed{\mathcal{B}_k^{\#}(Y) = \sum_{r=1}^{R} \mathcal{E}_k(D_r) + \mathcal{E}_k(U_Y)} \tag{18.10}$$

is exact. It converts RH into a hierarchy of total geometric-energy bounds over the actual excursions, without requiring the prediction of any individual value of X_m.

Under the uniform Catalan law at fixed length L, Brownian scaling predicts

$$\mathbb{E}_{\mathrm{Cat},L} \mathcal{E}_k(D) \asymp_k L^{k+1}. \tag{18.11}$$

This motivates the length budget

$$\Lambda_k(Y) = \sum_{r=1}^{R} L_r^{k+1} + t_Y^{k+1}. \tag{18.12}$$

Since the segment lengths sum to Y,

$$\Lambda_k(Y) \le Y^{k+1}. \tag{18.13}$$

Conjecture 18.5 (Aggregate Catalan domination)**.** *For every fixed $k \ge 1$ and every $\varepsilon > 0$,*

$$\mathcal{B}_k^{\#}(Y) \ll_{k,\varepsilon} Y^{\varepsilon} \Lambda_k(Y). \tag{18.14}$$

Proposition 18.6. *Conjecture 18.5 for all fixed k implies RH.*

Proof. Combine (18.14) with (18.13) and then apply (18.8). □

Warning 18.7. Conjecture 18.5 is a sufficient selector principle and may be strictly stronger than RH. RH controls the total moment $\mathcal{B}_k^{\#}(Y)$ but does not by itself compare it with the potentially much smaller random length budget $\Lambda_k(Y)$. The conjecture is proposed as a structured attack target, not as an equivalent reformulation.

Unlike a pointwise percentile bound, aggregate Catalan domination permits exceptional excursions: one excursion may have energy far above its Catalan mean provided the total energy remains within the global length budget. This is the precise sense in which the route avoids predicting the real Mertens path excursion by excursion.

18.3. **The arithmetic correlation target.** Write $a_j = \mu(q_j) \in \{-1, +1\}$. Expanding the positive moment and changing the order of summation gives the exact identity

$$\mathcal{B}_k^{\#}(Y) = \sum_{1 \le i_1, \dots, i_{2k} \le Y} a_{i_1} \cdots a_{i_{2k}} (Y - \max(i_1, \dots, i_{2k}) + 1). \tag{18.15}$$

The configurations in which every index occurs with even multiplicity make a contribution of order at most $O_k(Y^{k+1})$. The remaining configurations form an explicit weighted off-diagonal correlation sum. Thus the analytic problem is not to predict a_j but to prove cancellation in the aggregate off-diagonal part of (18.15).

Definition 18.8 (Off-diagonal correlation budget)**.** Let $\mathcal{O}_k(Y)$ denote the part of (18.15) indexed by tuples for which at least one index has odd multiplicity.

A concrete sufficient target is

$$\mathcal{O}_k(Y) = O_{k,\varepsilon}(Y^{k+1+\varepsilon}) \tag{18.16}$$

for every fixed k. This target retains ordering and high-order sign correlations; it cannot be replaced by one-point equidistribution or by a fixed-shift Chowla statement without an additional uniform inversion theorem.

18.4. **Finite evidence and its exact limitation.**

Experiment 18.9. For the supplied file through ordinary cutoff 455052511, compressed time is $Y = 276638778$. Direct blockwise summation gave the following ratios. The second column divides $\mathcal{B}_k^{\#}(Y)$ by the leading simple-walk moment $((2k-1)!!/(k+1))Y^{k+1}$; the third divides it by $\Lambda_k(Y)$, including the open terminal meander.

k	simple-walk ratio	$\mathcal{B}_k^{\#}(Y)/\Lambda_k(Y)$
1	3.70956×10^{-2}	0.404561
2	8.92436×10^{-4}	0.251180
3	1.64775×10^{-5}	0.188051
4	2.48322×10^{-7}	0.154931
5	3.16760×10^{-9}	0.135848
6	3.50903×10^{-11}	0.125099

These are finite measured values, not estimates uniform in Y. They support the scale in Conjecture 18.5 on this one range but provide no asymptotic control.

18.5. **A staged proof program.** The route now has intermediate statements that can fail independently:

(M1) Audit $R(m)/\sqrt{2m/\pi}$ and the extreme-gap survival curve at successive larger cutoffs; do not fit an asymptotic constant to one terminal value.

(M2) Attack the return identity (4.7) or an equivalent inverse-renewal estimate using the actual compressed Möbius signs and squarefree support.

(M3) Use Theorem 17.12 as the proved baseline and reproduce its finite operator identities without treating logarithmic saving as power saving.

(M4) Retain the complete Ramaré–Heath–Brown factor channels and measure their diagonal contribution in the first $H^{\delta/2}$ modes of $J_H^* J_H$.

(M5) Replace cardinality control of the rough set by a bound for its J_H-energy; a sparse coherent remainder can still store a mountain.

(M6) Prove a signed Type-II cross-channel estimate strong enough for (17.65). The first major analytic target is $\mathcal{E}_1(x, H) \ll H^{3-\delta}$, not the Brownian bound.

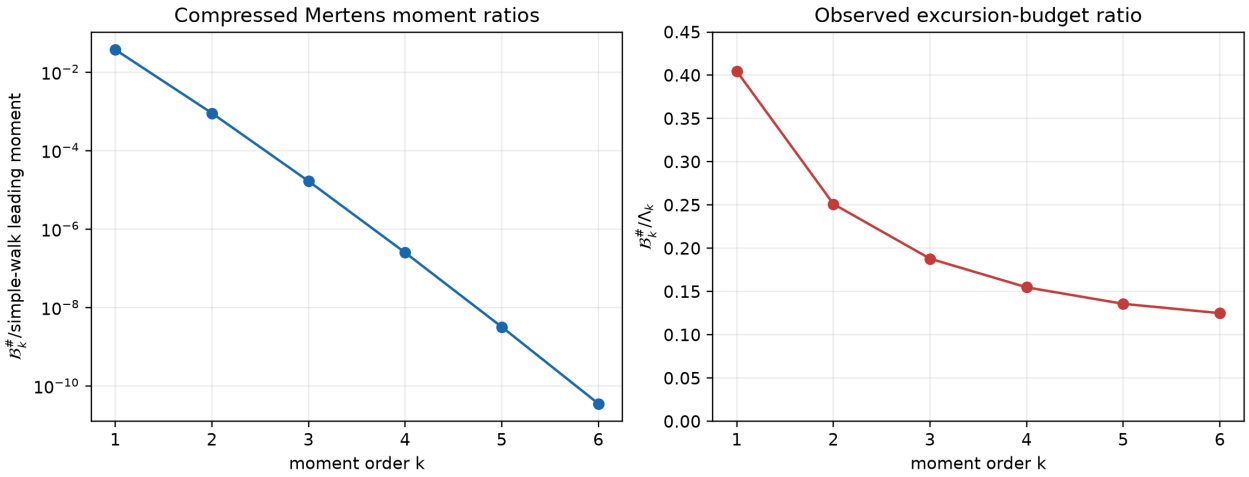


FIGURE 7. Finite compressed-time moment ratios through $k = 6$. The logarithmic left panel compares with the leading simple-walk moment; the right panel shows $\mathcal{B}_k^{\#}(Y)/\Lambda_k(Y)$.

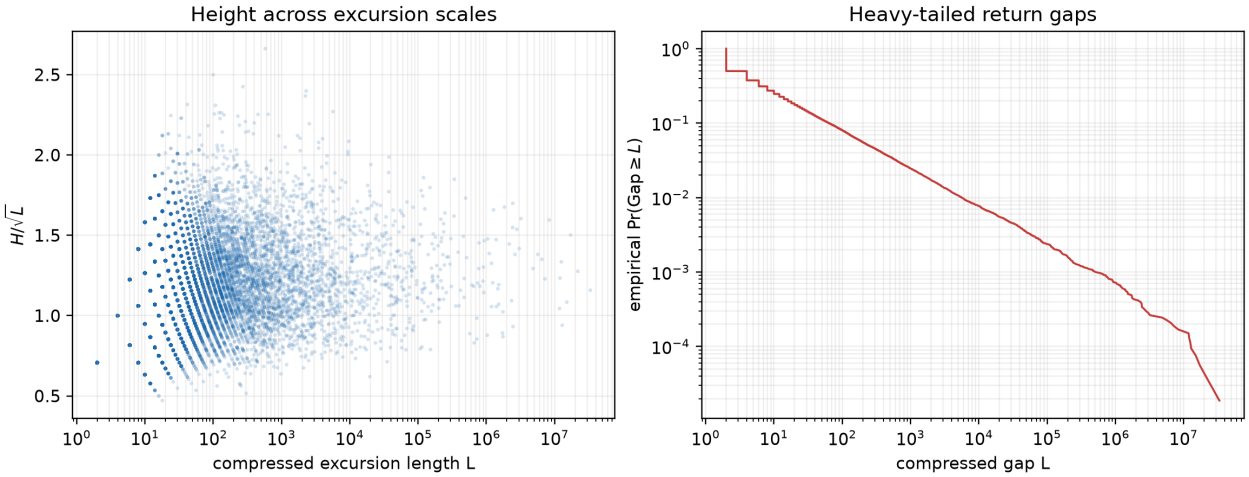


FIGURE 8. Observed excursion height and return-gap geometry through $N = 455{,}052{,}511$. Both panels are descriptive finite data only.

(M7) Only after a genuine power saving is proved should one construct a pinning, transfer, or hydrodynamic model intended to explain the estimate. A physical model fitted to the finite data is not an arithmetic proof.

The global aggregate hierarchy remains an exact criterion and a useful benchmark, but not the proposed first mechanism. The first open analytic frontier of Version 8 is the local, uniform power saving

$$\mathcal{E}_1(x, H) \ll H^{3-\delta}, \tag{18.17}$$

on a fixed mesoscopic range. Proposition 17.15 shows that even this first step would already improve the known global power exponent for Mertens.

19. Version 9: the Catalan fusion program

19.1. **Decision: fuse first, specialize to M"obius second.** The program proposed here does not apply a list of established techniques to M"obius one at a time. Its first stage is deliberately

arithmetic-free: construct a common algebra for weak restrictions of Catalan paths and discover which compositions can destroy a dangerous height sector even when the individual restrictions cannot. Only identities that survive this universal stage are then specialized to squarefreeness, prime-factor bands, and the multiplicative cutoff. This order is essential: it separates a new combinatorial mechanism from a restatement of the Mertens problem.

(19.1)

Catalan paths + prefix profiles + rank + restrictions
+ sectoral entropy + transfer operators + dual certificates
⇓
universal algebra of restricted Catalan classes
⇓
synergy, confinement, or exact sector annihilation
⇓
arithmetic specialization and independent proof of its hypotheses.

The last line is a gate, not a promise. A theorem about an abstract restricted class says nothing about the Mertens walk until the required restrictions are proved from factorization data without assuming an equivalent form of RH.

Provenance audit. The Catalan matroid, lattice-path matroids, minors and duality, the ordinary Tutte polynomial, deletion–contraction, and the rank characterization of a maximum intersection are established theory [LIT] [1, 2]. The prefix profile, sector gain, eight-count interaction formula, chain rules, and Boolean-lattice interaction expansion below are [DED] elementary consequences once the refined objects are defined. Their packaging may be useful, but no priority claim is made. Quantitative positivity of the interaction, Gaussian-scale lower bounds, exact annihilation before full revelation, and a M"obius-specific collective capacity deficit are [REC] open targets. A definitive novelty claim would require a systematic MathSciNet, zbMATH, arXiv, matroid, lattice-path, and sieve-literature review beyond the present search.

19.2. **Universal state space and the master spectrum.** Let $S_{r,j} \subseteq \{-1,+1\}$ be the set of increments still allowed at compressed time j after layer r. Write $\mathcal{P}_r(m)$ for all words $Y=(Y_j)_{0\le j\le m}$ with $Y_0=0$ and $Y_j-Y_{j-1}\in S_{r,j}$, together with any explicitly stated Catalan, meander, endpoint, or return constraints. The master polynomial is

$$\boxed{\mathfrak{Z}_{r,m}(v,z,u)=\sum_{Y\in\mathcal{P}_r(m)} v^{\mathcal{R}_m(Y)} z^{Y_m} u^{\max_{0\le j\le m}|Y_j|}}, \qquad \mathcal{R}_m(Y)=\#\{1\le j\le m : Y_j=0\}. \tag{19.2}$$

It contains, by specialization, the total number of completions, the endpoint support, the return spectrum, and the maximum-height spectrum. In particular,

$$N_r(m)=\mathfrak{Z}_{r,m}(1,1,1), \qquad H_r^*(m)=\deg_u \mathfrak{Z}_{r,m}(1,1,u). \tag{19.3}$$

The endpoint polynomial alone is computed by the exact dynamic recursion

$$Z_{r,0}(h)=\mathbf{1}_{\{h=0\}}, \qquad Z_{r,j}(h)=\sum_{\delta\in S_{r,j}} Z_{r,j-1}(h-\delta). \tag{19.4}$$

The return-refined recursion is

$$F_{r,0}(h;v)=\mathbf{1}_{\{h=0\}}, \qquad F_{r,j}(h;v)=v^{\mathbf{1}_{\{h=0\}}}\sum_{\delta\in S_{r,j}} F_{r,j-1}(h-\delta;v). \tag{19.5}$$

Both are [DED] identities. They do not insert the actual zero set of X_m; the return times emerge from the coefficients. Feeding the Mertens zero skeleton into the recurrence would make any attempted proof of its return law circular.

For a fixed partial word $\sigma = (\sigma_j)_{j\in F}$, with free positions $U = [m] \setminus F$, the unconstrained endpoint polynomial is simply

$$P_F(z) = z^{\sum_{j\in F}\sigma_j}(z + z^{-1})^{|U|}. \tag{19.6}$$

Thus $|\mathcal{P}| = 2^{|U|}$ and the information revealed by fixing F is $|F|\log 2$. These numbers contain no Catalan geometry; height enters only after the prefix inequalities are retained.

In ordinary, uncompressed time the allowed increment set is

$$\Sigma_r(n) = \begin{cases} \{0\}, & n \text{ is not squarefree,} \\ \{(-1)^{\omega(n)}\}, & n \text{ is squarefree and } \omega(n) \le r, \\ \{-1, +1\}, & n \text{ is squarefree and } \omega(n) > r. \end{cases} \tag{19.7}$$

The analogue of (19.4) sums over $\delta \in \Sigma_r(n)$. Before a nonnegativity constraint is imposed this is an unrestricted ternary walk, not a Motzkin path. Compression removes the zero waiting times but not their occupation weights.

19.3. **Prefix-profile polynomial and insertion algebra.** For a path encoded by an up-step set $B \subseteq [2n]$, define

$$S_B(t) = 2|B \cap I_t| - t, \qquad I_t = [1, t]. \tag{19.8}$$

The completely refined profile enumerator is

$$\boxed{\mathcal{G}_{\mathcal{C}}(\mathbf{u}) = \sum_{B\in\mathcal{C}} \prod_{t=1}^{2n} u_t^{S_B(t)}}. \tag{19.9}$$

For a general base-word class this is a Laurent polynomial: negative prefix heights are possible. The maximum-height polynomial is obtained by grouping monomials according to $\max_t S_B(t)$, rather than by substituting a single value for the u_t.

A deletion–contraction slogan is insufficient here because deleting a coordinate changes both the ground-set indexing and every later prefix. The correct exact statement uses insertion operators. If a word w of length $m-1$ is inserted at coordinate e, set

$$(\iota_e^+ w)_j = \begin{cases} w_j, & j < e, \\ +1, & j = e, \\ w_{j-1}, & j > e, \end{cases} \qquad (\iota_e^- w)_j = \begin{cases} w_j, & j < e, \\ -1, & j = e, \\ w_{j-1}, & j > e. \end{cases} \tag{19.10}$$

Then for every fixed-length path class $\mathcal{C}$ partitioned by the step at e,

$$\mathcal{G}_{\mathcal{C}} = \mathscr{I}_e^+ \mathcal{G}_{(\iota_e^+)^{-1}\mathcal{C}} + \mathscr{I}_e^- \mathcal{G}_{(\iota_e^-)^{-1}\mathcal{C}}, \tag{19.11}$$

where $\mathscr{I}_e^{\pm}$ applies the corresponding causal shift to all prefix exponents from e onward. This is [DED]. It becomes an ordinary matroidal deletion/contraction formula only after the relevant minor, rank, and reindexing maps have been proved to agree.

19.4. **Catalan and lattice-path matroid fusion.** Dyck paths of semilength n are bases of the Catalan matroid under the standard up-step-position encoding [LIT] [1, 2]. Fixing up and down positions produces minors or, more generally, lattice-path corridors. Let F^+ be the forced up positions, F^- the forced down positions, $E' = [2n] \setminus (F^+ \cup F^-)$, and let M' denote the induced minor on E'. The exact largest compatible height at time t is

$$\boxed{h_{max}(t) = 2|F^+ \cap I_t| + 2r_{M'}(E' \cap I_t) - t.} \tag{19.12}$$

Consequently

$$H^* = \max_t h_{\max}(t), \qquad \mathcal{B}_{\ge H} = \varnothing \iff h_{\max}(t) < H \quad \text{for every } t. \tag{19.13}$$

These are [DED] extremal identities. Their value is that a proposed arithmetic restriction can be audited as a concrete prefix-rank deficit. Without an independently proved deficit, they merely re-encode the desired height bound.

There is a second exact computation which does not require matroid ranks. Let $A, D \subseteq [2n]$ be disjoint forced-up and forced-down sets. Define $c_t(h; H)$ to be the number of compatible prefixes of length t which stay in $[0, H]$ and finish at h. With $c_0(0; H) = 1$ and zero outside $0 \le h \le H$,

$$c_t(h;H) = \begin{cases} c_{t-1}(h-1;H), & t \in A, \\ c_{t-1}(h+1;H), & t \in D, \\ c_{t-1}(h-1;H) + c_{t-1}(h+1;H), & t \notin A \cup D. \end{cases} \tag{19.14}$$

If $N(A, D; H) = c_{2n}(0; H)$ and $N(A, D) = N(A, D; 2n)$, then

$$\boxed{H^*(A,D) = \min\{H : N(A,D;H) = N(A,D)\}.} \tag{19.15}$$

This is [DED]: increasing H restores exactly the paths whose maximum was previously cut off. It gives an independent check of (19.12) and an exact algorithm for every finite layer.

The up-step set $B = \{b_1 < \cdots < b_n\}$ is a Dyck path if and only if

$$b_i \le 2i - 1 \qquad (1 \le i \le n). \tag{19.16}$$

Indeed, immediately before the ith up step there are $i-1$ up steps and $b_i - i$ down steps; nonnegativity is equivalent to $i - 1 \ge b_i - i$. This is the basis description used by the Catalan matroid.

Proposition 19.1 (Exact prefix-capacity loss of a new layer)**.** *Suppose $\mathcal{C}(A, D)$ is nonempty and a new layer fixes previously free sets P up and N down. Put $A' = A \cup P$, $D' = D \cup N$, let M and M' be the corresponding nonempty Catalan minors, and let E_t denote the old free positions in I_t. Define*

$$\Delta_t = r_M(E_t) - |P \cap I_t| - r_{M'}(E_t \setminus (P \cup N)). \tag{19.17}$$

Then $\Delta_t \ge 0$ and

$$\boxed{h^*_{A,D}(t) - h^*_{A',D'}(t) = 2\Delta_t.} \tag{19.18}$$

Proof. Every independent set of the new minor, together with the contracted elements P, is independent in the old minor. This proves $\Delta_t \ge 0$. Subtract formula (19.12) for the two layers; the forced-up term increases by $2|P \cap I_t|$ and the two rank terms give exactly $2\Delta_t$. □

For a nested layer chain define

$$\Delta_{r,t} = \frac{1}{2}(h^*_{r-1}(t) - h^*_r(t)). \tag{19.19}$$

Then the exact telescope is

$$\boxed{h^*_r(t) = h^*_0(t) - 2\sum_{j=1}^{r} \Delta_{j,t}.} \tag{19.20}$$

Consequently, if one independently proves, uniformly in $t \le 2n$,

$$\sum_{j \le r} \Delta_{j,t} \ge \frac{1}{2} h^*_0(t) - C_\varepsilon n^{1/2+\varepsilon}, \tag{19.21}$$

then $H^*_r \le 2C_\varepsilon n^{1/2+\varepsilon}$. The implication is immediate from (19.20); the premise (19.21) is the open arithmetic content. Since $\Delta_{r,t}$ was defined from the height difference, the telescope itself is not progress toward that premise. A useful theorem must express or bound $\Delta_{r,t}$ from the locations and collective divisibility relations of the newly revealed layer.

Proposition 19.2 (Cardinality-only no-go)**.** *There is no uniform sublinear upper bound for $H^*(\mathcal{C})$ which depends only on $|\mathcal{C}|$ over all nonempty conditioned Catalan classes of varying semilength.*

Proof. Fix every step of the mountain U^nD^n. The resulting conditioned class has cardinality 1 and maximum height n. Thus classes of the same cardinality 1 have unbounded, indeed linear, height as n varies. □

The base polytope gives the equivalent dual formulation. If $P(M) = \mathrm{conv}\{\mathbf{1}_B : B \in \mathcal{B}(M)\}$, then

$$\max_{x \in P(M)} \left(2 \sum_{j \le t} x_j - t \right) = 2r_M(I_t) - t. \tag{19.22}$$

A nonnegative combination of valid inequalities for $P(M)$ which bounds the left side is a finite certificate excluding a height sector. We denote the minimum chosen norm of such a certificate by $\mathfrak{d}_M(H)$. This is a discovery invariant; its usefulness for M"obius depends entirely on whether its inequalities have arithmetic proofs.

19.5. **Sectoral entropy and the zero-count boundary.** For a nested chain $\mathcal{C}_0 \supseteq \mathcal{C}_1 \supseteq \cdots$, put

$$N_r = |\mathcal{C}_r|, \qquad N_r^{\ge H} = |\{P \in \mathcal{C}_r : H(P) \ge H\}|. \tag{19.23}$$

Whenever all four counts involved are positive, define

$$I_r = \log \frac{N_{r-1}}{N_r}, \qquad \Delta_r(H) = \log \frac{N_{r-1}^{\ge H}}{N_r^{\ge H}}, \qquad \boxed{\Xi_r(H) = \Delta_r(H) - I_r = \log \frac{N_{r-1}^{\ge H}/N_{r-1}}{N_r^{\ge H}/N_r}}. \tag{19.24}$$

Hence $\Xi_r(H) > 0$ means preferential elimination of high paths. The exact telescoping law is

$$\log \frac{N_0^{\ge H}/N_0}{N_R^{\ge H}/N_R} = \sum_{r=1}^{R} \Xi_r(H), \tag{19.25}$$

but only while $N_r^{\ge H} > 0$. If a layer makes this integer count zero, the logarithm is not a finite invariant: sector annihilation is recorded separately as the stronger, exact event $N_r^{\ge H} = 0$.

The same invariant admits an order-independent interaction calculus. Let Ω be a finite path class, $E_H = \{P \in \Omega : H(P) \ge H\}$, and let a restriction A also denote its surviving subset of Ω. Put

$$p_H = \frac{|E_H|}{|\Omega|}, \qquad p_H(A) = \frac{|A \cap E_H|}{|A|}, \qquad \Xi_H(A) = \log \frac{p_H}{p_H(A)}. \tag{19.26}$$

When $A \cap E_H = \varnothing$, set $\Xi_H(A) = +\infty$; this convention records the exact equivalence

$$\Xi_H(A) = +\infty \iff H^*(A) < H. \tag{19.27}$$

For two restrictions define the *sector-entropy interaction*

$$\boxed{\Delta_H^{\mathrm{sec}}(A,B) = \Xi_H(A \cap B) - \Xi_H(A) - \Xi_H(B) = \log \frac{p_H(A)p_H(B)}{p_H p_H(A \cap B)}}. \tag{19.28}$$

This symbol is distinct from the prefix-rank loss $\Delta_{r,t}$ in (19.19). In terms of the eight integer counts $N = |\Omega|$, $N_H = |E_H|$, $N_A = |A|$, $N_{A,H} = |A \cap E_H|$, and their B, AB analogues,

$$\boxed{\Delta_H^{\mathrm{sec}}(A,B) = \log \frac{N\, N_{A,H} N_{B,H} N_{AB}}{N_H N_A N_B N_{AB,H}}}. \tag{19.29}$$

These are [DED] algebraic identities, not probability assumptions.

If P is uniform on Ω and

$$\mathrm{Assoc}(A,B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)\mathbb{P}(B)}, \qquad \mathrm{Assoc}_H(A,B) = \frac{\mathbb{P}(A \cap B \mid E_H)}{\mathbb{P}(A \mid E_H)\mathbb{P}(B \mid E_H)}, \tag{19.30}$$

then Bayes' identity gives

$$\boxed{\Delta_H^{\mathrm{sec}}(A,B) = \log \frac{\mathrm{Assoc}(A,B)}{\mathrm{Assoc}_H(A,B)}.} \tag{19.31}$$

Thus positive interaction is equivalent to the restrictions being relatively less compatible inside the high sector than in the whole class. No universal sign is asserted. If $N_{A,H}, N_{B,H} > 0$ but $N_{AB,H} = 0$, the joint restriction annihilates the sector and $\Delta_H^{\mathrm{sec}}(A,B) = +\infty$.

Conditional on a previous restriction C, replace Ω by C. Explicitly,

$$\Delta_H^{\mathrm{sec}}(A,B \mid C) = \log \frac{p_H(A\cap C)p_H(B\cap C)}{p_H(C)p_H(A\cap B\cap C)}. \tag{19.32}$$

This is the correct form for comparing, for example, prime and semiprime layers after squarefreeness has already been imposed.

For layers $A_1,\dots,A_R$ and $A_S = \bigcap_{r\in S} A_r$, define the pure interaction by Boolean-lattice M"obius inversion,

$$\Delta_H^{\mathrm{sec}}(S) = \sum_{T\subseteq S} (-1)^{|S|-|T|}\Xi_H(A_T), \qquad A_\varnothing = \Omega. \tag{19.33}$$

Whenever all quantities are finite, inversion gives

$$\Xi_H(A_1\cap\cdots\cap A_R) = \sum_{\varnothing\neq S\subseteq[R]} \Delta_H^{\mathrm{sec}}(S). \tag{19.34}$$

This separates individual effects, pair interactions, and higher interactions. If a sector becomes empty, the finite expansion stops at the preceding layer and the annihilation event is retained separately.

Finally, while $N_R^{\geq H} > 0$, exact rearrangement gives

$$N_R^{\geq H} = N_R \frac{N_H}{N} \exp(-\Xi_H(A_1\cap\cdots\cap A_R)). \tag{19.35}$$

Therefore any independently proved finite upper bound making the right-hand side strictly smaller than 1 forces $N_R^{\geq H} = 0$ by integrality. This is a useful stopping principle, but substituting the unknown exact Ξ_H into its own identity would be circular.

For two restrictions A and B applied to the same baseline class, define the height synergy by the non-telescoping comparison

$$\boxed{\mathrm{Syn}(A,B) = (H_0 - H_{A\cup B}) - (H_0 - H_A) - (H_0 - H_B).} \tag{19.36}$$

Sequential height drops always telescope, so calling them superadditive would be an algebraic error. Formula (19.36) instead compares the joint intervention with the two standalone interventions.

Proposition 19.3 (Exact finite certificate, not a general synergy theorem)**.** *Among Dyck paths of semilength* 6*, impose the baseline condition that step* 6 *is down. Let A additionally require step* 7 *to be up, and let B additionally require step* 8 *to be up. Then*

(19.37)
$$(N_0,H_0) = (66,5), \quad (N_A,H_A) = (42,5), \quad (N_B,H_B) = (32,5), \quad (N_{A\cup B},H_{A\cup B}) = (19,4).$$

Thus $\mathrm{Syn}(A,B) = 1$*. Moreover, the numbers of paths with height at least* 5 *are respectively* $5,1,1,0$*: neither restriction alone removes the dangerous sector, while their union annihilates it.*

Proof. Enumerate the $C_6 = 132$ Dyck words and retain the words satisfying each set of coordinate conditions. Their height distributions for heights 1 through 5 are

(19.38)

	1	2	3	4	5	total
0	1	15	33	12	5	66
A	1	15	18	7	1	42
B	0	8	16	7	1	32
$A \cup B$	0	8	7	4	0	19.

The finite enumeration, the prefix-rank check (19.12), and the machine-readable certificate are reproduced by `catalan_fusion_v9.py`. □

The unrestricted Catalan class has a unique height-n mountain, so two coordinate restrictions that each preserve that unique maximizer must also preserve it jointly. Proposition 19.3 therefore needs an already partially restricted baseline with several competing maximizers; this qualification is mathematically essential.

Warning 19.4 (Status of the interaction language). The quantities Ξ_H, Δ_H^{sec}, their Bayes formula, chain rule, and higher-order inversion are elementary finite identities. They may provide useful notation for this program, but they are not substantive new theorems. The finite witness above proves only that one chosen pair of restrictions can have a joint effect. In unrestricted families Δ_H^{sec} can be positive, zero, negative, arbitrarily large, or infinite. No general sign, Gaussian lower bound, superadditivity law, rank consequence, or arithmetic sector annihilation has been proved. The first genuine problem is to specify a nontrivial structural class of Catalan restrictions for which the sign or size of Δ_H^{sec} can be determined without restating the height maximum. No bibliographic priority is claimed for that open problem.

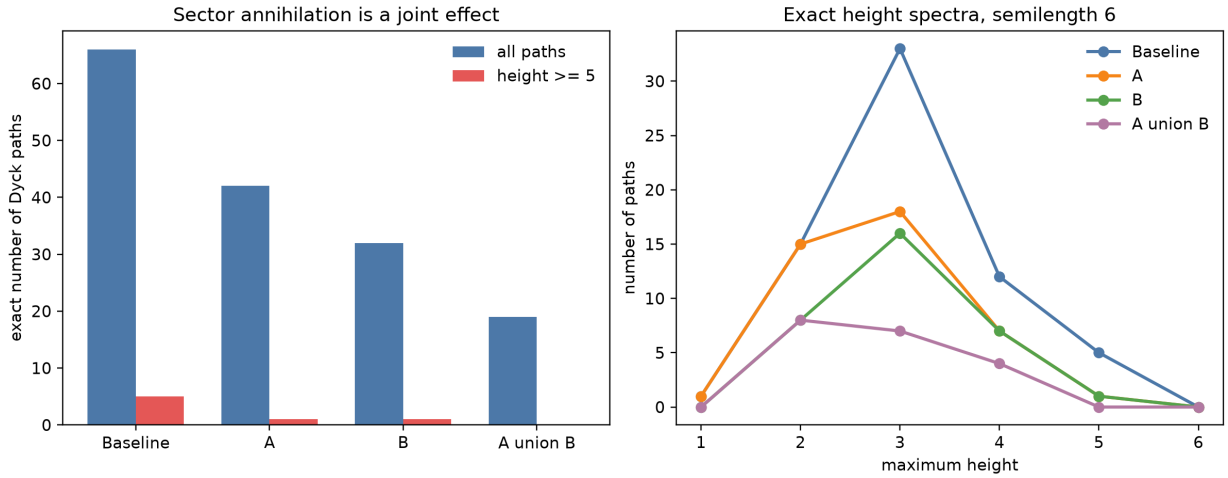


FIGURE 9. Exact Version 9 fusion audit. The left panel separates total class size from the dangerous sector $H \geq 5$; the right panel gives the complete maximum-height distributions. The union $A \cup B$ has 19 paths but no path of height 5, an exact joint effect rather than a probability fit.

19.6. **Arithmetic specialization and its exact limitations.** Let $q_1 < q_2 < \cdots$ be squarefree and reveal the signs with at most r prime factors. In compressed time,

$$S_{r,j} = \begin{cases} \{(-1)^{\omega(q_j)}\}, & \omega(q_j) \leq r, \\ \{-1, +1\}, & \omega(q_j) > r. \end{cases} \tag{19.39}$$

The true M"obius word belongs to every resulting class. Therefore a support bound for any partial layer implies the same bound for the true path. However,

$$\operatorname{supp}_u \mathfrak{Z}_{r,m} \subseteq [0, C_\varepsilon m^{1/2+\varepsilon}] \tag{19.40}$$

is stronger than RH whenever many completions remain: it confines every word compatible with the revealed information, not only M"obius. At the fully revealed layer the class is a singleton and (19.40) is exactly the Mertens criterion for RH, hence tautological as a route. A successful theorem must occur at a genuinely partial layer and its confinement must be proved from weaker arithmetic structure.

There is also a fixed-layer obstruction. For each fixed r, integers with $\omega(n) \le r$ have density zero [LIT] [3]; consequently the number of fixed positions among the first m squarefree integers is $o(m)$. By assigning the still-free signs coherently, one retains a terminal displacement $m - o(m)$ and hence a ballistic compatible path. Thus For the open unconstrained class this gives

$$H_r^*(m) = m - o(m) \qquad \text{for every fixed } r. \tag{19.41}$$

With a balanced Dyck endpoint at time $m = 2n$, the analogous construction still has height at least $n - o(n)$, hence remains ballistic. Similarly, the minimum number $R_r^-(m)$ of returns among compatible unconstrained completions can be kept bounded in terms of r by choosing a common sign on the free tail. The threshold must therefore grow with m, or the restrictions must be genuinely collective rather than pointwise.

The moving threshold

$$r(m) = \log\log q_m + \lambda(m)\sqrt{\log\log q_m} \tag{19.42}$$

is a legitimate [REC] starting point because Erdős–Kac and Sathe–Selberg theory locate the mass of ω on this scale. Yet a small unrevealed tail alone does not provide cancellation in the revealed sum. It must be coupled to rank deficits, capacity constraints, or exact sector annihilation.

Ordinary and compressed occupation energies are related without approximation. If $Q(x) = \#\{n \le x : \mu(n) \ne 0\}$ and $w_j(x) = \#\{n \le x : Q(n) = j\}$, then

$$\sum_{n \le x} |M(n)|^{2k} = \sum_{j \le Q(x)} w_j(x) |X_j|^{2k}. \tag{19.43}$$

This preserves the Catalan geometry of X_j together with the square-induced waiting times, without incorrectly calling the unrestricted ordinary path a Motzkin path.

19.7. **Collective restrictions: circuits, capacities, and prime bands.** Pointwise revelation is not the only admissible layer. A collective restriction has the form

$$|B \cap S| \le c(S), \tag{19.44}$$

and may arise from a circuit, a hyperedge of incompatible simultaneous up steps, a shadow inequality, or a colored prime-factor relation. Intersecting the Catalan base polytope with such inequalities need not remain a matroid; it may be a matroid intersection, greedoid, delta-matroid, or merely an integral independence system. The program will not assume matroidality where it has not been proved.

For prime bands $I_1, \dots, I_s$, label a squarefree integer by

$$\nu(n) = (\nu_1(n), \dots, \nu_s(n)), \qquad \mu(n) = (-1)^{\nu_1(n)+\cdots+\nu_s(n)}. \tag{19.45}$$

The order of revealing the bands is then an optimization problem: minimize information cost subject to a certified decrease of the high-height support. A fractional cover of prefixes by constraints (19.44) would yield a dual height certificate through (19.22). The desired bridge is therefore concrete:

$$\boxed{\begin{aligned} &\text{filtered multiplicative relation} \implies \text{proved prefix-capacity deficit} \\ &\qquad\qquad\qquad\qquad\qquad\quad \implies \text{Catalan height confinement.} \end{aligned}} \tag{19.46}$$

Only the second implication is currently universal; the first is the arithmetic problem.

19.8. **Operator, determinant, and causal-cutoff modules.** For a packet of restrictions $\mathscr{A}$, let $\mathcal{T}_{\mathscr{A}}$ act on the vector space of complete path profiles, and let $\Pi_{\geq H}$ project onto the dangerous sector. The sector norm

$$\kappa_H(\mathscr{A}) = \|\Pi_{\geq H}\mathcal{T}_{\mathscr{A}}\Pi_{\geq H}\| \tag{19.47}$$

tests whether compositions contract directions that every individual operator leaves untouched. Noncommutativity is potentially informative, but no general identity equating a commutator norm with height loss is asserted.

The same class can be counted by height-truncated transfer matrices,

$$Z_{\mathscr{R}}(m; h, H) = e_0^{\mathsf{T}} A_1^{(\mathscr{R})} \cdots A_m^{(\mathscr{R})} e_h. \tag{19.48}$$

Determinantal or inclusion–exclusion representations are valuable only if they force exact rank loss or zero coefficients in a forbidden sector; a generic determinant formula by itself is not confinement.

For prime operators $\mathscr{T}_p$, the multiplicative cutoff is causal. If Π_x retains only squarefree products at most x, the formal state has the shape

$$\mathfrak{Z}_x = \Pi_x \left(\prod_{p \leq x}^{\longrightarrow} \mathscr{T}_p \right) \mathfrak{Z}_\varnothing. \tag{19.49}$$

The defect

$$\mathscr{C}_x = \Pi_x \prod_p \mathscr{T}_p - \left(\prod_p \mathscr{T}_p \right) \Pi_x \tag{19.50}$$

records the temporal information lost when an Euler-product description is made independent of the size ordering. Formula (19.50) is a [REC] algebraic object, not yet an estimate.

19.9. **Fused work packages.** The many candidate routes are organized into five modules so that they do not become a catalogue of unrelated methods.

Module U: universal restriction algebra. Develop (19.2), (19.9), the insertion identity, rank profiles, open meanders, bilateral paths, and fixed-endpoint sectors. Determine which restriction operations preserve lattice-path matroid structure and which require a larger independence system.

Module S: sector invariants and exact certificates. Compute N, $N^{\geq H}$, H^*, $\Xi(H)$, $\mathfrak{d}_M(H)$, and the complete prefix-rank profile. Search for fractional-cover certificates and integer annihilation criteria. Proposition 19.3 is the first finite test case.

Module O: operators and algebraic counting. Construct transfer, inclusion–exclusion, determinant, and ordered-composition representations on the same profile space. Test whether compositions have a common-kernel loss in the high sector even when individual packets do not.

Module A: arithmetic specializations. Test moving ω thresholds, bilateral ω tails, prime-factor bands, colored divisibility classes, square waiting times, hypergraphs of incompatibilities, and filtered topological shadows. Every proposed capacity inequality must be proved arithmetically before it enters Module S.

Module D: discovery engines. Use exact dynamic programming, SAT/MaxSAT, integer and semidefinite relaxations, moment matrices, modular divisibility tests, and sign-reversing involutions to discover candidate constraints. Computational discovery creates conjectures and finite certificates, never an asymptotic theorem by itself.

19.10. **Milestones and fail gates.**

V9.1. Universal theorem. Prove the insertion/profile algebra and the rank-height/dual-certificate equivalence for the precise class of restrictions used later. *Fail gate:* if the statement only rewrites the maximum of the already known word, stop.

V9.2. Nontrivial synergy. Generalize Proposition 19.3 to a parametrized family with a quantitative joint height loss. *Fail gate:* total entropy loss without preferential loss of $N^{\geq H}$ does not count.

V9.3. Arithmetic constraint. Prove at least one collective capacity inequality from squarefreeness or factorization, without Mertens cancellation as an input. *Fail gate:* test the statement under $\mu \mapsto |\mu|$; if it survives unchanged, it has not yet used sign cancellation.

V9.4. Growing partial layer. Show confinement stronger than the ballistic bound while exponentially many completions remain. *Fail gate:* fixed r is excluded by (19.41); full revelation is the RH criterion in disguise.

V9.5. Scale escalation. Only after a genuine partial-layer theorem, seek $H^*(m) \ll m^{1/2+\varepsilon}$. The theorem must include every open meander, not only completed excursions or return times.

19.11. **Relation to the factor-four phenomenon.** The factor-four return observation remains the principal numerical target, but Version 9 does not assume the constant 4. The return polynomial (19.5) makes the stronger and cleaner question possible: determine the smallest and largest return exponents among all completions at a partial arithmetic layer, and show that some nontrivial return constraint emerges before the actual signs are fully revealed. Any theorem of the form

$$R(m) \sim c\sqrt{m} \quad (c > 0) \tag{19.51}$$

would be substantial, regardless of whether c equals the observed finite ratio. It would still control only an average completed-excursion length: at a return time,

$$\overline{L}(m) = \frac{m}{R(m)} \sim \frac{1}{c}\sqrt{m}. \tag{19.52}$$

It would not by itself bound the longest excursion or the final open meander. The fusion program therefore couples the return spectrum to the maximum-height and open-sector variables in the same polynomial (19.2).

19.12. **Logical boundary with RH.** Because $q_m \asymp m$ and $M(q_m) = X_m$,

$$M(x) = O_\varepsilon(x^{1/2+\varepsilon}) \iff X_m = O_\varepsilon(m^{1/2+\varepsilon}). \tag{19.53}$$

Therefore the statement "the actual M"obius path is not Catalan-exceptional", if *exceptional* is defined precisely as exceeding the RH scale, is exactly the Mertens criterion for RH. It is neither weaker nor a new lemma. The noncircular v9 target is different: prove that a class determined by strictly partial, independently established arithmetic restrictions has empty high sector. That target is stronger than RH as a conclusion about all compatible completions, but it may be approachable only if the restrictions create a new collective mechanism. This is the sole admissible route from the fusion theory to RH.

20. Research program retained from Version 8

The route can be advanced without assuming a probabilistic model for Möbius by attacking finite, exact questions.

20.1. **Exact finite combinatorics.** For each actual excursion or meander prefix, compute

$$(L, H, A, S_{\mathrm{Cat}}, \mathcal{I}_{\mathrm{Cat}}), \tag{20.1}$$

where L is duration, H is maximum height or current endpoint, A is area, S_{Cat} is Catalan entropy, and $\mathcal{I}_{\mathrm{Cat}}$ is the open-sector selection energy. The formulas for all ambient counts are exact or can be evaluated by stable dynamic programming/transfer matrices.

20.2. **Finite selector chambers.** For a finite set of primes and squarefree products, build the hyperplane arrangement (11.8). Determine which chambers generate Catalan excursions of a prescribed length and height. The first structural question is whether the total chamber volume of height-H words exhibits the same H^2/L rate as the uniform Catalan ensemble or a stronger one.

20.3. **Dynamic shadows.** Translate Björner's static shadow inequalities into statements on increments of face numbers over multiplicative windows

$$[x, xe^{\delta}]. \tag{20.2}$$

The target is a lower bound on opposite-parity arrivals conditional on a current Euler-characteristic imbalance.

20.4. **Transfer operators.** Construct a transfer operator whose states retain both Catalan height and a finite amount of arithmetic filtration data. The ordinary Dyck transfer matrix has spectral gap scale h^{-2}; an arithmetic deformation could reveal whether the prime-log selector adds a confining or restoring term.

20.5. **Hydrodynamic coarse graining.** If a local transport rule emerges, derive its conservation law and large-deviation functional. The TASEP example shows that Brownian-excursion fluctuations can coexist with Burgers hydrodynamics [20]. The corresponding Mertens question is whether the arithmetic selector generates an entropy condition that suppresses macroscopic parity shocks.

21. Relation to existing literature

The closest direct probabilistic work on Möbius located in this literature survey is the study of Mussardo and LeClair, which reorganizes the nonzero Möbius coefficients into a ± 1 walk and investigates Brownian/random-walk behaviour through statistical tests and block structures [23]. The present program differs in its primary invariant: the exact Catalan state space attached to each return excursion and its open meander sectors.

On the mathematical-probability side, conditioned random walks converging to Brownian excursion or meander are classical [13, 14, 15]; the maximum of a random Dyck path and its exponential moments and deviations are quantitatively understood [16, 17]. On the statistical-mechanics side, Dyck paths occur as vesicle/polymer configurations with genuine partition functions and free-energy differences [18, 19]. On the quantum side, Fredkin-type spin chains realize Dyck-path superpositions as physical states [21, 22]. On the non-equilibrium side, Brownian excursions arise in exact TASEP fluctuation calculations, with Burgers-type hydrodynamics at macroscopic scale [20].

On the arithmetic-topological side, Björner's complex gives the exact identity (11.2), shifted-complex structure, Betti asymptotics, and shadow inequalities, while explicitly leaving the growth problem for M unresolved [12]. The potential new intersection is therefore

$$\boxed{\text{filtered arithmetic selector} + \text{Catalan excursion state space} + \text{physical free-energy geometry}}. \tag{21.1}$$

The notebook treats the selector theorem as the decisive missing component rather than as an implicit randomness assumption.

22. Conclusions and open questions

The compressed Mertens function is an exact nearest-neighbour walk, and every complete zero-to-zero excursion is a primitive Dyck path counted by a Catalan number. This yields a canonical finite state space for each excursion and a natural open-boundary meander space for every point inside an excursion. Standard Catalan probability then supplies a quantitative geometric energy: large endpoint displacement or height is exponentially expensive on the scale H^2/L.

The central unresolved issue is selection. The arithmetic path is generated by a deterministic ordering of prime-subset energies and simultaneously by a filtered shifted simplicial complex whose Euler characteristic is Mertens. The project becomes viable only if one of these structures can be shown to bound the Catalan selection energy.

Version 8 retains the Version 5–Version 7 hidden-equivalence audits explicitly. The shell identity (16.2), transfer recursion (16.5), convolution product (16.7), and maximal Walsh coefficient (16.17) remain exact diagnostic identities. The Catalan-only no-go theorem, redundant-prime-lift proposition, and stability floor show why none of them currently supplies the missing cancellation.

The untwisted Matomäki–Teräväinen theorem supplies endpoint anti-ballisticism at every initial point, and the gentle-mountain construction shows why that chord information alone does not control the path interior. The twisted theorem supplies strictly more after the exact $J_H^* J_H$ diagonalization: Theorem 17.12 is an unconditional mesoscopic occupation-energy bound for all nested prefixes. The higher-uniformity theorem strengthens the logarithmic saving at a longer threshold. The one-sided Catalan tent inequality then yields the first explicit interior-peak bounds (17.46) and (17.47) in this notebook. Neither result gives a fixed power saving.

The finite return computation exposes a different and more intrinsically Catalan problem. The ratio $R(m)/\sqrt{2m/\pi}$ remains elevated across the computed scales and equals 3.9996 at the full cutoff, even though the length-two excursion frequency and the next-sign frequency are essentially unbiased. The exact entropy inequality (4.2) makes clear why this matters: returns are the multiplicative cuts of the Catalan class. The factor-four phenomenon is therefore the principal empirical target of Version 8. It is not promoted to an asymptotic law, and return abundance alone does not imply RH.

Version 9 adds a logically earlier objective. Before asking which arithmetic estimate proves confinement, it constructs a universal algebra in which weak Catalan restrictions can interact. The finite certificate of Proposition 19.3 exhibits one joint effect but establishes no general synergy principle and no bibliographic novelty. In particular, it does not show that prime-factor restrictions realize that effect. The next substantive theorem must therefore be an independently proved arithmetic capacity deficit that enters the prefix-rank or dual certificate identities. This dependency boundary prevents both the full-revelation tautology and the substitution of ambient Catalan rarity for a selector theorem.

The exact logical position is now sharper. The statement that the Mertens path is non-exceptional for the entire dyadic Catalan occupation-energy hierarchy is exactly equivalent to RH, by Theorem 17.7; it is neither weaker nor more restrictive. By contrast, demanding the Brownian energy scale uniformly in every shorter window is more restrictive and is not known to follow from RH. The route is therefore not closed. The signed low-mode Ramaré–Heath–Brown recombination in Conjecture 17.19 remains a calibrated analytic power-saving frontier, while the abstract dilation frame is explicitly ruled out. It is not treated as the explanation of the return anomaly; the distinct Catalan renewal problem is encoded exactly by (4.7).

The most concrete first targets, in updated priority order, are:

(i) extend the return-skeleton computation substantially beyond the present cutoff and decide whether the elevated local-time ratio and extreme-gap depletion persist;
(ii) seek a deterministic local-limit or renewal estimate for (4.7), or an arithmetic bound on the extreme tail in (4.5);

(iii) combine return abundance with an independent height or gap-maximum estimate—return abundance alone cannot control an open excursion;
(iv) prove any fixed power saving $\mathcal{E}_1(x,H) \ll H^{3-\delta}$ uniformly on a mesoscopic range;
(v) establish the rough-current energy bound and signed low-mode cross-channel estimate in Conjecture 17.19, keeping the polynomial-scale Heath–Brown factors visible;
(vi) test the sufficient numerical balance (17.65) for exact finite factor channels;
(vii) retain the Brownian CRSF (17.71)–(17.72) as a long-term target and the all-q dyadic hierarchy as an RH-complete criterion, not as the starting hypothesis;
(viii) derive filtered versions of Björner's shadow inequalities over multiplicative windows and compare their boundary terms with the rough Ramaré current;
(ix) prove nontrivial chamber-volume inequalities for parity words induced by positive additive subset weights;
(x) test whether persistent homology forces adjacent-parity compensation on the timescale of a Catalan excursion;
(xi) retain transfer-matrix, Fredkin, TASEP, q-Catalan, and polymer formalisms as diagnostic coordinates unless an independent arithmetic compatibility theorem is obtained;
(xii) subject every proposed bridge to Definition 17.29 before treating it as a route toward the uniform Catalan prefix bound.

A successful result need not identify Möbius with a random walk. Theorem 18.2 shows that the all-q dyadic occupation hierarchy is an exact RH-complete criterion: it avoids predicting individual signs, but it is not by itself a mechanism toward RH. Likewise the quadratic critical estimate $\sum_{m\le Y} X_m^2 \ll_\varepsilon Y^{2+\varepsilon}$ is an exact benchmark, not the first accessible lemma. The new noncircular milestone isolated in Version 8 is instead a fixed power saving

$$\mathcal{E}_1(x,H) \ll H^{3-\delta}x^{o(1)} \qquad (\delta>0) \tag{22.1}$$

on a polynomial mesoscopic range. It is strictly weaker than the Brownian scale $H^{2+o(1)}$, goes beyond the unconditional logarithmic savings of Theorem 17.12, and already improves the global exponent by Proposition 17.15.

The central conditional chain of Version 8 is

$$\begin{aligned} &\text{signed low-mode Ramaré–Heath–Brown estimate [open ADAPT]}\\ &\Longrightarrow \mathcal{E}_1(x,H) \ll H^{3-\delta+o(1)}\\ &\Longrightarrow |M(X)| \ll X^{1-\theta\delta/3+o(1)}. \end{aligned} \tag{22.2}$$

This chain does not yet imply RH; precisely for that reason it is not a restatement of RH. The remaining escalation, kept separately as the RH-complete long-term criterion, is

$$\begin{aligned} &\text{causal arithmetic occupation control for every fixed } q \text{ [open ADAPT]}\\ &\Longrightarrow \text{dyadic Catalan occupation hierarchy}\\ &\Longrightarrow |M(x)| \ll_\varepsilon x^{1/2+\varepsilon} \Longrightarrow \text{RH}. \end{aligned} \tag{22.3}$$

Theorem 17.7 supplies the second arrow and proves that the full dyadic hierarchy is itself equivalent to RH. No claim is made that the first arrow has been proved or reduced to a standard estimate. The separation between (22.2) and this RH-complete chain is deliberate: it prevents an exact criterion from being advertised as an independent advance toward RH.

Appendix A. Reproducible computations and archived outputs

The accompanying archive contains the Python programs printed below and their outputs. The binary input is not redistributed; it consists of $\mu(1),\dots,\mu(N)$ stored as signed bytes. For the supplied file with $N = 455{,}052{,}511$, the moment audit found 276,638,778 compressed steps, 53,078

returns, $M(N) = -5414$, and maximum observed absolute height 6695. The computed ratios $\mathcal{B}_k^{\#}/\Lambda_k$ for $k = 1, \dots, 6$ were respectively

$$0.4045606,\ 0.2511805,\ 0.1880514,\ 0.1549310,\ 0.1358477,\ 0.1250992. \tag{A.1}$$

These are finite measurements only and are not used as asymptotic premises.

The finite selector audit checked the prime-shell identity through 10^5 for $p = 2, 3, 5, 7, 11, 97$, the threshold-kernel identity through 30, the exact transfer factorization, and the Fourier product at four frequencies. All integer identities passed; the largest floating-point Fourier discrepancy was 9.54×10^{-14}.

Typical commands are

```
python3 code/mertens_moment_audit.py /path/to/moebius.bin \
  --output outputs/moment_audit.json
python3 code/prime_selector_identity_check.py \
  --output outputs/selector_check.json
python3 code/analiza_alturas_excursiones_mertens.py /path/to/moebius.bin \
  --output outputs/mertens_excursiones_alturas.csv
python3 code/tau_dephasing_audit.py \
  --output-png outputs/tau_dephasing.png \
  --output-json outputs/tau_dephasing.json
python3 code/plot_catalan_mertens_outputs.py \
  --moments outputs/moment_audit.json \
  --excursions outputs/mertens_excursiones_alturas.csv \
  --output-dir outputs
python3 code/catalan_no_go_audit_v5.py
python3 code/catalan_energy_ladder_v7.py
python3 code/jstarj_antiballistic_audit_v8.py
MPLBACKEND=Agg python3 code/catalan_fusion_v9.py \
  --json outputs/catalan_fusion_v9.json \
  --png outputs/catalan_fusion_v9.png
```

Appendix B. Python source: moment hierarchy audit

```python
#!/usr/bin/env python3
"""Blockwise audit of compressed-time Mertens moments and excursion budgets.

Input format: one signed int8 value mu(n) for each n=1,...,N.
The program never materializes the entire Mertens walk in memory.
"""

from __future__ import annotations

import argparse
import json
import math
from pathlib import Path

import numpy as np


def odd_double_factorial(k: int) -> int:
    value = 1
    for j in range(1, k + 1):
        value *= 2 * j - 1
    return value


def parse_args() -> argparse.Namespace:
    parser = argparse.ArgumentParser()
    parser.add_argument("filename", type=Path)
    parser.add_argument("--block", type=int, default=5_000_000)
```

```
    parser.add_argument("--max-k", type=int, default=6)
    parser.add_argument("--output", type=Path, default=Path("moment_audit.json"))
    return parser.parse_args()


def main() -> None:
    args = parse_args()
    if args.max_k < 1:
        raise ValueError("--max-k must be positive")

    size = args.filename.stat().st_size
    data = np.memmap(args.filename, dtype=np.int8, mode="r", shape=(size,))
    moments = np.zeros(args.max_k, dtype=np.float64)
    length_budgets = [0] * args.max_k

    mertens = 0
    compressed = 0
    previous_zero = 0
    returns = 0
    max_abs = 0

    for lo in range(0, size, args.block):
        hi = min(lo + args.block, size)
        block = np.asarray(data[lo:hi])
        nz = np.flatnonzero(block)
        if nz.size == 0:
            continue

        steps = block[nz].astype(np.int64, copy=False)
        walk = np.cumsum(steps, dtype=np.int64)
        walk += mertens
        absolute = np.abs(walk).astype(np.float64, copy=False)
        max_abs = max(max_abs, int(absolute.max(initial=0.0)))

        power = absolute * absolute
        for k in range(1, args.max_k + 1):
            if k > 1:
                power = power * absolute * absolute
            moments[k - 1] += float(np.sum(power, dtype=np.float64))

        hits = np.flatnonzero(walk == 0)
        for hit in hits:
            zero = compressed + int(hit) + 1
            gap = zero - previous_zero
            for k in range(1, args.max_k + 1):
                length_budgets[k - 1] += gap ** (k + 1)
            previous_zero = zero
            returns += 1

        mertens = int(walk[-1])
        compressed += int(nz.size)

    tail = compressed - previous_zero
    records = []
    for k in range(1, args.max_k + 1):
        budget = length_budgets[k - 1] + tail ** (k + 1)
        iid_leading = (
            odd_double_factorial(k) / (k + 1) * float(compressed) ** (k + 1)
        )
        records.append(
            {
                "k": k,
                "B_k": float(moments[k - 1]),
                "Lambda_k": str(budget),
                "B_k_over_Lambda_k": float(moments[k - 1] / budget),
```

```
                "B_k_over_iid_leading": float(moments[k - 1] / iid_leading),
            }
        )

    result = {
        "input": str(args.filename.resolve()),
        "N": size,
        "compressed_steps": compressed,
        "M_N": mertens,
        "returns_to_zero": returns,
        "last_zero_compressed": previous_zero,
        "open_tail_compressed": tail,
        "max_abs_M": max_abs,
        "moments": records,
        "interpretation": "finite computation only; no asymptotic inference",
    }
    args.output.parent.mkdir(parents=True, exist_ok=True)
    args.output.write_text(json.dumps(result, indent=2) + "\n", encoding="utf-8")
    print(json.dumps(result, indent=2))


if __name__ == "__main__":
    main()
```

Appendix C. Python source: exact selector identities

```
#!/usr/bin/env python3
"""Finite exact checks of the prime-shell, transfer, and Fourier identities."""

from __future__ import annotations

import argparse
import cmath
import json
import math
from pathlib import Path


def mobius_sieve(limit: int) -> list[int]:
    mu = [1] * (limit + 1)
    prime = [True] * (limit + 1)
    prime[0:2] = [False, False]
    primes: list[int] = []
    mu[0] = 0
    for n in range(2, limit + 1):
        if prime[n]:
            primes.append(n)
            mu[n] = -1
        for p in primes:
            if n * p > limit:
                break
            prime[n * p] = False
            if n % p == 0:
                mu[n * p] = 0
                break
            mu[n * p] = -mu[n]
    return mu


def primes_up_to(limit: int) -> list[int]:
    sieve = [True] * (limit + 1)
    sieve[0:2] = [False, False]
    for p in range(2, int(math.isqrt(limit)) + 1):
```

```
        if sieve[p]:
            for n in range(p * p, limit + 1, p):
                sieve[n] = False
    return [p for p in range(2, limit + 1) if sieve[p]]


def parse_args() -> argparse.Namespace:
    parser = argparse.ArgumentParser()
    parser.add_argument("--limit", type=int, default=100_000)
    parser.add_argument("--kernel-x", type=int, default=30)
    parser.add_argument("--output", type=Path, default=Path("selector_check.json"))
    return parser.parse_args()


def main() -> None:
    args = parse_args()
    mu = mobius_sieve(args.limit)
    mertens = [0] * (args.limit + 1)
    for n in range(1, args.limit + 1):
        mertens[n] = mertens[n - 1] + mu[n]

    test_primes = [p for p in (2, 3, 5, 7, 11, 97) if p <= args.limit]
    shell_failures = []
    for p in test_primes:
        filtered = [0] * (args.limit + 1)
        for n in range(1, args.limit + 1):
            filtered[n] = filtered[n - 1] + (mu[n] if n % p else 0)
        for x in range(1, args.limit + 1):
            shell = filtered[x] - filtered[x // p]
            if shell != mertens[x]:
                shell_failures.append({"p": p, "x": x})
                break

    kernel_primes = primes_up_to(args.kernel_x)
    atoms = [(1, 1)]
    for p in kernel_primes:
        atoms += [(product * p, -sign) for product, sign in atoms]
    atoms.sort()
    kernel_failures = []
    for x in range(1, args.kernel_x + 1):
        value = sum(sign for product, sign in atoms if product <= x)
        if value != mertens[x]:
            kernel_failures.append({"x": x, "K": value, "M": mertens[x]})

    convolution = {1: 1}
    for p in kernel_primes:
        updated = dict(convolution)
        for product, sign in convolution.items():
            updated[product * p] = updated.get(product * p, 0) - sign
        convolution = updated
    transfer_ok = sorted(convolution.items()) == atoms

    fourier_errors = []
    for tau in (0.125, 0.5, 1.0, 2.75):
        direct = sum(sign * cmath.exp(-1j * tau * math.log(product)) for product, sign in atoms)
        product = 1.0 + 0.0j
        for p in kernel_primes:
            product *= 1.0 - cmath.exp(-1j * tau * math.log(p))
        fourier_errors.append(abs(direct - product))

    result = {
        "limit": args.limit,
        "tested_shell_primes": test_primes,
        "shell_failures": shell_failures,
        "kernel_x": args.kernel_x,
```

```
        "kernel_primes": kernel_primes,
        "kernel_failures": kernel_failures,
        "transfer_factorization_ok": transfer_ok,
        "max_fourier_factorization_error": max(fourier_errors),
        "status": "PASS" if not shell_failures and not kernel_failures and transfer_ok else "FAIL",
        "interpretation": "finite verification of exact identities, not an RH test",
    }
    args.output.parent.mkdir(parents=True, exist_ok=True)
    args.output.write_text(json.dumps(result, indent=2) + "\n", encoding="utf-8")
    print(json.dumps(result, indent=2))


if __name__ == "__main__":
    main()
```

Appendix D. Python source: excursion heights and Catalan tails

```
#!/usr/bin/env python3
"""Extrae longitudes, alturas y areas de las excursiones de Mertens.

El fichero de entrada contiene mu(1),...,mu(N) como int8. Las posiciones
mu(n)=0 se eliminan y el paseo comprimido tiene pasos +/-1.
"""

from __future__ import annotations

import argparse
import csv
import math
import time
from pathlib import Path

import numpy as np


def parse_args() -> argparse.Namespace:
    parser = argparse.ArgumentParser()
    parser.add_argument("filename", type=Path)
    parser.add_argument("--block", type=int, default=5_000_000)
    parser.add_argument(
        "--output",
        type=Path,
        default=Path("mertens_excursiones_alturas.csv"),
    )
    return parser.parse_args()


def catalan_height_upper_tail(length: int, height: int) -> float:
    """P(maximo >= height) en la excursion uniforme de longitud ``length``.

    Se quitan el primer paso ascendente y el ultimo descendente. Queda un
    Dyck path de longitud 2*n=length-2 y altura maxima height-1. El conteo
    acotado se obtiene del espectro del grafo camino de alturas 0,...,a.
    """
    if length == 2:
        return 1.0
    n = length // 2 - 1
    a = height - 2
    if a < 0:
        return 1.0
    j = np.arange(1, a + 2, dtype=np.float64)
    theta = math.pi * j / (a + 2)
    eigen_abs = np.abs(2.0 * np.cos(theta))
```

```
    with np.errstate(divide="ignore"):
        terms = (
            math.log(2.0 / (a + 2))
            + 2.0 * np.log(np.sin(theta))
            + 2.0 * n * np.log(eigen_abs)
        )
    peak = float(np.max(terms))
    log_bounded = peak + math.log(float(np.exp(terms - peak).sum()))
    log_catalan = (
        math.lgamma(2 * n + 1) - math.lgamma(n + 1) - math.lgamma(n + 2)
    )
    log_cdf = min(0.0, log_bounded - log_catalan)
    return min(1.0, max(0.0, -math.expm1(log_cdf)))


def main() -> None:
    args = parse_args()
    size = args.filename.stat().st_size
    data = np.memmap(args.filename, dtype=np.int8, mode="r", shape=(size,))

    rows: list[tuple[int, int, int, int, int, int, int, int, int, float]] = []
    mertens = 0
    compressed = 0
    previous_zero_n = 0
    previous_zero_m = 0
    open_height = 0
    open_area = 0
    open_sign = 0
    started = time.monotonic()

    for lo in range(0, size, args.block):
        hi = min(lo + args.block, size)
        block = np.asarray(data[lo:hi])
        nz = np.flatnonzero(block)
        if nz.size:
            steps = block[nz].astype(np.int64, copy=False)
            walk = np.cumsum(steps, dtype=np.int64)
            walk += mertens
            hits = np.flatnonzero(walk == 0)

            start = 0
            for hit in hits:
                segment = walk[start : hit + 1]
                if segment.size:
                    if open_sign == 0:
                        open_sign = int(np.sign(segment[0]))
                    open_height = max(open_height, int(np.max(np.abs(segment))))
                    open_area += int(np.sum(np.abs(segment), dtype=np.int64))

                end_m = compressed + int(hit) + 1
                end_n = lo + int(nz[hit]) + 1
                length = end_m - previous_zero_m
                rows.append(
                    (
                        len(rows) + 1,
                        previous_zero_n,
                        end_n,
                        previous_zero_m,
                        end_m,
                        length,
                        open_sign,
                        open_height,
                        open_area,
                        open_height / math.sqrt(length),
                    )
```

```
                )
                previous_zero_n = end_n
                previous_zero_m = end_m
                open_height = 0
                open_area = 0
                open_sign = 0
                start = int(hit) + 1

            if start < walk.size:
                segment = walk[start:]
                if open_sign == 0:
                    open_sign = int(np.sign(segment[0]))
                open_height = max(open_height, int(np.max(np.abs(segment))))
                open_area += int(np.sum(np.abs(segment), dtype=np.int64))

            mertens = int(walk[-1])
            compressed += int(nz.size)

        block_number = lo // args.block + 1
        if block_number % 10 == 0 or hi == size:
            elapsed = time.monotonic() - started
            print(
                f"bloque={block_number:3d} n={hi:12d} m={compressed:12d} "
                f"M={mertens:7d} retornos={len(rows):7d} t={elapsed:7.1f}s",
                flush=True,
            )

    args.output.parent.mkdir(parents=True, exist_ok=True)
    columns = (
        "Return",
        "Start_n",
        "End_n",
        "Start_m",
        "End_m",
        "Gap",
        "ExcursionSign",
        "MaxHeight",
        "Area",
        "HeightOverSqrtGap",
        "ReducedHeightOverSqrtLength",
        "CatalanHeightUpperTailExact",
        "HeightFreeEnergyNats",
        "HeightFreeEnergyBits",
    )
    with args.output.open("w", newline="") as handle:
        writer = csv.writer(handle)
        writer.writerow(columns)
        for row in rows:
            length = row[5]
            height = row[7]
            reduced_scaled = (
                (height - 1) / math.sqrt(length - 2) if length > 2 else 0.0
            )
            upper_tail = catalan_height_upper_tail(length, height)
            free_nats = -math.log(upper_tail)
            writer.writerow(
                row
                + (
                    reduced_scaled,
                    upper_tail,
                    free_nats,
                    free_nats / math.log(2.0),
                )
            )

```

```
    gaps = np.fromiter((row[5] for row in rows), dtype=np.int64)
    heights = np.fromiter((row[7] for row in rows), dtype=np.int64)
    scaled = heights / np.sqrt(gaps)
    print("\nRESUMEN")
    print(f"N={size}")
    print(f"m={compressed}")
    print(f"M(N)={mertens}")
    print(f"retornos={len(rows)}")
    print(f"ultimo_retorno_n={previous_zero_n}")
    print(f"ultimo_retorno_m={previous_zero_m}")
    print(f"cola_m={compressed-previous_zero_m}")
    print(f"gap_max={int(gaps.max())}")
    print(f"altura_max_absoluta={int(heights.max())}")
    print(f"max_altura_sobre_raiz_gap={float(scaled.max()):.12g}")
    for q in (0.5, 0.9, 0.95, 0.99, 0.999):
        print(f"q{q:g}_altura_sobre_raiz_gap={np.quantile(scaled, q):.12g}")
    print(f"csv={args.output.resolve()}")


if __name__ == "__main__":
    main()
```

Appendix E. Python source: finite τ-dephasing illustration

```
#!/usr/bin/env python3
"""Illustrate the exact Cesaro dephasing law for finite prime products.

The plot is numerical evidence only.  The limiting formula used for the
horizontal reference lines is proved algebraically in the accompanying paper.
"""

from __future__ import annotations

import argparse
import json
import math
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np


PRIMES = np.array([2, 3, 5, 7, 11, 13, 17, 19], dtype=np.float64)


def parse_args() -> argparse.Namespace:
    parser = argparse.ArgumentParser()
    parser.add_argument("--tau-max", type=float, default=10_000.0)
    parser.add_argument("--points", type=int, default=200_001)
    parser.add_argument("--output-png", type=Path, default=Path("tau_dephasing.png"))
    parser.add_argument("--output-json", type=Path, default=Path("tau_dephasing.json"))
    return parser.parse_args()


def running_cesaro(values: np.ndarray) -> np.ndarray:
    return np.cumsum(values, dtype=np.float64) / np.arange(1, values.size + 1)


def main() -> None:
    args = parse_args()
    tau = np.linspace(0.0, args.tau_max, args.points)
    fig, axes = plt.subplots(1, 2, figsize=(11.0, 4.2), constrained_layout=True)
    records = []
```

```

    for j, color in ((4, "#1967ad"), (6, "#c43c39"), (8, "#4b8f29")):
        normalized = np.prod(
            np.abs(np.sin(0.5 * tau[:, None] * np.log(PRIMES[:j])[None, :])),
            axis=1,
        )
        square_mean = running_cesaro(normalized * normalized)
        target = 2.0 ** (-j)
        axes[0].plot(tau[:4001], normalized[:4001], color=color, lw=0.8, label=f"j={j}")
        axes[1].plot(tau[1:], square_mean[1:], color=color, lw=1.0, label=f"j={j}")
        axes[1].axhline(target, color=color, ls="--", lw=0.8, alpha=0.8)
        records.append(
            {
                "j": j,
                "tau_max": args.tau_max,
                "sample_cesaro_square": float(square_mean[-1]),
                "exact_infinite_cesaro_square": target,
            }
        )

    axes[0].set_title("Normalized prime sine product")
    axes[0].set_xlabel(r"$\tau$ (initial window)")
    axes[0].set_ylabel(r"$2^{-j}|\widehat{dK_j}(\tau)|$")
    axes[0].legend(frameon=False)
    axes[0].grid(alpha=0.2)

    axes[1].set_title("Running Cesaro mean of the square")
    axes[1].set_xscale("log")
    axes[1].set_yscale("log")
    axes[1].set_xlabel(r"frequency cutoff $T$")
    axes[1].set_ylabel("running mean; dashed = exact limit")
    axes[1].legend(frameon=False)
    axes[1].grid(alpha=0.2, which="both")

    args.output_png.parent.mkdir(parents=True, exist_ok=True)
    fig.savefig(args.output_png, dpi=220)
    plt.close(fig)
    result = {
        "records": records,
        "interpretation": "finite illustration; the limit is an exact theorem",
    }
    args.output_json.parent.mkdir(parents=True, exist_ok=True)
    args.output_json.write_text(json.dumps(result, indent=2) + "\n", encoding="utf-8")
    print(json.dumps(result, indent=2))


if __name__ == "__main__":
    main()
```

Appendix F. Python source: publication figures

```
#!/usr/bin/env python3
"""Create publication PNGs from the archived moment and excursion outputs."""

from __future__ import annotations

import argparse
import csv
import json
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
```

```python


def parse_args() -> argparse.Namespace:
    parser = argparse.ArgumentParser()
    parser.add_argument("--moments", type=Path, required=True)
    parser.add_argument("--excursions", type=Path, required=True)
    parser.add_argument("--output-dir", type=Path, required=True)
    return parser.parse_args()


def main() -> None:
    args = parse_args()
    args.output_dir.mkdir(parents=True, exist_ok=True)
    moment_data = json.loads(args.moments.read_text(encoding="utf-8"))
    records = moment_data["moments"]
    k = np.array([row["k"] for row in records])
    iid = np.array([row["B_k_over_iid_leading"] for row in records])
    budget = np.array([row["B_k_over_Lambda_k"] for row in records])

    fig, axes = plt.subplots(1, 2, figsize=(10.5, 4.0), constrained_layout=True)
    axes[0].semilogy(k, iid, "o-", color="#1967ad")
    axes[0].set_xlabel("moment order k")
    axes[0].set_ylabel(r"$\mathcal{B}_k^{\#}$/simple-walk leading moment")
    axes[0].set_title("Compressed Mertens moment ratios")
    axes[0].grid(alpha=0.25, which="both")
    axes[1].plot(k, budget, "o-", color="#c43c39")
    axes[1].set_xlabel("moment order k")
    axes[1].set_ylabel(r"$\mathcal{B}_k^{\#}/\Lambda_k$")
    axes[1].set_ylim(0.0, max(0.45, 1.08 * budget.max()))
    axes[1].set_title("Observed excursion-budget ratio")
    axes[1].grid(alpha=0.25)
    fig.savefig(args.output_dir / "moment_ratios.png", dpi=220)
    plt.close(fig)

    gaps: list[int] = []
    scaled: list[float] = []
    with args.excursions.open(newline="") as handle:
        for row in csv.DictReader(handle):
            gaps.append(int(row["Gap"]))
            scaled.append(float(row["HeightOverSqrtGap"]))
    gap = np.asarray(gaps, dtype=np.float64)
    height_scaled = np.asarray(scaled, dtype=np.float64)
    ordered = np.sort(gap)
    ccdf = (ordered.size - np.arange(ordered.size)) / ordered.size

    fig, axes = plt.subplots(1, 2, figsize=(10.5, 4.0), constrained_layout=True)
    axes[0].scatter(gap, height_scaled, s=5, alpha=0.18, color="#1967ad", edgecolors="none")
    axes[0].set_xscale("log")
    axes[0].set_xlabel("compressed excursion length L")
    axes[0].set_ylabel(r"$H/\sqrt{L}$")
    axes[0].set_title("Height across excursion scales")
    axes[0].grid(alpha=0.2, which="both")
    axes[1].loglog(ordered, ccdf, color="#c43c39", lw=1.2)
    axes[1].set_xlabel("compressed gap L")
    axes[1].set_ylabel(r"empirical $\Pr(\mathrm{Gap}\geq L)$")
    axes[1].set_title("Heavy-tailed return gaps")
    axes[1].grid(alpha=0.2, which="both")
    fig.savefig(args.output_dir / "excursion_geometry.png", dpi=220)
    plt.close(fig)

    print(args.output_dir / "moment_ratios.png")
    print(args.output_dir / "excursion_geometry.png")


if __name__ == "__main__":
```

```
    main()
```

Appendix G. Python source: Version 5 no-go audit retained in Version 6

```
#!/usr/bin/env python3
"""Finite audit for the Catalan--Mertens v5 criterion/mechanism note."""

from __future__ import annotations

import json
import math
from pathlib import Path

def catalan(n: int) -> int:
    return math.comb(2 * n, n) // (n + 1)


def primitive_dyck_paths(semilength: int):
    length = 2 * semilength

    def rec(pos: int, height: int, word: list[int]):
        if pos == length:
            if height == 0:
                yield tuple(word)
            return
        for step in (1, -1):
            new_height = height + step
            if new_height < 0:
                continue
            if pos + 1 < length and new_height == 0:
                continue
            if pos + 1 == length and new_height != 0:
                continue
            word.append(step)
            yield from rec(pos + 1, new_height, word)
            word.pop()

    yield from rec(0, 0, [])


def heights(word):
    out = [0]
    for step in word:
        out.append(out[-1] + step)
    return out


def mobius_sieve(limit: int):
    mu = [1] * (limit + 1)
    prime = [True] * (limit + 1)
    prime[0:2] = [False, False]
    for p in range(2, limit + 1):
        if not prime[p]:
            continue
        for n in range(p, limit + 1, p):
            prime[n] = False if n != p else prime[n]
            mu[n] *= -1
        square = p * p
        for n in range(square, limit + 1, square):
            mu[n] = 0
    mu[0] = 0
    return mu

```

```

def prime_dilation_error(mu, limit: int):
    mertens = [0] * (limit + 1)
    for n in range(1, limit + 1):
        mertens[n] = mertens[n - 1] + mu[n]
    primes = [p for p in range(2, 50) if all(p % d for d in range(2, int(p**0.5) + 1))]
    maximum = 0
    for p in primes:
        ap = [0] * (limit + 1)
        for n in range(1, limit + 1):
            ap[n] = ap[n - 1] + (mu[n] if n % p else 0)
        for x in range(1, limit + 1):
            maximum = max(maximum, abs(mertens[x] - (ap[x] - ap[x // p])))
    return maximum


def main():
    root = Path(__file__).resolve().parents[1]
    output = root / "outputs"
    output.mkdir(parents=True, exist_ok=True)

    rows = []
    for n in range(1, 8):
        paths = list(primitive_dyck_paths(n))
        maxima = [max(heights(path)) for path in paths]
        rows.append(
            {
                "semilength": n,
                "enumerated_primitive_paths": len(paths),
                "expected_catalan": catalan(n - 1),
                "largest_possible_height": max(maxima),
            }
        )

    mu = mobius_sieve(5000)
    identity_error = prime_dilation_error(mu, 5000)
    report = {
        "purpose": "audit identities and demonstrate the Catalan-only obstruction",
        "primitive_excursions": rows,
        "prime_dilation_identity_max_abs_error": identity_error,
        "conclusion": (
            "Primitive Catalan structure permits mountain excursions of height L/2; "
            "the prime-dilation average reproduces M exactly and supplies no bound."
        ),
    }
    (output / "catalan_no_go_audit_v5.json").write_text(
        json.dumps(report, indent=2), encoding="utf-8"
    )

    print(json.dumps(report, indent=2))


if __name__ == "__main__":
    main()
```

Appendix H. Python source: Version 6 mesoscopic Catalan audit

```
#!/usr/bin/env python3
"""Finite audit of the gentle-mountain construction in Version 6."""

from __future__ import annotations

import json
```

```python
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np


def gentle_mountain(r: int, blocks: int) -> np.ndarray:
    """Return the heights of W followed by the path retracing W."""
    up_block = np.r_[np.ones(r + 1, dtype=int), -np.ones(r, dtype=int)]
    first_half = np.tile(up_block, blocks)
    second_half = -first_half[::-1]
    steps = np.r_[first_half, second_half]
    return np.r_[0, np.cumsum(steps)]


def chord_profile(height: np.ndarray) -> np.ndarray:
    """Maximum absolute increment among all chords of each length."""
    length = height.size - 1
    profile = np.zeros(length + 1, dtype=int)
    for span in range(1, length + 1):
        profile[span] = np.max(np.abs(height[span:] - height[:-span]))
    return profile


def main() -> None:
    root = Path(__file__).resolve().parents[1]
    output_dir = root / "outputs"
    output_dir.mkdir(parents=True, exist_ok=True)

    r = 12
    blocks = 80
    height = gentle_mountain(r, blocks)
    profile = chord_profile(height)
    length = height.size - 1
    delta = 1.0 / (2 * r + 1)
    spans = np.arange(length + 1)
    deterministic_envelope = delta * spans + 4 * r

    assert height[0] == 0 and height[-1] == 0
    assert np.all(height[1:-1] > 0)
    assert np.all(profile <= deterministic_envelope + 1e-12)

    fig, axes = plt.subplots(2, 1, figsize=(9.2, 6.4), constrained_layout=True)
    axes[0].plot(height, color="#243b72", linewidth=1.25)
    axes[0].set_title("Primitive Catalan gentle mountain")
    axes[0].set_xlabel("compressed step")
    axes[0].set_ylabel("height")
    axes[0].grid(alpha=0.25)

    axes[1].plot(spans[1:], profile[1:] / spans[1:], color="#b33a3a",
                 label="maximum chord slope")
    axes[1].plot(spans[1:], deterministic_envelope[1:] / spans[1:],
                 color="#202020", linestyle="--",
                 label=r"$\delta+4r/s$")
    axes[1].axhline(delta, color="#238b57", linestyle=":",
                    label=r"block drift $\delta$")
    axes[1].set_xscale("log")
    axes[1].set_xlabel("chord length s")
    axes[1].set_ylabel("maximum |increment| / s")
    axes[1].set_ylim(0, 1.05)
    axes[1].grid(alpha=0.25)
    axes[1].legend(frameon=False)

    png_path = output_dir / "mt_catalan_gentle_mountain_v6.png"
    fig.savefig(png_path, dpi=220)
```

```
    plt.close(fig)

    audit = {
        "r": r,
        "blocks": blocks,
        "length": int(length),
        "maximum_height": int(height.max()),
        "block_drift_delta": delta,
        "primitive": bool(np.all(height[1:-1] > 0) and height[-1] == 0),
        "envelope_verified": bool(np.all(profile <= deterministic_envelope + 1e-12)),
        "png": str(png_path),
    }
    json_path = output_dir / "mt_catalan_gentle_mountain_v6.json"
    json_path.write_text(json.dumps(audit, indent=2) + "\n", encoding="utf-8")
    print(json.dumps(audit, indent=2))


if __name__ == "__main__":
    main()
```

Appendix I. Python source: Version 6 causal Ramaré prefix audit

```
#!/usr/bin/env python3
"""Finite exact audit of the prime-coloured Ramare prefix decomposition.

This is a falsification tool for Version 6, not evidence for an asymptotic
bound.  The default prime band is deliberately computational and is not
claimed to reproduce the optimized band in Matomaki--Teravainen.
"""

from __future__ import annotations

import argparse
import json
import math
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np


def mobius_and_spf(limit: int) -> tuple[np.ndarray, np.ndarray]:
    mu = np.ones(limit + 1, dtype=np.int8)
    spf = np.zeros(limit + 1, dtype=np.int32)
    primes: list[int] = []
    mu[0] = 0
    for n in range(2, limit + 1):
        if spf[n] == 0:
            spf[n] = n
            primes.append(n)
            mu[n] = -1
        for p in primes:
            value = p * n
            if value > limit:
                break
            spf[value] = p
            if n % p == 0:
                mu[value] = 0
                break
            mu[value] = -mu[n]
    return mu, spf


```

```python
def distinct_factors(n: int, spf: np.ndarray) -> list[int]:
    factors: list[int] = []
    while n > 1:
        p = int(spf[n])
        factors.append(p)
        while n % p == 0:
            n //= p
    return factors


def main() -> None:
    parser = argparse.ArgumentParser()
    parser.add_argument("--x", type=int, default=100_000)
    parser.add_argument("--theta", type=float, default=0.56)
    parser.add_argument("--P", type=int, default=3)
    parser.add_argument("--Q", type=int, default=97)
    parser.add_argument("--output", type=Path, default=None)
    args = parser.parse_args()

    h = max(2, int(round(args.x ** args.theta)))
    limit = args.x + h
    mu, spf = mobius_and_spf(limit)

    block_increments: dict[int, np.ndarray] = {}
    remainder_increment = np.zeros(h, dtype=float)
    direct_increment = mu[args.x + 1 : args.x + h + 1].astype(float)

    for offset, n in enumerate(range(args.x + 1, args.x + h + 1)):
        if mu[n] == 0:
            continue
        eligible = [p for p in distinct_factors(n, spf) if args.P < p <= args.Q]
        if not eligible:
            remainder_increment[offset] = float(mu[n])
            continue
        weight = 1.0 / len(eligible)
        for p in eligible:
            block = int(math.floor(math.log2(p)))
            block_increments.setdefault(block, np.zeros(h, dtype=float))
            # -mu(n/p)/nu_P(n) = mu(n)/nu_P(n) for squarefree n.
            block_increments[block][offset] += -float(mu[n // p]) * weight

    block_prefix = {
        block: np.cumsum(increments) for block, increments in block_increments.items()
    }
    remainder_prefix = np.cumsum(remainder_increment)
    direct_prefix = np.cumsum(direct_increment)
    reconstructed = remainder_prefix.copy()
    for values in block_prefix.values():
        reconstructed += values

    reconstruction_error = float(np.max(np.abs(reconstructed - direct_prefix)))
    diagonal = float(sum(np.dot(v, v) for v in block_prefix.values()))
    blocks = sorted(block_prefix)
    signed_off_diagonal = 0.0
    absolute_off_diagonal = 0.0
    for i, left in enumerate(blocks):
        for right in blocks[i + 1 :]:
            inner = float(np.dot(block_prefix[left], block_prefix[right]))
            signed_off_diagonal += 2.0 * inner
            absolute_off_diagonal += 2.0 * abs(inner)
    remainder_energy = float(np.dot(remainder_prefix, remainder_prefix))
    projected_energy = float(np.dot(direct_prefix, direct_prefix))
    crsf_absolute_total = remainder_energy + diagonal + absolute_off_diagonal

    root = Path(__file__).resolve().parents[1]
```

```
    output_dir = root / "outputs"
    output_dir.mkdir(parents=True, exist_ok=True)
    output_path = args.output or output_dir / "ramare_prefix_square_function_v6.json"

    result = {
        "finite_audit_only": True,
        "x": args.x,
        "H": h,
        "theta": args.theta,
        "prime_band": [args.P, args.Q],
        "dyadic_prime_blocks": blocks,
        "reconstruction_max_abs_error": reconstruction_error,
        "remainder_energy": remainder_energy,
        "diagonal_block_energy": diagonal,
        "signed_off_diagonal": signed_off_diagonal,
        "absolute_off_diagonal": absolute_off_diagonal,
        "crsf_absolute_total_over_H2": crsf_absolute_total / (h * h),
        "projected_prefix_energy_over_H2": projected_energy / (h * h),
        "maximum_prefix_height_over_sqrtH": float(
            np.max(np.abs(direct_prefix)) / math.sqrt(h)
        ),
    }
    output_path.write_text(json.dumps(result, indent=2) + "\n", encoding="utf-8")

    fig, axes = plt.subplots(2, 1, figsize=(9.2, 6.4), constrained_layout=True)
    axes[0].plot(direct_prefix, color="#202060", label="direct Mertens prefix")
    axes[0].plot(reconstructed, "--", color="#d05432", label="Ramare reconstruction")
    axes[0].set_ylabel("prefix sum")
    axes[0].grid(alpha=0.25)
    axes[0].legend(frameon=False)
    for block in blocks:
        axes[1].plot(block_prefix[block], linewidth=1.0,
                     label=fr"$2^{block}\leq p<2^{block + 1}$")
    axes[1].plot(remainder_prefix, color="black", linewidth=1.2,
                 label="rough remainder")
    axes[1].set_xlabel("prefix length")
    axes[1].set_ylabel("coloured current")
    axes[1].grid(alpha=0.25)
    axes[1].legend(frameon=False, ncol=2, fontsize=8)
    png_path = output_dir / "ramare_prefix_square_function_v6.png"
    fig.savefig(png_path, dpi=220)
    plt.close(fig)

    result["png"] = str(png_path)
    output_path.write_text(json.dumps(result, indent=2) + "\n", encoding="utf-8")
    print(json.dumps(result, indent=2))


if __name__ == "__main__":
    main()
```

Appendix J. Python source: Version 7 Catalan-energy exponent ladder

```
#!/usr/bin/env python3
"""Exact exponent audit for the Version 7 Catalan-energy hierarchy.

This program visualizes algebraic consequences of the proposed bounds.
It is not numerical evidence that any Catalan-energy estimate holds for
the Mobius function.
"""

from __future__ import annotations

```

```python
import json
from fractions import Fraction
from pathlib import Path

import matplotlib.pyplot as plt


def main() -> None:
    root = Path(__file__).resolve().parents[1]
    out_dir = root / "outputs"
    out_dir.mkdir(parents=True, exist_ok=True)

    theta = Fraction(11, 20)
    rows = []
    for q in range(1, 41):
        dyadic = Fraction(q + 1, 2 * q + 1)
        mesoscopic = Fraction(1, 1) - theta * Fraction(q, 2 * q + 1)
        rows.append(
            {
                "q": q,
                "dyadic_fraction": str(dyadic),
                "dyadic_decimal": float(dyadic),
                "critical_block_fraction": str(mesoscopic),
                "critical_block_decimal": float(mesoscopic),
            }
        )

    payload = {
        "status": "exact_formula_audit_not_arithmetic_evidence",
        "theta": str(theta),
        "dyadic_formula": "(q+1)/(2q+1)",
        "dyadic_limit": "1/2",
        "critical_block_formula": "1-theta*q/(2q+1)",
        "critical_block_limit": "29/40",
        "rows": rows,
    }
    json_path = out_dir / "catalan_energy_ladder_v7.json"
    json_path.write_text(json.dumps(payload, indent=2) + "\n", encoding="utf-8")

    q_values = [row["q"] for row in rows]
    dyadic_values = [row["dyadic_decimal"] for row in rows]
    mesoscopic_values = [row["critical_block_decimal"] for row in rows]

    fig, ax = plt.subplots(figsize=(9.2, 5.2))
    ax.plot(q_values, dyadic_values, "o-", lw=2, ms=4,
            label=r"dyadic: $(q+1)/(2q+1)$")
    ax.plot(q_values, mesoscopic_values, "s-", lw=2, ms=3,
            label=r"critical blocks: $1-\frac{11}{20}\frac{q}{2q+1}$")
    ax.axhline(0.5, color="tab:blue", ls="--", alpha=0.7,
               label=r"dyadic limit $1/2$")
    ax.axhline(29 / 40, color="tab:orange", ls="--", alpha=0.7,
               label=r"critical-block limit $29/40$")
    ax.set_xlabel(r"fixed moment index $q$")
    ax.set_ylabel("global exponent forced by the energy bound")
    ax.set_title("Catalan occupation-energy exponent ladder (exact algebra)")
    ax.grid(alpha=0.25)
    ax.legend(loc="best")
    fig.tight_layout()

    png_path = out_dir / "catalan_energy_ladder_v7.png"
    fig.savefig(png_path, dpi=220)
    plt.close(fig)

    print(f"Wrote {json_path}")
    print(f"Wrote {png_path}")
```

```
76
77
78 if __name__ == "__main__":
79     main()
```

Appendix K. Python source: Version 8 spectrum, Narayana, frame, and return audit

```
1  #!/usr/bin/env python3
2  """Finite audits for the exact Version 8 identities.
3
4  This program checks algebraic/combinatorial statements only.  It does not
5  test the open arithmetic power-saving conjecture and it uses no Mobius data.
6  """
7
8  from __future__ import annotations
9
10 import argparse
11 import csv
12 import itertools
13 import json
14 import math
15 from pathlib import Path
16
17 import matplotlib
18 import numpy as np
19
20 matplotlib.use("Agg")
21 import matplotlib.pyplot as plt
22
23
24 def occupation_kernel(h: int) -> tuple[np.ndarray, np.ndarray]:
25     j = np.tril(np.ones((h, h), dtype=float))
26     return j, j.T @ j
27
28
29 def closed_spectrum(h: int) -> tuple[np.ndarray, np.ndarray, np.ndarray]:
30     k = np.arange(1, h + 1, dtype=float)
31     theta = (2 * k - 1) * np.pi / (2 * h + 1)
32     eigenvalues = 1.0 / (4.0 * np.sin(theta / 2.0) ** 2)
33     n = np.arange(1, h + 1, dtype=float)[:, None]
34     vectors = 2.0 / np.sqrt(2 * h + 1) * np.sin((h + 1 - n) * theta)
35     return theta, eigenvalues, vectors
36
37
38 def spectrum_audit(h_values: list[int]) -> dict:
39     records = []
40     for h in h_values:
41         j, kernel = occupation_kernel(h)
42         theta, lam, vectors = closed_spectrum(h)
43         numerical = np.linalg.eigvalsh(kernel)[::-1]
44         rng = np.random.default_rng(1701 + h)
45         a = rng.choice([-1.0, 0.0, 1.0], size=h)
46         energy_direct = float(np.dot(j @ a, j @ a))
47         coeff = vectors.T @ a
48         energy_spectral = float(np.sum(lam * coeff**2))
49
50         n = np.arange(1, h + 1, dtype=float)
51         twists = np.array(
52             [np.sum(a * np.exp(-1j * th * n)) for th in theta],
53             dtype=complex,
54         )
55         coeff_from_twist = 2.0 / np.sqrt(2 * h + 1) * np.imag(
```

```
            np.exp(1j * (h + 1) * theta) * twists
        )
        finite_twist_rhs = (
            2.0 * h * (h + 1) / (2 * h + 1) * np.max(np.abs(twists)) ** 2
        )
        records.append(
            {
                "H": h,
                "kernel_formula_error": float(
                    np.max(
                        np.abs(
                            kernel
                            - np.fromfunction(
                                lambda r, s: h - np.maximum(r, s), (h, h)
                            )
                        )
                    )
                ),
                "orthonormality_error": float(
                    np.max(np.abs(vectors.T @ vectors - np.eye(h)))
                ),
                "eigen_equation_error": float(
                    np.max(np.abs(kernel @ vectors - vectors * lam))
                ),
                "eigenvalue_error": float(np.max(np.abs(numerical - lam))),
                "energy_reconstruction_error": abs(
                    energy_direct - energy_spectral
                ),
                "twist_coefficient_error": float(
                    np.max(np.abs(coeff - coeff_from_twist))
                ),
                "energy_over_finite_twist_bound": (
                    energy_direct / finite_twist_rhs if finite_twist_rhs else 0.0
                ),
            }
        )
    return {"records": records}


def shifted_dilation_audit() -> dict:
    rng = np.random.default_rng(8128)
    records = []
    for p in [2, 3, 5, 11, 17]:
        for length in [3, 8, 15]:
            for residue in range(min(p, 4)):
                ell = 1 + ((3 * length + residue) % p)
                b = rng.choice([-1.0, 1.0], size=length)
                h = residue + p * (length - 1) + ell
                lifted = np.zeros(h)
                lifted[residue + p * np.arange(length)] = b
                lhs = float(np.sum(np.cumsum(lifted) ** 2))
                child = float(np.sum(np.cumsum(b) ** 2))
                endpoint = float(np.sum(b) ** 2)
                rhs = p * child - (p - ell) * endpoint
                records.append(abs(lhs - rhs))
    return {
        "cases": len(records),
        "maximum_identity_error": float(max(records)),
    }


def high_mode_audit(h: int = 128) -> dict:
    rng = np.random.default_rng(314159)
    _, lam, vectors = closed_spectrum(h)
    worst_ratio = 0.0
```

```
    cases = 0
    for _ in range(40):
        a = rng.choice([-1.0, 1.0], size=h)
        coeff = vectors.T @ a
        norm2 = float(np.dot(a, a))
        for cutoff in [2, 4, 8, 16, 32, 64]:
            tail = float(np.sum(lam[cutoff:] * coeff[cutoff:] ** 2))
            exact_bound = float(lam[cutoff] * norm2)
            worst_ratio = max(worst_ratio, tail / exact_bound)
            cases += 1
    return {
        "H": h,
        "cases": cases,
        "maximum_tail_over_lambda_Kplus1_norm2": worst_ratio,
    }


def abstract_frame_counterexample(h: int = 24000) -> dict:
    r_values = [2, 4, 8, 16, 32, 48]
    ratios = []
    d0 = 170
    n = np.arange(1, h + 1)
    for channels in r_values:
        lifted = []
        for d in range(d0, d0 + channels):
            lifted.append(np.cumsum((n % d == 0).astype(float)))
        matrix = np.vstack(lifted)
        numerator = float(np.sum(np.sum(matrix, axis=0) ** 2))
        denominator = float(np.sum(matrix**2))
        ratios.append(numerator / denominator)
    return {"H": h, "channels": r_values, "ratios": ratios}


def narayana(n: int, k: int) -> int:
    if not (1 <= k <= n):
        return 0
    return math.comb(n, k) * math.comb(n, k - 1) // n


def dyck_paths(n: int):
    for up_positions in itertools.combinations(range(2 * n), n):
        up = set(up_positions)
        steps = tuple(1 if j in up else -1 for j in range(2 * n))
        heights = np.cumsum(steps)
        if np.min(heights) >= 0 and heights[-1] == 0:
            yield steps, heights


def narayana_audit(max_n: int = 8) -> dict:
    records = []
    max_identity_error = 0.0
    for n in range(1, max_n + 1):
        all_counts: dict[int, int] = {}
        primitive_counts: dict[int, int] = {}
        for steps, heights in dyck_paths(n):
            peaks = sum(
                steps[j] == 1 and steps[j + 1] == -1
                for j in range(2 * n - 1)
            )
            local_formula = (
                2 * n + 1
                - sum(steps[j] * steps[j + 1] for j in range(2 * n - 1))
            ) / 4
            max_identity_error = max(max_identity_error, abs(peaks - local_formula))
            all_counts[peaks] = all_counts.get(peaks, 0) + 1
```

```
            if n == 1 or np.min(heights[:-1]) > 0:
                primitive_counts[peaks] = primitive_counts.get(peaks, 0) + 1
        expected_all = {k: narayana(n, k) for k in range(1, n + 1)}
        if n == 1:
            expected_primitive = {1: 1}
        else:
            expected_primitive = {
                k: narayana(n - 1, k) for k in range(1, n)
            }
        records.append(
            {
                "n": n,
                "all_counts": all_counts,
                "expected_all": expected_all,
                "primitive_counts": primitive_counts,
                "expected_primitive": expected_primitive,
                "all_distribution_pass": all_counts == expected_all,
                "primitive_distribution_pass": (
                    primitive_counts == expected_primitive
                ),
            }
        )
    return {
        "maximum_peak_identity_error": max_identity_error,
        "records": records,
    }


def return_local_time_audit(excursions_csv: Path, moment_json: Path) -> dict:
    end_m = []
    return_index = []
    gaps = []
    signs = []
    with excursions_csv.open(newline="", encoding="utf-8") as handle:
        for row in csv.DictReader(handle):
            return_index.append(int(row["Return"]))
            end_m.append(int(row["End_m"]))
            gaps.append(int(row["Gap"]))
            signs.append(int(row["ExcursionSign"]))
    moment = json.loads(moment_json.read_text(encoding="utf-8"))
    total_m = int(moment["compressed_steps"])
    total_returns = int(moment["returns_to_zero"])
    iid_constant = math.sqrt(2.0 / math.pi)
    ratio = total_returns / math.sqrt(total_m)
    factor = ratio / iid_constant
    asymptotic_half_normal_tail = math.erfc(ratio / math.sqrt(2.0))
    flips = sum(signs[j] != signs[j - 1] for j in range(1, len(signs)))
    targets = [
        10**3,
        10**4,
        10**5,
        10**6,
        10**7,
        5 * 10**7,
        10**8,
        2 * 10**8,
        total_m,
    ]
    scale_table = []
    cursor = 0
    for target in targets:
        while cursor < len(end_m) and end_m[cursor] <= target:
            cursor += 1
        returns = cursor
        local_ratio = returns / math.sqrt(target)
```

```
        scale_table.append(
            {
                "m": target,
                "returns": returns,
                "R_over_sqrt_m": local_ratio,
                "factor_over_iid_mean": local_ratio / iid_constant,
            }
        )
    gap_survival_table = []
    for threshold in [2, 10, 100, 1000, 10_000, 100_000, 1_000_000, 10_000_000]:
        empirical = sum(gap > threshold for gap in gaps) / len(gaps)
        iid_asymptotic = math.sqrt(2.0 / (math.pi * threshold))
        gap_survival_table.append(
            {
                "threshold": threshold,
                "empirical_survival": empirical,
                "iid_asymptotic": iid_asymptotic,
                "ratio": empirical / iid_asymptotic,
            }
        )
    return {
        "compressed_steps": total_m,
        "returns": total_returns,
        "last_return_compressed": end_m[-1],
        "R_over_sqrt_m": ratio,
        "iid_mean_constant": iid_constant,
        "factor_over_iid_mean": factor,
        "half_normal_limit_tail_at_observed_ratio": asymptotic_half_normal_tail,
        "gap_2_count": sum(gap == 2 for gap in gaps),
        "gap_2_fraction": sum(gap == 2 for gap in gaps) / len(gaps),
        "gap_at_most_20_fraction": sum(gap <= 20 for gap in gaps) / len(gaps),
        "sign_flips": flips,
        "sign_flip_fraction": flips / (len(signs) - 1),
        "scale_table": scale_table,
        "gap_survival_table": gap_survival_table,
        "return_end_m": end_m,
        "return_index": return_index,
    }


def make_figure(report: dict, output: Path) -> None:
    output.parent.mkdir(parents=True, exist_ok=True)
    fig, axes = plt.subplots(2, 2, figsize=(12.8, 9.0))
    axes = axes.ravel()

    h = 96
    _, lam, _ = closed_spectrum(h)
    _, kernel = occupation_kernel(h)
    numerical = np.linalg.eigvalsh(kernel)[::-1]
    axes[0].loglog(range(1, h + 1), lam, lw=2.2, label="closed form")
    axes[0].scatter(range(1, h + 1), numerical, s=10, label="numeric")
    axes[0].set_title(r"Exact spectrum of $J^*J$")
    axes[0].set_xlabel("mode k")
    axes[0].set_ylabel(r"$\lambda_k$")
    axes[0].legend(frameon=False)

    counter = report["abstract_frame_counterexample"]
    axes[1].plot(counter["channels"], counter["ratios"], "o-", lw=2.2)
    axes[1].plot(counter["channels"], counter["channels"], "--", alpha=0.6)
    axes[1].set_title("Abstract frame obstruction")
    axes[1].set_xlabel("number of positive channels R")
    axes[1].set_ylabel("combined / diagonal energy")

    record = report["narayana"]["records"][6]
    ks = sorted(int(k) for k in record["all_counts"])
```

```
    observed = [record["all_counts"][str(k)] for k in ks]
    expected = [record["expected_all"][str(k)] for k in ks]
    axes[2].bar(np.array(ks) - 0.17, observed, width=0.34, label="enumerated")
    axes[2].bar(np.array(ks) + 0.17, expected, width=0.34, label="Narayana")
    axes[2].set_title("Dyck peaks, semilength 7")
    axes[2].set_xlabel("number of peaks")
    axes[2].set_ylabel("paths")
    axes[2].legend(frameon=False)

    local = report["return_local_time"]
    end_m = np.asarray(local["return_end_m"], dtype=float)
    return_index = np.asarray(local["return_index"], dtype=float)
    factor_curve = return_index / np.sqrt(end_m) / local["iid_mean_constant"]
    axes[3].semilogx(end_m, factor_curve, lw=1.25, color="#7137a8")
    axes[3].axhline(1.0, color="black", ls="--", lw=1.2, label="iid mean")
    axes[3].scatter(
        [local["compressed_steps"]],
        [local["factor_over_iid_mean"]],
        color="#d24b40",
        zorder=5,
        label="full cutoff",
    )
    axes[3].set_title("Catalan return local time")
    axes[3].set_xlabel("compressed time m")
    axes[3].set_ylabel(r"$R(m)/\sqrt{2m/\pi}$")
    axes[3].legend(frameon=False)

    fig.suptitle("Version 8 finite identity audit (not arithmetic evidence)")
    fig.tight_layout()
    fig.savefig(output, dpi=190, bbox_inches="tight")
    plt.close(fig)


def make_local_time_figure(report: dict, output: Path) -> None:
    output.parent.mkdir(parents=True, exist_ok=True)
    local = report["return_local_time"]
    end_m = np.asarray(local["return_end_m"], dtype=float)
    return_index = np.asarray(local["return_index"], dtype=float)
    factor_curve = return_index / np.sqrt(end_m) / local["iid_mean_constant"]
    fig, ax = plt.subplots(figsize=(9.4, 5.2))
    ax.semilogx(end_m, factor_curve, lw=1.45, color="#7137a8")
    ax.axhline(1.0, color="black", ls="--", lw=1.2, label="iid mean")
    ax.scatter(
        [local["compressed_steps"]],
        [local["factor_over_iid_mean"]],
        s=58,
        color="#d24b40",
        zorder=5,
        label=f"full cutoff: {local['factor_over_iid_mean']:.4f}",
    )
    ax.set_title("Compressed Mertens return local time")
    ax.set_xlabel("compressed time m")
    ax.set_ylabel(r"$R(m)/\sqrt{2m/\pi}$")
    ax.legend(frameon=False)
    ax.grid(alpha=0.18)
    fig.tight_layout()
    fig.savefig(output, dpi=200, bbox_inches="tight")
    plt.close(fig)


def main() -> None:
    parser = argparse.ArgumentParser()
    parser.add_argument(
        "--output-json",
        type=Path,
```

```
        default=Path("outputs/jstarj_antiballistic_audit_v8.json"),
    )
    parser.add_argument(
        "--output-png",
        type=Path,
        default=Path("outputs/jstarj_antiballistic_audit_v8.png"),
    )
    parser.add_argument(
        "--excursions-csv",
        type=Path,
        default=Path("outputs/mertens_excursiones_alturas.csv"),
    )
    parser.add_argument(
        "--moment-json",
        type=Path,
        default=Path("outputs/moment_audit.json"),
    )
    parser.add_argument(
        "--output-localtime-png",
        type=Path,
        default=Path("outputs/catalan_return_factor4_v8.png"),
    )
    args = parser.parse_args()

    report = {
        "scope": (
            "Finite checks of exact identities only; no test of the open "
            "Mobius power-saving estimate."
        ),
        "spectrum": spectrum_audit([3, 8, 17, 41, 96]),
        "shifted_dilation": shifted_dilation_audit(),
        "high_mode_tail": high_mode_audit(),
        "abstract_frame_counterexample": abstract_frame_counterexample(),
        "narayana": narayana_audit(),
        "return_local_time": return_local_time_audit(
            args.excursions_csv, args.moment_json
        ),
    }
    args.output_json.parent.mkdir(parents=True, exist_ok=True)
    args.output_json.write_text(json.dumps(report, indent=2), encoding="utf-8")
    # Reloading normalizes integer dictionary keys to strings for plotting.
    normalized = json.loads(args.output_json.read_text(encoding="utf-8"))
    make_figure(normalized, args.output_png)
    make_local_time_figure(normalized, args.output_localtime_png)

    spectrum_records = report["spectrum"]["records"]
    print("V8 exact-identity audit")
    print("max eigenvalue error:", max(r["eigenvalue_error"] for r in spectrum_records))
    print("max dilation error:", report["shifted_dilation"]["maximum_identity_error"])
    print("high-mode inequality ratio:", report["high_mode_tail"]["maximum_tail_over_lambda_Kplus1_norm2"])
    print("Narayana identities pass:", all(r["all_distribution_pass"] and r["primitive_distribution_pass"] for
      r in report["narayana"]["records"]))
    print("return factor over iid mean:", report["return_local_time"]["factor_over_iid_mean"])
    print("wrote", args.output_json)
    print("wrote", args.output_png)
    print("wrote", args.output_localtime_png)


if __name__ == "__main__":
    main()
```

Appendix L. Python source: Version 9 Catalan fusion and synergy certificate

```python
#!/usr/bin/env python3
"""Exact finite audit for the Version 9 Catalan-fusion example.

The script enumerates Dyck paths of semilength 6 and verifies that two
additional weak restrictions can have a genuinely joint effect on the
maximum height after a baseline restriction has created several maximizers.
It writes a machine-readable JSON certificate and a publication PNG.
"""

from __future__ import annotations

import argparse
import json
import math
from collections import Counter
from pathlib import Path

import matplotlib.pyplot as plt


def dyck_words(n: int) -> list[tuple[int, ...]]:
    words: list[tuple[int, ...]] = []

    def extend(word: tuple[int, ...], height: int, up: int, down: int) -> None:
        if up == n and down == n:
            words.append(word)
            return
        if up < n:
            extend(word + (1,), height + 1, up + 1, down)
        if down < n and height > 0:
            extend(word + (-1,), height - 1, up, down + 1)

    extend((), 0, 0, 0)
    return words


def height(word: tuple[int, ...]) -> int:
    level = 0
    maximum = 0
    for step in word:
        level += step
        maximum = max(maximum, level)
    return maximum


def select(words: list[tuple[int, ...]], forced: dict[int, int]) -> list[tuple[int, ...]]:
    # Positions are 1-based, as in the paper.
    return [w for w in words if all(w[position - 1] == step for position, step in forced.items())]


def summarize(words: list[tuple[int, ...]]) -> dict[str, object]:
    distribution = Counter(height(w) for w in words)
    h_star = max(distribution) if distribution else None
    high_count = distribution[h_star] if h_star is not None else 0
    return {
        "total": len(words),
        "maximum_height": h_star,
        "maximizer_count": high_count,
        "height_distribution": {str(h): distribution.get(h, 0) for h in range(1, 7)},
    }


def prefix_rank_formula(words: list[tuple[int, ...]]) -> list[int]:
    """Return max_t height from the exact prefix rank profile.

```

```
    For each prefix I_t, the rank-like quantity is the maximum number of up
    steps in I_t among the surviving paths. Thus 2*r(I_t)-t is exactly the
    largest compatible height at time t.
    """
    profile = []
    for t in range(1, 13):
        rank = max(sum(1 for step in w[:t] if step == 1) for w in words)
        profile.append(2 * rank - t)
    return profile


def bounded_dyck_count(n: int, forced: dict[int, int], ceiling: int) -> int:
    states = {0: 1}
    for position in range(1, 2 * n + 1):
        allowed = (forced[position],) if position in forced else (-1, 1)
        updated: dict[int, int] = {}
        for old_height, count in states.items():
            for step in allowed:
                new_height = old_height + step
                if 0 <= new_height <= ceiling:
                    updated[new_height] = updated.get(new_height, 0) + count
        states = updated
    return states.get(0, 0)


def main() -> None:
    parser = argparse.ArgumentParser()
    parser.add_argument("--json", default="../outputs/catalan_fusion_v9.json")
    parser.add_argument("--png", default="../outputs/catalan_fusion_v9.png")
    args = parser.parse_args()

    all_words = dyck_words(6)
    constraints = {
        "baseline": {6: -1},
        "A": {6: -1, 7: 1},
        "B": {6: -1, 8: 1},
        "A_union_B": {6: -1, 7: 1, 8: 1},
    }
    classes = {name: select(all_words, forced) for name, forced in constraints.items()}
    summaries = {name: summarize(words) for name, words in classes.items()}

    baseline_height = summaries["baseline"]["maximum_height"]
    assert len(all_words) == 132
    assert summaries["baseline"]["total"] == 66
    assert [summaries[key]["total"] for key in ("A", "B", "A_union_B")] == [42, 32, 19]
    assert [summaries[key]["maximum_height"] for key in ("baseline", "A", "B", "A_union_B")] == [5, 5, 5, 4]
    assert [summaries[key]["maximizer_count"] for key in ("baseline", "A", "B", "A_union_B")] == [5, 1, 1, 4]

    # Exact prefix-rank identity for each nonempty restricted class.
    rank_profiles = {name: prefix_rank_formula(words) for name, words in classes.items()}
    for name, profile in rank_profiles.items():
        assert max(profile) == summaries[name]["maximum_height"]

    bounded_dp = {
        name: {
            str(ceiling): bounded_dyck_count(6, constraints[name], ceiling)
            for ceiling in range(1, 7)
        }
        for name in classes
    }
    for name in classes:
        cumulative = 0
        for ceiling in range(1, 7):
            cumulative += summaries[name]["height_distribution"][str(ceiling)]
            assert bounded_dp[name][str(ceiling)] == cumulative
```

```python

    base_total = summaries["baseline"]["total"]
    base_high = summaries["baseline"]["maximizer_count"]
    sector_xi: dict[str, float | None] = {}
    for name in ("A", "B", "A_union_B"):
        total = summaries[name]["total"]
        high = sum(height(w) >= baseline_height for w in classes[name])
        if high == 0:
            sector_xi[name] = None
        else:
            sector_xi[name] = math.log((base_high / base_total) / (high / total))

    sector_interactions: dict[str, object] = {}
    for threshold in range(2, baseline_height + 1):
        counts = {
            name: sum(height(w) >= threshold for w in classes[name])
            for name in classes
        }
        p0 = counts["baseline"] / summaries["baseline"]["total"]
        p_a = counts["A"] / summaries["A"]["total"]
        p_b = counts["B"] / summaries["B"]["total"]
        p_ab = counts["A_union_B"] / summaries["A_union_B"]["total"]
        if p_ab == 0:
            interaction = "infinity"
        else:
            interaction = math.log((p_a * p_b) / (p0 * p_ab))
        sector_interactions[str(threshold)] = {
            "high_counts": counts,
            "delta_sec_A_B": interaction,
        }

    height_drop = {
        name: baseline_height - summaries[name]["maximum_height"]
        for name in ("A", "B", "A_union_B")
    }
    synergy = height_drop["A_union_B"] - height_drop["A"] - height_drop["B"]
    assert synergy == 1

    certificate = {
        "semilength": 6,
        "unrestricted_dyck_count": len(all_words),
        "constraints_1_based": constraints,
        "classes": summaries,
        "prefix_rank_height_profiles": rank_profiles,
        "bounded_height_dp_counts": bounded_dp,
        "sector_threshold": baseline_height,
        "sector_xi": sector_xi,
        "sector_interactions_by_threshold": sector_interactions,
        "sector_annihilated_by_union": sum(height(w) >= baseline_height for w in classes["A_union_B"]) == 0,
        "height_synergy": synergy,
    }

    json_path = Path(args.json)
    png_path = Path(args.png)
    json_path.parent.mkdir(parents=True, exist_ok=True)
    png_path.parent.mkdir(parents=True, exist_ok=True)
    json_path.write_text(json.dumps(certificate, indent=2, sort_keys=True) + "\n", encoding="utf-8")

    labels = ["Baseline", "A", "B", "A union B"]
    keys = ["baseline", "A", "B", "A_union_B"]
    totals = [summaries[k]["total"] for k in keys]
    high = [sum(height(w) >= baseline_height for w in classes[k]) for k in keys]
    distributions = [[summaries[k]["height_distribution"][str(h)] for h in range(1, 7)] for k in keys]

    fig, axes = plt.subplots(1, 2, figsize=(10.8, 4.2), constrained_layout=True)
```

```
    x = range(len(keys))
    axes[0].bar([i - 0.18 for i in x], totals, width=0.36, label="all paths", color="#4C78A8")
    axes[0].bar([i + 0.18 for i in x], high, width=0.36, label="height >= 5", color="#E45756")
    axes[0].set_xticks(list(x), labels)
    axes[0].set_ylabel("exact number of Dyck paths")
    axes[0].set_title("Sector annihilation is a joint effect")
    axes[0].legend(frameon=False)

    colors = ["#4C78A8", "#F58518", "#54A24B", "#B279A2"]
    for key, label, color, values in zip(keys, labels, colors, distributions):
        axes[1].plot(range(1, 7), values, marker="o", linewidth=2, label=label, color=color)
    axes[1].set_xlabel("maximum height")
    axes[1].set_ylabel("number of paths")
    axes[1].set_title("Exact height spectra, semilength 6")
    axes[1].set_xticks(range(1, 7))
    axes[1].legend(frameon=False)

    fig.savefig(png_path, dpi=220)
    print(json.dumps(certificate, indent=2, sort_keys=True))


if __name__ == "__main__":
    main()
```

References

[1] F. Ardila, *The Catalan matroid*, J. Combin. Theory Ser. A **104** (2003), no. 1, 49–62. arXiv:math/0209354.

[2] J. Bonin, A. de Mier, and M. Noy, *Lattice path matroids: enumerative aspects and Tutte polynomials*, J. Combin. Theory Ser. A **104** (2003), no. 1, 63–94. arXiv:math/0211188.

[3] G. H. Hardy and S. Ramanujan, *The normal number of prime factors of a number n*, Quart. J. Math. **48** (1917), 76–92.

[4] K. Matomäki and J. Teräväinen, *On the Möbius function in all short intervals*, J. Eur. Math. Soc. **25** (2023), no. 4, 1207–1225. DOI: 10.4171/JEMS/1205. arXiv:1911.09076.

[5] K. Matomäki, X. Shao, T. Tao, and J. Teräväinen, *Higher uniformity of arithmetic functions in short intervals I. All intervals*, Forum Math. Pi **11** (2023), e29. DOI: 10.1017/fmp.2023.28. arXiv:2204.03754.

[6] K. Matomäki, M. Radziwiłł, and T. Tao, *Sign patterns of the Liouville and Möbius functions*, arXiv:1509.01545v2, 2015.

[7] T. Tao, *Sign patterns of the Möbius and Liouville functions*, What's New, 6 September 2015, `https://terrytao.wordpress.com/2015/09/06/sign-patterns-of-the-mobius-and-liouville-functions/`.

[8] D. Hackmann, *Karhunen–Loève expansions of Lévy processes*, arXiv:1603.00677, 2016.

[9] C. Döbler, *Stein's method for the half-normal distribution with applications to limit theorems related to the simple symmetric random walk*, ALEA Lat. Am. J. Probab. Math. Stat. **12** (2015), no. 1, 171–191. arXiv:1303.4592.

[10] A. Hildebrand, *On consecutive values of the Liouville function*, L'Enseignement Math. **32** (1986), 219–226. DOI: 10.5169/seals-55088.

[11] S. N. Majumdar, *Brownian functionals in physics and computer science*, Current Science **89** (2005), 2076. arXiv:cond-mat/0510064.

[12] A. Björner, *A cell complex in number theory*, Adv. Appl. Math. **46** (2011), 71–85. DOI: 10.1016/j.aam.2010.09.007. arXiv:1101.5704.

[13] W. D. Kaigh, *An invariance principle for random walk conditioned by a late return to zero*, Ann. Probab. **4** (1976), 115–121. DOI: 10.1214/aop/1176996189.

[14] R. T. Durrett, D. L. Iglehart, and D. R. Miller, *Weak convergence to Brownian meander and Brownian excursion*, Ann. Probab. **5** (1977), 117–129. DOI: 10.1214/aop/1176995895.

[15] E. Csáki and S. G. Mohanty, *Excursion and meander in random walk*, Canad. J. Statist. **9** (1981), 57–70. DOI: 10.2307/3315296.

[16] O. Khorunzhiy and J.-F. Marckert, *Uniform bounds for exponential moment of maximum of a Dyck path*, Electron. Commun. Probab. **14** (2009), 327–333. DOI: 10.1214/ECP.v14-1486. arXiv:0902.1229.

[17] T. Kousha, *Asymptotic behavior and the moderate deviation principle for the maximum of a Dyck path*, Statist. Probab. Lett. **82** (2012), 340–347. DOI: 10.1016/j.spl.2011.10.015. arXiv:1008.0606.

[18] A. L. Owczarek and T. Prellberg, *Pressure exerted by a vesicle on a surface*, J. Phys. A: Math. Theor. **47** (2014), 215001. DOI: 10.1088/1751-8113/47/21/215001. arXiv:1311.2174.

[19] N. Haug and T. Prellberg, *Uniform asymptotics of area-weighted Dyck paths*, J. Math. Phys. **56** (2015), 043301. DOI: 10.1063/1.4917052.
[20] B. Derrida, C. Enaud, and J. L. Lebowitz, *The asymmetric exclusion process and Brownian excursions*, J. Stat. Phys. **115** (2004), 365–382. DOI: 10.1023/B:JOSS.0000019833.35328.b4. arXiv:cond-mat/0306078.
[21] L. Dell'Anna, O. Salberger, L. Barbiero, A. Trombettoni, and V. E. Korepin, *Violation of cluster decomposition and absence of light cones in local integer and half-integer spin chains*, Phys. Rev. B **94** (2016), 155140. DOI: 10.1103/PhysRevB.94.155140. arXiv:1604.08281.
[22] O. Salberger and V. Korepin, *Entangled spin chain*, Rev. Math. Phys. **29** (2017), 1750031. DOI: 10.1142/S0129055X17500313.
[23] G. Mussardo and A. LeClair, *Randomness of Möbius coefficients and Brownian motion: growth of the Mertens function and the Riemann hypothesis*, J. Stat. Mech. Theory Exp. (2021), 113106. DOI: 10.1088/1742-5468/ac22fb. arXiv:2101.10336.
[24] N. Ng, *The distribution of the summatory function of the Möbius function*, Proc. London Math. Soc. (3) **89** (2004), 361–389. DOI: 10.1112/S0024611504014721. arXiv:math/0310381.
[25] A. Verjovsky, *How random is the Möbius function? Smoothing, probability, and the Riemann hypothesis*, arXiv:2607.25002v2, 2026.
[26] K. Matulef, R. O'Donnell, R. Rubinfeld, and R. A. Servedio, *Testing halfspaces*, SIAM J. Comput. **39** (2010), 2004–2047. DOI: 10.1137/070707890.